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# Does national culture affect corporate innovation? International evidence<sup>☆</sup>

Narjess Boubakri<sup>a,\*</sup>, Imed Chkir<sup>b,1</sup>, Samir Saadi<sup>b,d</sup>, Hui Zhu<sup>c</sup><sup>a</sup> Bank of Sharjah Chair, School of Business Administration, American University of Sharjah, University City, PO Box 26666, Sharjah, United Arab Emirates<sup>b</sup> Telfer School of Management, University of Ottawa, 55 Laurier E., Ottawa, Ontario K1N 6N5, Canada<sup>c</sup> Ontario Tech University, 2000 Simcoe Street North, Oshawa, Ontario L1G 0C5, Canada<sup>d</sup> IPAG Business School, 184 Boulevard Saint-Germain, 75006 Paris, France

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## ABSTRACT

We examine whether and how national culture influences corporate innovation using a newly available comprehensive database on innovation around the world. After controlling for the impacts of formal institutions, and firm-level and country-level variables, we find that culture has relevance for innovation: The probability of a firm innovating is higher in individualistic, indulgent, and long-term oriented societies, as well as in cultures with less power distance, less uncertainty avoidance, and less masculine cultures. In the innovative firms subsample, we continue to find the same significant impact of culture on firms' innovation performance/quality. Our results are robust to endogeneity concerns, different model specifications, alternative measures of innovation and culture, and different subsample analyses.

## 1. Introduction

This paper examines whether and how national culture influences corporate innovation. In his pioneering work, Porter (1992) points out that firms must innovate to enhance their competitive advantage and survive in an increasingly globalized environment. Understanding the drivers of corporate innovation has since drawn extensive attention among academicians over time, particularly with respect to managerial incentives and skills, corporate governance, firm competitiveness, and size as key determinants of innovation. For instance, evidence in Manso (2011), Hirshleifer et al. (2012), Chang et al. (2015), Chen et al. (2015), and Cho et al. (2016) shows that innovation is higher with managerial contracts focused on rewarding long-term success, CEO overconfidence, non-executive employee stock options, managerial ability, and latent firm and manager characteristics, respectively. Focusing on

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\* Corresponding author.

E-mail addresses: [nboubakri@aus.edu](mailto:nboubakri@aus.edu) (N. Boubakri), [chkir@telfer.uottawa.ca](mailto:chkir@telfer.uottawa.ca) (I. Chkir), [samir.saadi@telfer.uottawa.ca](mailto:samir.saadi@telfer.uottawa.ca) (S. Saadi), [hui.zhu@utoriatechu.ca](mailto:hui.zhu@utoriatechu.ca) (H. Zhu).

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corporate governance mechanisms as determinants of innovation, other studies find that firms with greater analyst followings produce less innovation (i.e., fewer patents and patents with less impact) (He and Tian 2013), while firms with higher levels of institutional ownership and venture capital backing exhibit more innovation (Aghion et al. 2013; Chemmanur et al. 2014). In the same line of inquiry, Aghion et al. (2005) provide evidence of an inverted-U shaped relationship between product market competition and corporate innovation, arguing that high levels and low levels of competition hinder innovation.

The commonality of these previous studies is that they are conducted in single country settings. As such, they do not enable us to identify the country-level drivers of innovation and explain from where the observed cross-country variations in the level of corporate innovation could stem. Anecdotal evidence suggests that neither religion nor level of economic development provides a satisfying explanation for the observed cross-country variation in innovation rates and innovation impact.<sup>2</sup> In other words, beyond the firm- and industry-level drivers of innovation, it is likely that other potential characteristics of a country's institutions (formal, such as laws and regulations, or informal, such as culture and beliefs), may also be at play. Against this backdrop, we propose in this paper to focus on the particular impact of culture on corporate innovation. We follow Hofstede (1980, p. 25) and define culture as "the collective programming of the mind which distinguishes the members of one human group from another . . . the interactive aggregate of common characteristics that influence a human group's response to its environment." Specifically, we examine the relationships between Hofstede's cultural dimensions of power distance, individualism, uncertainty avoidance, masculinity, time orientation, and indulgence with national rates of innovation in a large set of 29 institutionally and culturally diverse countries, using an exhaustive international dataset on innovation: Derwent Innovation, operated by Clarivate Analytics.<sup>3</sup>

Our interest in culture as a determinant of innovation is motivated by the following considerations. First, few studies have suggested that culture may affect innovation. For instance, Saha (1998, p. 499) argues that differences in technological capabilities and innovation "can be explained satisfactorily only when cultural factors are considered." In the same line of thought, Herbig and Dunphy (1998, p. 14) further emphasize that "existing cultural conditions determine whether, when, how and under what form a new innovation will be adopted". In the same spirit, Jones and Davis (2000) claim that, when multinational firms decide to internationalize their R&D expenses, one important criterion to consider in the choice of the location must be national culture. These conjectures have been tested empirically at the firm or the individual level, but there is limited evidence on the impact of national culture at the country level (e.g., Shane 1992, 1993; Suzuki et al. 2002; Lee et al. 2000; Chen et al. 2017; Murswieck et al. 2020). Second, innovation is important in its own right as a driver of entrepreneurship and, hence, economic growth. The contribution of innovation to national economic growth has been well-established in the economic literature, on both theoretical (Solow 1956; Romer 1986) and empirical grounds (Gorodnichenko and Roland 2016). In fact, several countries nowadays consider innovation as an economic development strategy.<sup>4</sup> From a corporate strategy point of view, innovation is the engine of competitiveness and, thus, of conditions firms' survival in the long run.

To test the impact of culture on corporate innovation, we develop our hypotheses based on the six cultural attributes identified by Hofstede (1980), Hofstede and Bond (1988), and Hofstede et al. (2010). First, considering that power distance values reflect the importance of hierarchy, the centralization of power, and vertical communication patterns, we posit that these values/beliefs will deter innovation. Second, we hypothesize that individualistic cultural values (as opposed to collectivism) stressing freedom of entrepreneurship, rewards for success and creativity, competitiveness, and sustained individual efforts are likely to be positively related to innovation. Third, the uncertainty avoidance cultural dimension captures a society's tolerance for uncertainty and ambiguity. Societies with high levels of uncertainty avoidance are heavily structured with strict laws and rules to minimize the possibility of ambiguous situations. Given that innovation involves a long uncertain process with a high probability of failure (Holmstrom 1989), we expect such values to constrain the behavior of individuals and their innovation capabilities, leading to less innovation. Fourth, masculine cultural values capture assertiveness and strong levels of competitiveness (as opposed to care, supportive social climates, trust, and modesty in more feminine societies). Although one can expect great amounts of value placed on masculinity (i.e., high masculinity values) to enhance innovation capabilities, the opposing conjecture that innovation can be better nurtured in environments with low levels of masculinity cannot be excluded. We thus leave this issue to be resolved empirically. Fifth, we hypothesize that firms in long-term oriented societies, where individuals focus on future challenges and seek future growth and prosperity, are likely to be more innovative than firms in short-term oriented societies, where people prioritize immediate success. Finally, indulgent, as opposed to restrained, societies give more freedom to individuals in satisfying their desires, and are generally more optimistic. Accordingly, we expect indulgence to be associated with more innovation. Following the recommendations discussed in Beugelsdijk et al. (2017), we also control for the 11 cultural zones identified by Ronen and Shenkar (2013).

<sup>2</sup> Anecdotal evidence shows that some countries are indeed more innovative than others. For example, in its May 15th, 2015 edition, the Economist reports that more religious countries tend to be less innovative (The Economist 2015a). Although the article does not claim that religion causes an innovation deficit, it sustains that "theoretical models of government, in which political leaders are strongly influenced by religious institutions, may provide a channel for anti-scientific views to influence public policy", thus deterring innovation. This assertion, however, cannot explain why the US, which is classified as religious, is also a top innovative country. In a follow-up edition, the Economist (dated September 17th, 2015) ranks the world's most innovative countries using the Global Innovation Index. The findings show that, "in innovation, many countries in Africa punch above their economic weight measured by the level of economic development. Surprisingly, China is still behind American and other rich countries when it comes to innovation quality despite the fact it is producing a lot of patents" (The Economist 2015b).

<sup>3</sup> Previously known as Thomson Reuters Innovation, this database was sold in October 2016 by Thomson Reuters as part of its IP and science business to private equity firms. This division now operates under the name Clarivate Analytics.

<sup>4</sup> Please refer to the surveys on the determinants of economic growth conducted by Jones (2005), and Aghion and Howitt (2009).

Using a sample of 55,272 firm-year observations (27,636 with innovations and 27,636 without innovations) from 27 countries over the period from 1992 to 2016, we find support for our main conjectures. Particularly, when we compare innovative and non-innovative firms by running a logit analysis of the probability of innovating, after controlling for firm- and country-level variables, we find that all six cultural dimensions have a significant impact on the probability of a firm having patents or citations. Firms are more likely to innovate in individualist societies, less masculine societies, societies with low levels of power distance and low levels of uncertainty avoidance, and in indulgent and long-term oriented societies. The results continue to hold when the cultural dimensions are considered jointly. Our results are not only statistically significant, but are also economically sizable. All six cultural dimensions exhibit a high level of economic significance; power distance and individualism exhibit the highest levels. An increase in the level of individualism from the minimum to the maximum (which is from the level of individualism in South Korea to that in the US) would lead to a 23.7% decrease in the probability of a firm being innovative.

Next, we focus on the subsample of innovative firms and confirm that innovation rates are positively (negatively) related to individualism, long-term orientation, and indulgence (masculinity, high power distance, and high uncertainty avoidance).

One possible concern in our analysis is the potential endogeneity of culture. Although reverse causality is unlikely to run from innovation to culture, most culture studies maintain that values, beliefs, and preferences are durable and are thus very slow to change (Guiso et al. 2006). We run a two-stage least square (2SLS) regression model using the prevalence of infectious diseases as an instrument for culture. The findings remain qualitatively unchanged. Our results continue to hold when we implement an instrumental generalized method of moments (GMM) approach to further alleviate concerns with potential endogeneity problems. Our evidence also survives a battery of robustness checks that includes alternative measures of innovation, alternative measures of culture (namely, Tang and Koveos's (2008) revised cultural values), different model specifications, country-level regressions, and different subsample analyses, and using an alternative widely-used database on innovation (i.e., the NBER patent database).

To our knowledge, this is the first study to investigate the effects of all six cultural dimensions on firms' innovation and to follow Beugelsdijk et al.'s (2017) recommendation to control for cultural zones. Our results add to the existing literature on innovation on several grounds. First, we provide robust evidence that culture affects innovation at the country level, thus adding to the literature on the corporate outcomes of culture (e.g., Chen et al. 2017; Kanagaretnam et al. 2011; Li et al. 2013). Second, rather than using R&D, as in Chen et al. (2017), a common measure of (input to) innovation in the literature, we focus on measures of innovation output through patents, innovation impact through citations. By doing so, we put aside concerns that input to innovation may not necessarily lead to innovation. In fact, the traditional view that R&D is the driver of a firm's innovation ability and competitiveness has been increasingly questioned (e.g., Bound et al. 1984; Cohen et al. 1987; Arundel et al. 2008; Rammer et al. 2009). For instance, Arundel et al. (2008) show that about half of the innovative firms in Europe surveyed in the Community Innovation Survey (CIS) do not engage in R&D. Rammer et al. (2009) report similar results for the German manufacturing industry. According to the 2010 OECD Innovation Strategy report (OECD 2010), a large number of highly innovative firms do not conduct R&D at all. A comprehensive study conducted by IBM, (Global CEO Study 2006) shows that the CEOs of a great majority of organizations consider internal R&D as a small part of the entire innovation process; they identify business relationships as the main source of a firm's innovation prowess. The premise that higher expenditures on R&D do not necessarily translate into more innovative ability is also evidenced by the international innovation study conducted by PricewaterhouseCoopers (2016), which compares firms based on their R&D expenses. While Volkswagen tops the 2016 list of R&D spenders (\$13.2 billion), it only ranks 35 among innovative companies. In comparison, Apple leads the list as the most innovative company in the world, with "only" \$8.1 billion invested in R&D.

Third, our sample period is long enough to capture the emergence of new innovators, changes in economic cycles, and potential effects that might appear in the long run. In doing so, we go beyond prior studies and control for a large set of institutional, legal, and economic variables likely to affect national innovation rates, thus eliminating potential omitted variables bias. Finally, we add to the previous literature by examining the impact of culture on the probability of a firm innovating by comparing innovative and non-innovative firms. This contrasts with prior evidence from samples of innovative firms only (e.g., Chen et al. 2017), which is likely to lead to selection bias. The overall evidence from our study also adds to the economic growth literature by showing that the effect of culture on economic growth can be channeled through innovation.

The main implications of our results for managers and policy-makers are twofold. First, we show that culture indeed matters to innovation. As a consequence, increasing rates of innovation cannot be addressed by merely increasing investment in R&D. We argue that attitudes, cultural beliefs in society, and the programming of the mind need to be changed through government policies and programs in order to foster innovative behavior in the long run. Emphasizing individual efforts, enhancing competitiveness, putting in place reward systems, minimizing hierarchy, and improving tolerance for change and risk will likely lead to high innovation rates. This applies at both the government and the corporate level, since innovation is a primary driver of corporate and economic growth. Second, while Shane (1992) finds that the strength of the relationship between innovation and two cultural values (individualism and lack of power distance) were stronger in 1975 than in 1980, we show that, over a span of two and a half decades, six cultural dimensions pervasively impact innovation. We thus provide support for Hofstede et al.'s (2004, p. 195) assertion that "societies with

large power distance, stronger uncertainty avoidance, more bureaucracy, more corruption ... do not provide a promotional environment for innovation and entrepreneurship”.

The rest of the paper is organized as follows. In the next section, we review the literature and draw our hypotheses. In Section 3, we describe the data and variables, before we move on to Section 4 to discuss our empirical results and robustness checks. Section 5 concludes this paper.

## 2. Literature review and hypothesis development

In the first available study on the link between culture and innovation, [Shane \(1993\)](#) argues that organizational behavior reflects the cultural values of people who establish the organizations, and notes: “Organizational behavior reflects societal attitudes toward authority, trust, loyalty, commitment, motivation, control, discipline, communication, consultation, participation, coordination, and uncertainty”. Because differences in organizational behavior are shown to be associated with cross-country differences in terms of rates of innovation, we posit, following [Shane \(1993\)](#), that “national culture that explains the differences in human behavior should also be reflected in national differences in the country rates of innovation”. Although previous studies show that culture impacts innovation rates at the individual ([Islam and Zein 2020](#); [Wei et al. 2019](#); [McGrath et al. 1992](#)) and firm levels ([O'Regan et al. 2006](#); [Irani et al. 2004](#); [Stock and McDermott 2001](#)), evidence at the country level remains sparse. [Shane \(1993\)](#), for instance, examines a sample of 33 countries in 1975 and 1980, and finds that rates of innovation (measured as the per capita number of trademarks) are predominately influenced by weak levels of uncertainty avoidance and, to a lesser extent, by power distance and individualism. [Taylor and Wilson \(2012\)](#) use more recent data to test whether [Shane's \(1993\)](#) findings regarding individualism still hold, and call for more research that examines other cultural dimensions, besides individualism. In a sample of 28 European countries, [Murswieck et al. \(2020\)](#) find that indulgence, as a cultural dimension, helps to improve innovation measured by the European Innovation Scoreboard. It is noteworthy that findings from [Shane \(1993\)](#), [Taylor and Wilson \(2012\)](#), and [Murswieck et al. \(2020\)](#) are based on (global) country-level data regarding innovation and country-level controls, which explains the very small samples they employed. In this study, however, we use firm-level innovation data and we control not only for country-level but also for firm-specific factors that matter to corporate innovation. Our approach allows us to better draw out the influence of national culture on corporate innovation, and provides more power to our empirical analyses, due to the much larger sample employed. [Chen et al. \(2017\)](#) focus on individualism and uncertainty avoidance, and show that R&D expenditures scaled by total assets (as a measure of input to innovation) increase with individualism and decrease with uncertainty avoidance. However, as stated above, recent evidence suggests that R&D expenditures do not necessarily translate into more innovative prowess. [Griffin et al. \(2019\)](#) investigate the effect of board gender diversity on corporate innovation. They find that boards are more likely to include women in less masculine countries, which has a positive effect on firms' innovation.

[Hofstede \(1980\)](#) claims that innovation is directly related to entrepreneurship and that the cultural dimensions that influence entrepreneurial activities are also likely to determine innovation. [Hofstede's \(1980\)](#) cultural dimension have been widely used in the economics and finance literature ([Griffin et al. 2019](#); [Cheol et al., 2015](#); [Li et al. 2013](#); [Zheng et al. 2012](#); [Shao et al., 2010](#)). In what follows, we discuss the six cultural attributes developed by [Hofstede \(1980 and 2001\)](#), [Hofstede and Bond \(1988\)](#), and [Hofstede et al. \(2010\)](#), and their potential impact on innovation, drawing from their discussions related to entrepreneurship.

### 2.1. Power distance

[Hofstede's \(1980\)](#) power distance captures the extent to which the less powerful members of organizations and institutions accept that power is unequally distributed. Power distance is thus reflected in the importance of hierarchy, vertical communication patterns, the centralization of power, and the acceptance of the power distribution at hand. As a result, societies with high levels of power distance are characterized by centralized decision-making structures, the extensive use of formal rules, and limited information sharing. In this context, individuals have limited access to information sharing, which is detrimental to innovation ([Shane 1993](#)), and their actions are constrained by the hierarchy ([Van Everdingen and Waarts 2003](#)), leading [Shane \(1992\)](#) to conclude that “detailed instructions make employees passive and eliminate creative thinking”. In contrast, in societies with low levels of power distance, individuals have more freedom and more incentives to innovate ([Herbig and Dunphy 1998](#)). Classic studies by [Thompson \(1967\)](#), and [Burns and Stalker \(1961\)](#) indeed show that less hierarchy and freer communication improve innovation.

We thus draw our first hypothesis, as follows.

**H<sub>1</sub> :** *High (low) power distance is associated with less (more) innovation.*

It is noteworthy, however, that high power distance might have positive effects in the late stages of the innovation process, when it is time to make a decision on whether to adopt an original idea or to abandon an inconclusive process ([Zhang and Begley, 2011](#)).

## 2.2. Individualism

On one hand, individualism, as opposed to collectivism, captures the degree to which individuals are dissociated and disconnected from groups. In individualistic societies, values of individual freedom, personal decision-making, personal efforts, and initiative are encouraged and rewarded. In collectivist societies, people are integrated into strong, cohesive in-groups that disapprove of self-enrichment and prioritize collective interests. Consequently, innovation is fostered in individualistic societies, which provide a more dynamic and incentivizing environment for creativity, and is hindered in collectivist societies that constrain innovative behavior. In addition, individuals are more likely to expect recognition, compensation, and rewards in individualistic societies that encourage personal freedom. [Taylor and Wilson \(2012\)](#) find that “high-levels of cultural individualism correlate with national innovation rates”, implying that “individualism generally helps (and collectivism generally hurts) rates of technology patenting and scientific research publication”.

On the other hand, several studies in sociology and psychology suggest that the innovation process requires the sharing of knowledge, ideas, and information within diversified groups of individuals from different backgrounds ([Erez 1992](#); [Nemeth et al. 2004](#)). For example, brainstorming is a widely-used technique to generate new ideas. In fact, the larger the group is, the higher the number and quality of the generated ideas ([Osborn 1957](#)). Moreover, to be successful, original ideas need to be accepted and adopted collectively ([Stein 1953](#)). In this context, using brainstorming experiments, [Saad et al. \(2015\)](#) find that the quantity of generated ideas was substantially higher in their individualistic Canadian sample, whereas the quality (originality) of generated ideas was higher in their collectivistic Taiwanese sample.

Furthermore, the old stereotype describing collectivist Asian countries as imitators of the West, lacking innovative ideas, has now disappeared. Countries such as Japan and South Korea are becoming leaders in generating new patents.

Hence, it is not clear whether it is individualism or collectivism that fosters innovation. Thus, our non-directional second hypothesis is as follows.

**H<sub>2</sub> :** *Individualism (collectivism) is associated with innovation.*

## 2.3. Uncertainty avoidance

The cultural value of uncertainty avoidance reflects a society's tolerance for uncertainty and ambiguity. It indicates the extent to which a culture programs its members to feel either comfortable or uncomfortable in unstructured situations. [Hofstede \(1980\)](#) describes the uncertainty avoidance value as “what is different, is dangerous”. Uncertainty avoiding cultures shun uncertainty through strict laws and rules, regulations, and the enforcement of institutions. This, however, constrains opportunities to develop new innovations. Additionally, in uncertainty avoiding countries, societies are reluctant and less tolerant in the face of change, which is a characteristic of innovation.

Given that innovation is a highly uncertain process that involves a high probability of failure, and based on the arguments above, we hypothesize the following.

**H<sub>3</sub> :** *High (low) uncertainty avoidance is associated with less (more) innovation.*

It is worth noting, however, that some studies suggest that, in some circumstances, uncertainty avoidance might have a positive effect on innovation. For example, [Efrat \(2014\)](#) states that technological innovation is a mechanism that helps high uncertainty avoidance societies to maintain order. [Kedia et al. \(1992\)](#) argue that academic innovation is higher in countries with a high uncertainty avoidance index (UAI), as evidenced by the fact that great theoreticians and philosophers, such as Kant, Marx, Freud, Weber, and Popper, all came from countries with high UAIs, such as Germany and Austria. [Nakata and Sivakumar \(1996\)](#) assert that high uncertainty avoidance, as expressed by the penchant for planning, could facilitate new product development.

## 2.4. Masculinity

[Hofstede \(1980\)](#) argues that the cultural attribute of masculinity reflects assertiveness, competitiveness, rewards, and recognition for performance. Values that are seen as being masculine, such as assertiveness and competitiveness, contrast with values that are seen as being feminine, such as modesty, care, trust, and a lack of conflict. These views of femininity may seem too traditional in the 21st century. However, in a recent study, [Hofstede \(2011\)](#) asserts that these are more societal rather than individual cultural characteristic, which are based on the distance between men's and women's values in each country. For example, in many countries with a high masculinity score, such as Germany, women are assertive and competitive, but not as much as men.

A culture dominated by such masculine values as the orientation toward achievement and competition should thus be more conducive in terms of innovation than a more feminine society that is based on discretion, modesty, tolerance, and solidarity ([Nakata](#)

and Sivakumar 1996). This expectation is, however, not overwhelmingly supported by empirical evidence. Shane (1993), for instance, finds that masculinity has no effect on country-level innovation, measured by the number of trademarks per capita. In the same vein, Nakata and Sivakumar (1996) argue that low masculinity (i.e., soft values) “expressed by managerial attention to people and their inter-relations is conducive to new product development” (p. 65). Griffin et al. (2019) find that firms in less masculine countries include more woman in their boards, which translates into higher levels of innovation.

Accordingly, it is not clear whether the degree of masculinity in a given society is positively or negatively related to innovation. Thus, our non-directional fourth hypothesis is as follows.

**H<sub>4</sub>.** : *Masculine (feminine) cultural values are associated with innovation.*

## 2.5. Time orientation

Time orientation is the fifth cultural dimension, which was later added to the four previous dimensions by Hofstede and Bond (1988). Originally defined as ‘Pragmatic versus Normative’, the time orientation dimension captures the long-term versus short-term orientations of societies. Countries with a high score in this dimension are long-term oriented societies and typically prioritize the future. They focus on future challenges, and seek future growth and prosperity. Countries with a low score are short-term oriented and are typically more connected to traditions and social obligations. They prioritize short-term over long-term success. Individuals in long-term oriented societies tend to be more pragmatic, perseverant, and persistent (Hofstede 2011). They accept that results take time to achieve and tend to build long-term relationships with their employers.

In their theoretical paper, Lumpkin et al. (2010) argue that, the higher a business’s long-term orientation is, the higher levels of innovativeness it will achieve. In the same vein, Nakata and Sivakumar (1996) state that a long-term orientation should promote innovation in both the initiation and implementation stages of the development of a new product, whereas short-term orientation impedes such innovation by focusing on the preservation of past and present realities. Waarts and van Everdingen (2005) find that, within 10 European countries, firms with high levels of long-term orientation are significantly more likely to adopt new innovative products. Waarts and van Everdingen argue that, by focusing on future results, cultures with a long-term orientation are more receptive to long-term changes to the firm.

Because innovation is a long-term process that requires a high level of perseverance, and based on the above discussion, we hypothesize the following.

**H<sub>5</sub>.** : *A long (short)-term orientation is associated with more (less) innovation.*

## 2.6. Indulgence

Indulgence, as opposed to restraint, is a relatively new dimension of culture added by Hofstede et al. (2010). Indulgence captures the extent to which societies give freedom to their people in terms of satisfying their human desires, enjoying life, and having fun. In contrast, restrained societies impose strict social norms to control and regulate the gratification of desires and impulses. Hofstede (2011) argues that individuals living in indulgent societies are generally optimistic, encourage debate and dialogue in meetings or decision-making processes, and prioritize feedback, coaching, and mentoring. These characteristics may be considered as good assets to the complex innovation process. Indulgence prevails mostly in South and North America, and Western Europe, whereas restraint is more dominant in Eastern Europe and Asia.

Handoyo (2018) finds that the national innovative capacity of 77 countries, as measured by the Pillar Innovation of Global Competitiveness Index released by the World Economic Forum, is positively and significantly related to indulgence. In the same vein, a recent study by Murswieck et al. (2020) shows that indulgence, as a cultural dimension, is one of the main drivers of the leverage of national innovation performance of European Union countries.

Based on the discussion above, we hypothesize the following.

**H<sub>6</sub>.** : *Indulgence (restraint) is associated with more (less) innovation.*

## 2.7. Cultural zones

In addition to Hofstede’s six cultural dimensions, we control for the cultural zone to which each firm belongs.<sup>5</sup> Ronen and Shenkar (2013) identify 11 ecocultural global clusters covering 96 countries: African, Anglo, Arabic, Nordic, Germanic, Latin America, Near East, Latin Europe, East Europe, Far East, and Confucian. Their mapping focuses on three ecocultural variables (namely, geography, language, and religion), along with other economic variables. Ronen and Shenkar (2013) highlight the importance of clustering in theoretical and empirical international business studies on culture, as it renders the members of each cluster discernible, understood, and differentiated, while controlling for cultural, economic, and institutional similarities within the clusters.

Beugelsdijk et al. (2017) remark that cultural differences may be more region- than country-specific. They stress that, in addition to exploring intra-country cultural diversity, it is crucial for country-level cultural research to distinguish between country and culture by emphasizing supranational cultural zones.

<sup>5</sup> We are thankful to an anonymous reviewer for suggesting these additional controls.

In light of the discussion above, and following Cumming and Zhang (2019), we control for the 11 cultural zones identified by Ronen and Shenkar (2013), together with Hofstede's six cultural dimensions. This allows us to capture country-level differences in culture while controlling for integrations in cultural, institutional, economic, and other country-level omitted variables within each zone.

### 3. Data and variables

#### 3.1. Sample selection

The Derwent Innovation database, operated by Clarivate Analytics, provides around 2.5 million documents per year, covering more than 30 million published international patents from 59 patent-issuing authorities. It includes annual patent and citation information from 1963 on patent assignee names, the number of patents, the number of citations received by each patent, and each patent's application and grant year.<sup>6</sup> Following Hall et al. (2005), Fang et al. (2014), and Cumming et al. (2020a), we use the application year of an eventually granted patent instead of its grant year. Prior studies document that it is a patent's application year that generally has significant innovative influence in the market, as it better represents the actual time of the innovation (Griliches et al. 1987). In general, there are two truncation problems associated with the patent database. The first results from the time lag between the application year and grant year. For instance, an application filed in 2019 may still not appear in the database that year, since it can take up to two years to be granted. The second stems from the citation counts, as a patent can continue to receive citations even after the end year of the sample. To eliminate these two potential truncation problems, we follow Hall et al. (2001) and Hirshleifer et al. (2012), and end our sample period in 2016, to allow for a large post-patent granting window; we also scale each patent's citation counts by the average citation count of all patents in the same industry and year.<sup>7</sup>

We employ the Compustat North America and Compustat Global databases to obtain detailed firm-specific characteristics, such as firm size, R&D expenditures, return on assets, property, plant, and equipment expenditures, leverage, capital expenditures, cash holdings, and common dividends. We then match the unique key from the patent citation database with the GVKEY in the Compustat database.

We use the nation code from the Derwent Innovation database to merge our sample with country-level measures of (i) market size and financial development, retrieved from World Development Indicators (WDI), (ii) the corruption and bureaucracy index, retrieved from International Country Risk Guide (ICRG), and other country-specific corporate governance characteristics. To be included in the sample, a country must have more than three firms and have all country-level variables available.

Finally, using a propensity score matching approach, we match innovative firms with non-innovative firms with very similar corporate attributes. In particular, for each company with granted patents, we identify a matching firm from the pool of non-innovative companies based on the following criteria: i) both firms must be in the same country, industry class, and year; ii) they must have the same value of total assets; iii) they must have the same level of profitability; and iv) both firms must have close employment sizes. We use a non-repetitive matching scheme, so that non-innovative firms can be used only once in the matching procedure. Our matched sample has 55,272 firm-year observations from 27 countries over the period from 1992 to 2016. Of these observations, 27,636 firm-years are innovative, reporting over one million patents that have collected over seven million citations.

##### 3.1.1. Measures of corporate innovative activities

Following the recent innovation literature, we use the number of eventually granted patents and the number of received citations as our main measures of corporate innovation quantity and quality (Chkir et al., 2020; Cumming et al. 2020a; Cumming et al. 2020b). In particular, we use two measures to proxy for a firm's innovative activities. The first measure is  $IN\_DUMMY_{t+1}$ , which is a dummy variable that equals one if the firm has patents filled (and eventually granted) in year  $t + 1$ , and zero otherwise. This measure, however, does not differentiate firms with only one patent from those with multiple patents, which can bias the probability of innovation, as it is only relative to firms with no patents at all. Thus, in the first part of our multivariate analysis (logit model), following Cumming et al. (2020a), and Griffin et al. (2019), we also adopt an alternative three-year (then five-year) rolling period dummy variable ( $ROLL\_DUMMY$ ) to select a sample of active, high-quality, innovative firms. This dummy variable equals one for firms with patents and at least one citation during the three (then five) subsequent years, and zero otherwise.

To obtain our second measure,  $LNCITEPAT_{t+1}$ , we first define  $PATENTS$  as a firm's total number of patents filed (and eventually granted) in year  $t + 1$  and  $CITATIONS$  as the firm's total number of citations received in year  $t + 1$ . Following Hirshleifer et al. (2012), we scale each patent's citation counts by the average citation count of all patents in the same industry and year (hereafter, adjusted-citations) in an attempt to further deal with the truncation problem caused by the citations counts. Then,  $LNCITEPAT_{t+1}$  is calculated as the natural logarithm of one plus a firm's total number of adjusted-citations received, scaled by the average patent count (filed and eventually granted) of all patents in the same country and industry class in year  $t + 1$ .<sup>8</sup>

To ensure that our findings are not sensitive to the choice of measure of innovation, we augment our analysis by considering three

<sup>6</sup> It is worth noting that the Derwent Innovation database provides a wider coverage than the NBER patent database widely used in the innovation literature. First, the NBER database ends in 2006. Second, it provides information on patents granted in the US only. As pointed out by Chen et al. (2017), relying solely on a US patent database can provide skewed results. For example, companies in uncertainty-avoidance countries may be more willing to patent their inventions internationally, rather than in the US only.

<sup>7</sup> We also truncated our sample at years 2008 and then 2005 as an additional robustness check.

<sup>8</sup> We also used the average patent count at the firm level as an alternative denominator and obtained similar results.

**Table 1**

Frequency distribution of total number of patents and citations by year.

| Year  | N      | Total Number of Patents | Total Number of Citations |
|-------|--------|-------------------------|---------------------------|
| 1992  | 594    | 14,184                  | 233,729                   |
| 1993  | 586    | 11,163                  | 183,569                   |
| 1994  | 711    | 15,023                  | 242,381                   |
| 1995  | 884    | 21,562                  | 428,817                   |
| 1996  | 935    | 24,566                  | 497,973                   |
| 1997  | 1063   | 32,703                  | 607,727                   |
| 1998  | 1065   | 34,223                  | 594,871                   |
| 1999  | 1016   | 27,515                  | 578,106                   |
| 2000  | 1367   | 42,540                  | 608,359                   |
| 2001  | 1597   | 53,278                  | 566,337                   |
| 2002  | 1532   | 56,410                  | 531,555                   |
| 2003  | 1539   | 62,721                  | 434,072                   |
| 2004  | 1592   | 65,081                  | 370,411                   |
| 2005  | 1518   | 63,878                  | 280,565                   |
| 2006  | 1381   | 59,034                  | 186,261                   |
| 2007  | 1210   | 56,056                  | 155,866                   |
| 2008  | 1098   | 54,898                  | 101,576                   |
| 2009  | 902    | 48,986                  | 62,127                    |
| 2010  | 781    | 46,393                  | 39,602                    |
| 2011  | 527    | 44,246                  | 10,674                    |
| 2012  | 868    | 53,959                  | 45,629                    |
| 2013  | 1166   | 74,976                  | 139,323                   |
| 2014  | 1284   | 74,783                  | 119,285                   |
| 2015  | 1283   | 64,642                  | 68,682                    |
| 2016  | 1137   | 41,392                  | 42,427                    |
| Total | 27,636 | 1,144,212               | 7,129,924                 |

Note: This table reports the frequency distribution of total number of patents and citations by year. Column N shows the number of innovative firms in the year.

additional proxies for innovation. (1)  $LNCITE_{t+1}$ , defined as the natural logarithm of one plus a firm's total number of adjusted-citations received, scaled by the average citation count of all patents in the same industry class and year in year  $t + 1$ . (2)  $LNCITEPAT\_NADJ_{t+1}$ , defined in the same way as  $LNCITEPAT_{t+1}$ , except that each patent's citation count is not adjusted. (3) Finally, we use the latest version of the NBER patents and citations database widely adopted in the innovation literature to ensure the robustness of our results. Using the NBER patents and citations database, we define  $LNCITEPAT\_NBER_{t+1}$  as the natural logarithm of one plus a firm's total number of citations received, scaled by the number of patents filed (and eventually granted) in year  $t + 1$ .

### 3.1.2. Measures of culture

To measure a country's cultural values, we use Hofstede's (1980 and 2001), Hofstede and Bond's (1988), and Hofstede et al.'s (2010) six cultural dimensions: uncertainty avoidance (UAI), individualism (IND), masculinity (MAS), power distance (PDI), time orientation (TOR), and indulgence (IDG). The dimension of uncertainty avoidance measures the level of acceptance of uncertainty and ambiguity within a society. A country with a high uncertainty avoidance score is more risk averse. The dimension of individualism captures the degree to which individuals are disconnected from groups. Individualistic societies reward personal efforts and initiative. A high masculinity score indicates that the masculine values of assertiveness and competitiveness dominate in society. Likewise, the dimension of power distance focuses on the degree of equality/inequality and hierarchy between people within a society. High power distance suggests that a country accepts and perpetuates inequalities between individuals. The dimension of time orientation classifies societies into a long-term or short-term orientation category. Long-term oriented societies prioritize the future by focusing on future challenges and seeking future growth and prosperity. The dimension of indulgence captures the extent to which societies give freedom to their people in satisfying their human desires. Individuals in indulgent societies are typically optimistic and prioritize dialogue, feedback, and coaching in decision-making.

Following Cumming and Zhang (2019), and in line with the recommendations made by Kirkman et al. (2006), which are also discussed in Beugelsdijk et al. (2017), we control for the cultural zones identified by Ronen and Shenkar (2013) by adding 10 dummy variables representing the following zones: African, Anglo, Arabic, Nordic, Germanic, Latin America, Near East, Latin Europe, East

**Table 2**

Firm characteristics by country for innovative and non-innovative firms.

| COUNTRY                                                 | N       | PATENTS | CITATIONS | SIZE   | R&D   | ROA    | PPE    | LEV   | CAPIT | CASH  | DIV   |
|---------------------------------------------------------|---------|---------|-----------|--------|-------|--------|--------|-------|-------|-------|-------|
| Panel A: Innovative firms sample                        |         |         |           |        |       |        |        |       |       |       |       |
| Argentina                                               | 18      | 83      | 0         | 8.250  |       | 0.000  | 0.193  | 0.647 | 0.312 | 0.066 | 0.051 |
| Australia                                               | 94      | 202     | 49        | 6.836  |       | 0.057  | 0.062  | 0.358 | 0.215 | 0.068 | 0.169 |
| Austria                                                 | 54      | 103     | 33        | 6.632  |       | 0.008  | 0.126  | 0.324 | 0.251 | 0.063 | 0.091 |
| Belgium                                                 | 19      | 33      | 1         | 6.474  |       | 0.010  | 0.134  | 0.322 | 0.251 | 0.053 | 0.085 |
| Brazil                                                  | 26      | 48      | 0         | 9.422  |       | 0.003  | 0.120  | 0.427 | 0.300 | 0.072 | 0.137 |
| Canada                                                  | 225     | 34,483  | 2303      | 7.304  |       | 0.053  | 0.071  | 0.444 | 0.224 | 0.080 | 0.152 |
| China                                                   | 63      | 1739    | 568       | 9.865  |       | 0.011  | 0.137  | 0.439 | 0.197 | 0.092 | 0.173 |
| Denmark                                                 | 334     | 1521    | 26        | 6.946  |       | 0.029  | 0.095  | 0.359 | 0.258 | 0.070 | 0.146 |
| Finland                                                 | 72      | 606     | 61        | 5.986  |       | 0.050  | 0.123  | 0.317 | 0.225 | 0.074 | 0.130 |
| France                                                  | 152     | 2799    | 173       | 8.338  |       | 0.034  | 0.117  | 0.220 | 0.195 | 0.046 | 0.131 |
| Germany                                                 | 510     | 16,190  | 16,235    | 7.696  |       | 0.035  | 0.127  | 0.261 | 0.183 | 0.059 | 0.126 |
| Indonesia                                               | 61      | 635     | 0         | 12.132 |       | 0.001  | 0.115  | 0.457 | 0.417 | 0.081 | 0.145 |
| India                                                   | 9       | 9       | 0         | 9.744  |       | 0.004  | 0.207  | 0.259 | 0.130 | 0.063 | 0.150 |
| Italy                                                   | 19      | 150     | 5         | 8.003  |       | 0.013  | 0.180  | 0.286 | 0.189 | 0.053 | 0.102 |
| Japan                                                   | 6718    | 662,199 | 106,528   | 11.468 |       | 0.024  | 0.091  | 0.292 | 0.202 | 0.041 | 0.156 |
| Malaysia                                                | 1110    | 7729    | 128       | 5.938  |       | 0.001  | 0.094  | 0.389 | 0.204 | 0.048 | 0.142 |
| Netherlands                                             | 49      | 180     | 93        | 7.059  |       | 0.024  | 0.125  | 0.252 | 0.252 | 0.050 | 0.116 |
| New Zealand                                             | 22      | 54      | 49        | 7.956  |       | 0.002  | 0.136  | 0.546 | 0.292 | 0.066 | 0.077 |
| Norway                                                  | 106     | 512     | 27        | 8.180  |       | 0.012  | 0.134  | 0.305 | 0.157 | 0.078 | 0.140 |
| Singapore                                               | 46      | 394     | 0         | 7.094  |       | 0.023  | 0.082  | 0.328 | 0.197 | 0.075 | 0.248 |
| South Korea                                             | 204     | 5641    | 6436      | 10.970 |       | 0.034  | 0.156  | 0.357 | 0.194 | 0.090 | 0.145 |
| Spain                                                   | 54      | 251     | 6         | 9.563  |       | 0.002  | 0.121  | 0.326 | 0.332 | 0.055 | 0.074 |
| Sweden                                                  | 125     | 1305    | 271       | 8.613  |       | 0.034  | 0.098  | 0.308 | 0.196 | 0.052 | 0.157 |
| Switzerland                                             | 84      | 165     | 107       | 7.403  |       | 0.026  | 0.146  | 0.324 | 0.224 | 0.052 | 0.141 |
| Turkey                                                  | 28      | 336     | 6         | 8.116  |       | 0.008  | 0.182  | 0.312 | 0.185 | 0.053 | 0.129 |
| United Kingdom                                          | 377     | 1466    | 178       | 6.246  |       | 0.041  | 0.146  | 0.329 | 0.198 | 0.062 | 0.141 |
| United States                                           | 17,057  | 405,379 | 6996,641  | 5.691  |       | 0.123  | -0.013 | 0.199 | 0.186 | 0.051 | 0.299 |
| Panel B: Non-innovative firms                           |         |         |           |        |       |        |        |       |       |       |       |
| Argentina                                               | 395     | 0       | 0         | 7.066  | 0.000 | 0.162  | 0.462  | 0.221 | 0.073 | 0.101 | 0.026 |
| Australia                                               | 5836    | 0       | 0         | 5.560  | 0.011 | 0.097  | 0.301  | 0.201 | 0.061 | 0.124 | 0.043 |
| Austria                                                 | 375     | 0       | 0         | 6.532  | 0.006 | 0.121  | 0.368  | 0.250 | 0.074 | 0.105 | 0.019 |
| Belgium                                                 | 489     | 0       | 0         | 6.420  | 0.021 | 0.134  | 0.333  | 0.243 | 0.066 | 0.111 | 0.028 |
| Brazil                                                  | 1627    | 0       | 0         | 7.570  | 0.003 | 0.139  | 0.318  | 0.269 | 0.063 | 0.149 | 0.032 |
| Canada                                                  | 11,779  | 0       | 0         | 4.648  | 0.027 | -0.049 | 0.427  | 0.248 | 0.096 | 0.152 | 0.008 |
| China                                                   | 25,484  | 0       | 0         | 7.521  | 0.005 | 0.082  | 0.330  | 0.206 | 0.067 | 0.205 | 0.009 |
| Denmark                                                 | 477     | 0       | 0         | 6.701  | 0.019 | 0.094  | 0.345  | 0.261 | 0.075 | 0.150 | 0.015 |
| Finland                                                 | 304     | 0       | 0         | 5.316  | 0.036 | 0.081  | 0.301  | 0.227 | 0.073 | 0.155 | 0.031 |
| France                                                  | 1752    | 0       | 0         | 6.127  | 0.017 | 0.106  | 0.189  | 0.210 | 0.045 | 0.158 | 0.021 |
| Germany                                                 | 3126    | 0       | 0         | 6.024  | 0.020 | 0.107  | 0.245  | 0.205 | 0.050 | 0.145 | 0.023 |
| Indonesia                                               | 881     | 0       | 0         | 12.435 | 0.000 | 0.130  | 0.394  | 0.298 | 0.072 | 0.134 | 0.023 |
| India                                                   | 15,084  | 0       | 0         | 8.274  | 0.005 | 0.143  | 0.351  | 0.264 | 0.082 | 0.101 | 0.017 |
| Italy                                                   | 1200    | 0       | 0         | 6.873  | 0.007 | 0.094  | 0.276  | 0.243 | 0.047 | 0.115 | 0.017 |
| Japan                                                   | 41,410  | 0       | 0         | 10.331 | 0.011 | 0.088  | 0.294  | 0.193 | 0.035 | 0.189 | 0.010 |
| Malaysia                                                | 5182    | 0       | 0         | 5.935  | 0.001 | 0.099  | 0.355  | 0.199 | 0.049 | 0.149 | 0.023 |
| Netherlands                                             | 1158    | 0       | 0         | 6.708  | 0.025 | 0.099  | 0.230  | 0.242 | 0.052 | 0.128 | 0.023 |
| New Zealand                                             | 617     | 0       | 0         | 5.752  | 0.003 | 0.147  | 0.470  | 0.253 | 0.067 | 0.054 | 0.047 |
| Norway                                                  | 1342    | 0       | 0         | 7.385  | 0.008 | 0.101  | 0.322  | 0.280 | 0.074 | 0.167 | 0.027 |
| Singapore                                               | 4609    | 0       | 0         | 5.486  | 0.002 | 0.095  | 0.299  | 0.186 | 0.056 | 0.194 | 0.026 |
| South Korea                                             | 10,036  | 0       | 0         | 12.168 | 0.012 | 0.105  | 0.334  | 0.191 | 0.058 | 0.174 | 0.013 |
| Spain                                                   | 653     | 0       | 0         | 7.573  | 0.003 | 0.126  | 0.351  | 0.271 | 0.055 | 0.100 | 0.027 |
| Sweden                                                  | 1091    | 0       | 0         | 6.923  | 0.029 | 0.039  | 0.264  | 0.209 | 0.058 | 0.173 | 0.020 |
| Switzerland                                             | 2366    | 0       | 0         | 6.777  | 0.023 | 0.121  | 0.316  | 0.217 | 0.046 | 0.154 | 0.022 |
| Turkey                                                  | 341     | 0       | 0         | 8.158  | 0.004 | 0.149  | 0.301  | 0.153 | 0.066 | 0.168 | 0.046 |
| United Kingdom                                          | 11,751  | 0       | 0         | 5.019  | 0.017 | 0.115  | 0.343  | 0.194 | 0.062 | 0.128 | 0.029 |
| United States                                           | 164,264 | 0       | 0         | 4.625  | 0.049 | -0.084 | 0.270  | 0.310 | 0.057 | 0.173 | 0.009 |
| Panel C: Non-innovative propensity score matching firms |         |         |           |        |       |        |        |       |       |       |       |
| Argentina                                               | 18      | 0       | 0         | 8.171  | 0.000 | 0.196  | 0.575  | 0.380 | 0.082 | 0.089 | 0.025 |
| Australia                                               | 94      | 0       | 0         | 6.873  | 0.011 | 0.090  | 0.360  | 0.217 | 0.058 | 0.095 | 0.030 |
| Austria                                                 | 54      | 0       | 0         | 6.751  | 0.010 | 0.126  | 0.374  | 0.266 | 0.065 | 0.095 | 0.017 |
| Belgium                                                 | 19      | 0       | 0         | 6.442  | 0.025 | 0.120  | 0.344  | 0.258 | 0.050 | 0.096 | 0.022 |
| Brazil                                                  | 26      | 0       | 0         | 9.232  | 0.003 | 0.109  | 0.347  | 0.384 | 0.050 | 0.142 | 0.013 |
| Canada                                                  | 225     | 0       | 0         | 7.298  | 0.020 | 0.074  | 0.478  | 0.271 | 0.082 | 0.116 | 0.008 |
| China                                                   | 63      | 0       | 0         | 9.807  | 0.009 | 0.136  | 0.445  | 0.241 | 0.079 | 0.171 | 0.022 |
| Denmark                                                 | 334     | 0       | 0         | 6.938  | 0.023 | 0.091  | 0.350  | 0.272 | 0.071 | 0.141 | 0.014 |
| Finland                                                 | 72      | 0       | 0         | 5.577  | 0.035 | 0.113  | 0.288  | 0.230 | 0.077 | 0.167 | 0.032 |
| France                                                  | 152     | 0       | 0         | 8.088  | 0.025 | 0.115  | 0.214  | 0.238 | 0.041 | 0.140 | 0.017 |

(continued on next page)

Table 2 (continued)

| COUNTRY        | N      | PATENTS | CITATIONS | SIZE   | R&D   | ROA   | PPE   | LEV   | CAPIT | CASH  | DIV   |
|----------------|--------|---------|-----------|--------|-------|-------|-------|-------|-------|-------|-------|
| Germany        | 510    | 0       | 0         | 7.549  | 0.023 | 0.121 | 0.251 | 0.223 | 0.052 | 0.116 | 0.019 |
| Indonesia      | 61     | 0       | 0         | 12.477 | 0.000 | 0.116 | 0.373 | 0.405 | 0.081 | 0.105 | 0.021 |
| India          | 9      | 0       | 0         | 10.112 | 0.006 | 0.186 | 0.374 | 0.225 | 0.121 | 0.175 | 0.031 |
| Italy          | 19     | 0       | 0         | 7.831  | 0.009 | 0.152 | 0.243 | 0.220 | 0.052 | 0.106 | 0.028 |
| Japan          | 6718   | 0       | 0         | 11.513 | 0.014 | 0.089 | 0.321 | 0.217 | 0.039 | 0.159 | 0.009 |
| Malaysia       | 1110   | 0       | 0         | 5.942  | 0.000 | 0.093 | 0.370 | 0.215 | 0.045 | 0.135 | 0.019 |
| Netherland     | 49     | 0       | 0         | 7.140  | 0.031 | 0.119 | 0.230 | 0.229 | 0.057 | 0.149 | 0.014 |
| New Zealand    | 22     | 0       | 0         | 7.054  | 0.008 | 0.157 | 0.586 | 0.265 | 0.073 | 0.033 | 0.044 |
| Norway         | 106    | 0       | 0         | 8.381  | 0.004 | 0.098 | 0.345 | 0.280 | 0.066 | 0.149 | 0.021 |
| Singapore      | 46     | 0       | 0         | 7.106  | 0.004 | 0.090 | 0.304 | 0.177 | 0.060 | 0.228 | 0.027 |
| South Korea    | 204    | 0       | 0         | 11.076 | 0.021 | 0.148 | 0.362 | 0.178 | 0.072 | 0.184 | 0.017 |
| Spain          | 54     | 0       | 0         | 9.083  | 0.000 | 0.110 | 0.365 | 0.299 | 0.052 | 0.113 | 0.014 |
| Sweden         | 125    | 0       | 0         | 8.606  | 0.030 | 0.081 | 0.286 | 0.231 | 0.058 | 0.124 | 0.018 |
| Switzerland    | 84     | 0       | 0         | 7.351  | 0.028 | 0.149 | 0.299 | 0.194 | 0.046 | 0.179 | 0.025 |
| Turkey         | 28     | 0       | 0         | 7.713  | 0.004 | 0.156 | 0.339 | 0.201 | 0.080 | 0.135 | 0.040 |
| United Kingdom | 377    | 0       | 0         | 6.252  | 0.016 | 0.145 | 0.367 | 0.218 | 0.063 | 0.123 | 0.032 |
| United States  | 17,057 | 0       | 0         | 5.719  | 0.039 | 0.026 | 0.269 | 0.270 | 0.052 | 0.155 | 0.009 |

Note: This table reports firm characteristics by country for innovative and non-innovative firms. The samples covers 27 countries over the period 1992–2016. Appendix presents the detailed descriptions and data sources for firm characteristics.

Europe, Far East, and Confucian.

Tang and Koveos (2008) develop an integrated empirical model to update five of the Hofstede cultural dimensions, based on changes in economic conditions. Their main findings are that individualism, long-term orientation, and power distance scores vary over time with national wealth, while uncertainty avoidance and masculinity remain relatively stable. Thus, as alternative measures of national culture, we use Tang and Koveos's (2008) five revised cultural dimensions, which we define as uncertainty avoidance ( $UAI_{TK}$ ), collectivism ( $CLT_{TK}$ ), masculinity ( $MAS_{TK}$ ), power distance ( $PDI_{TK}$ ), and time orientation ( $TOR_{TK}$ ).

### 3.1.3. Firm- and country-specific characteristics

To better understand the impact of a country's cultural values on corporate innovative activities, we control for relevant firm- and country-specific variables that are commonly used in the literature (e.g., Cornaggia et al. 2015; Fang et al. 2014; He and Tian 2013; Hirshleifer et al. 2012; Hsu et al. 2014). In particular, we control for the following.  $SIZE_t$  is defined as the logarithm of a firm's total book assets ( $AT$ ) in year  $t$ . We expect larger firms to be more innovative.  $R\&D_t$  is the ratio of research and development expenditures to total book assets in year  $t$ . We expect that more investment in R&D will lead to more innovation.  $ROA_t$  is the ratio of operating income before depreciation to total book assets in year  $t$ .  $ROA$  is a measure of profitability, and we expect more profitable firms to be more innovative.  $PPE_t$  is the ratio of property, plant, and equipment expenditures to total book assets in year  $t$ . We expect this ratio of asset tangibility to be negatively related to innovation.  $LEV$  measures the ratio of the book value of debt (long-term debt plus debt in current liabilities) to total book assets in year  $t$ .  $CAPIT_t$  is defined as the ratio of capital expenditures to total book assets in year  $t$ .  $CASH_t$  is the ratio of cash and short-term investments to total book assets in year  $t$ .  $DIV_t$  measures the ratio of common dividends to total book assets in year  $t$ . We also add a year and industry fixed effect to control for potential omitted variables.<sup>9</sup>

In addition to firm-specific variables, we also control for country-specific determinants of innovation. Specifically, we control for economic growth using a country's annual GDP growth,  $GDPG$ . To capture a country's level of financial development, we use: (1)  $STKTURNOVER$ , defined as the stocks turnover ratio traded in a country's equity markets (i.e., the volume of transactions by market capitalization); and (2)  $CREDIT$ , defined as the total domestic credit through financial sector scaled by a country's GDP. We also introduce the variable  $EDUCATION$  to capture a country's educational attainment, measured by the percentage of its population that has completed tertiary education (Barro and Lee 2013). To control for a country's adoption of national patent laws and the composition of patent rights, we use the international patent protection index, which we label  $PTY\_RIGHT$  (Ginarte and Park 1997; Park 2008). The protection scores range from 0 to 5, with higher values indicating the adoption of stronger patent laws.  $EXP\_RISK$  is the risk of expropriation, measuring the quality of market institutions, ranging from 0 to 10, with higher scores representing better institutions and lower scores indicating the risk of confiscation and forced nationalization (McArthur and Sachs, 2001; Tebaldi and Elmslie 2008).  $NANTI\_DIR$  is the revised anti-director rights index (Djankov et al. 2008) that is formed by summing: (1) votes by mail; (2) shares not deposited; (3) cumulative voting; (4) oppressed minorities; (5) pre-emptive rights; and (6) capital required to call a meeting.  $CORRUPTION$  measures political and financial corruption under a country's juridical system. It ranges from zero to six, with higher scores representing lower levels of actual or potential corruption.  $BUREAUCRACY$  measures a country's quality of bureaucracy and institutions. It ranges from zero to six, with higher values reflecting a better quality of institutions. In the appendix, we provide detailed definitions and data sources for all the variables used in our study.

<sup>9</sup> Omitted variables at the firm, as well as at the country level, may still be an issue, despite the extensive list of control variables we have used. Controlling for cultural zones alleviates the country-level omitted variables issue. We also run our main regression at the country level using the average values of corporate innovation (Table 15), which should attenuate the effect of omitted variables at the firm level.

**Table 3**  
Descriptive statistics.

| Variables                            | N       | P25    | Mean    | Median  | P75     | S.D.     |
|--------------------------------------|---------|--------|---------|---------|---------|----------|
| Panel A: Innovative firms sample     |         |        |         |         |         |          |
| PATENTS                              | 27,636  | 1.000  | 41.403  | 4.000   | 16.000  | 180.341  |
| CITATIONS                            | 27,636  | 1.000  | 257.994 | 17.000  | 91.500  | 1533.613 |
| IDV                                  | 27,636  | 46.000 | 57.442  | 46.000  | 86.000  | 20.917   |
| CLT                                  | 27,636  | 46.000 | 74.918  | 91.000  | 91.000  | 22.675   |
| MAS                                  | 27,636  | 62.000 | 68.040  | 62.000  | 66.000  | 17.414   |
| PDI                                  | 27,636  | 40.000 | 46.008  | 40.000  | 54.000  | 14.305   |
| TOR                                  | 27,636  | 12.091 | 55.312  | 74.307  | 74.307  | 27.304   |
| IDG                                  | 27,636  | 41.741 | 60.003  | 68.080  | 68.080  | 12.160   |
| Anglo                                | 27,636  | 0.000  | 0.643   | 1.000   | 1.000   | 0.479    |
| FarEast                              | 27,636  | 0.000  | 0.043   | 0.000   | 0.000   | 0.202    |
| Gemanic                              | 27,636  | 0.000  | 0.023   | 0.000   | 0.000   | 0.151    |
| LatinAmerica                         | 27,636  | 0.000  | 0.002   | 0.000   | 0.000   | 0.040    |
| LatinEurope                          | 27,636  | 0.000  | 0.009   | 0.000   | 0.000   | 0.094    |
| Nordic                               | 27,636  | 0.000  | 0.025   | 0.000   | 0.000   | 0.156    |
| Confucian                            | 27,636  | 0.000  | 0.254   | 0.000   | 1.000   | 0.436    |
| NearEast                             | 27,636  | 0.000  | 0.001   | 0.000   | 0.000   | 0.032    |
| Arabic                               | 27,636  | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| EastEurope                           | 27,636  | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| SIZE                                 | 27,636  | 4.760  | 7.315   | 6.841   | 9.968   | 3.246    |
| R&D                                  | 27,636  | 0.008  | 0.085   | 0.033   | 0.103   | 0.133    |
| ROA                                  | 27,636  | 0.034  | 0.030   | 0.096   | 0.152   | 0.273    |
| PPE                                  | 27,636  | 0.103  | 0.242   | 0.210   | 0.345   | 0.175    |
| LEV                                  | 27,636  | 0.019  | 0.194   | 0.151   | 0.297   | 0.205    |
| CAPIT                                | 27,636  | 0.020  | 0.050   | 0.038   | 0.065   | 0.046    |
| CASH                                 | 27,636  | 0.060  | 0.242   | 0.155   | 0.347   | 0.240    |
| DIV                                  | 27,636  | 0.000  | 0.009   | 0.002   | 0.012   | 0.017    |
| GDPG                                 | 27,636  | 1.495  | 2.412   | 2.661   | 3.799   | 2.021    |
| STKTURNOVER                          | 27,636  | 82.135 | 122.732 | 112.469 | 151.711 | 63.008   |
| CREDIT                               | 27,636  | 48.968 | 70.930  | 53.349  | 100.028 | 30.422   |
| EDUCATION                            | 27,636  | 11.634 | 15.544  | 19.241  | 19.241  | 5.210    |
| PTY_RIGHTS                           | 27,636  | 4.670  | 4.729   | 4.880   | 4.880   | 0.311    |
| EXP_RISK                             | 27,636  | 9.740  | 9.808   | 10.000  | 10.000  | 0.455    |
| NANTL_DIR                            | 27,636  | 3.000  | 3.529   | 3.000   | 4.500   | 0.752    |
| CORRUPTION                           | 27,636  | 3.917  | 4.133   | 4.000   | 4.500   | 0.727    |
| BUREAUCRACY                          | 27,636  | 4.000  | 3.931   | 4.000   | 4.000   | 0.274    |
| Panel B: Non Innovative firms sample |         |        |         |         |         |          |
| VARIABLES                            | N       | P25    | Mean    | Median  | P75     | S.D.     |
| PATENTS                              | 313,629 | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| CITATIONS                            | 313,629 | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| UAI                                  | 313,629 | 46.000 | 51.788  | 46.000  | 48.000  | 19.498   |
| IDV                                  | 313,629 | 46.000 | 70.641  | 91.000  | 91.000  | 27.037   |
| MAS                                  | 313,629 | 62.000 | 64.142  | 62.000  | 66.000  | 14.551   |
| PDI                                  | 313,629 | 40.000 | 49.096  | 40.000  | 54.000  | 16.021   |
| TOR                                  | 313,629 | 17.128 | 53.319  | 74.307  | 74.307  | 27.131   |
| IDG                                  | 313,629 | 41.741 | 56.545  | 68.080  | 68.080  | 16.816   |
| Anglo                                | 313,629 | 0.000  | 0.619   | 1.000   | 1.000   | 0.486    |
| FarEast                              | 313,629 | 0.000  | 0.067   | 0.000   | 0.000   | 0.251    |
| Gemanic                              | 313,629 | 0.000  | 0.019   | 0.000   | 0.000   | 0.135    |
| LatinAmerica                         | 313,629 | 0.000  | 0.006   | 0.000   | 0.000   | 0.080    |
| LatinEurope                          | 313,629 | 0.000  | 0.013   | 0.000   | 0.000   | 0.114    |
| Nordic                               | 313,629 | 0.000  | 0.014   | 0.000   | 0.000   | 0.117    |
| Confucian                            | 313,629 | 0.000  | 0.260   | 0.000   | 1.000   | 0.439    |
| NearEast                             | 313,629 | 0.000  | 0.001   | 0.000   | 0.000   | 0.033    |
| Arabic                               | 313,629 | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| EastEurope                           | 313,629 | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| SIZE                                 | 313,629 | 3.910  | 6.234   | 6.201   | 8.515   | 3.269    |
| R&D                                  | 313,629 | 0.000  | 0.031   | 0.000   | 0.009   | 0.099    |
| ROA                                  | 313,629 | 0.029  | -0.002  | 0.089   | 0.147   | 0.480    |
| PPE                                  | 313,629 | 0.090  | 0.297   | 0.243   | 0.449   | 0.243    |
| LEV                                  | 313,629 | 0.043  | 0.263   | 0.198   | 0.372   | 0.309    |
| CAPIT                                | 313,629 | 0.013  | 0.057   | 0.035   | 0.074   | 0.069    |
| CASH                                 | 313,629 | 0.031  | 0.169   | 0.098   | 0.229   | 0.196    |
| DIV                                  | 313,629 | 0.000  | 0.012   | 0.002   | 0.014   | 0.024    |
| GDPG                                 | 313,629 | 1.742  | 3.261   | 2.861   | 4.177   | 2.841    |
| STKTURNOVER                          | 313,629 | 70.689 | 121.112 | 105.444 | 152.065 | 74.813   |
| CREDIT                               | 313,629 | 49.811 | 75.901  | 56.858  | 102.027 | 33.962   |
| EDUCATION                            | 313,629 | 9.363  | 13.922  | 19.241  | 19.241  | 6.447    |

(continued on next page)

Table 3 (continued)

| Variables                                | N       | P25    | Mean    | Median  | P75     | S.D.     |
|------------------------------------------|---------|--------|---------|---------|---------|----------|
| PTY_RIGHTS                               | 313,629 | 4.540  | 4.615   | 4.880   | 4.880   | 0.380    |
| EXP_RISK                                 | 313,629 | 9.740  | 9.554   | 10.000  | 10.000  | 0.745    |
| NANTI_DIR                                | 313,629 | 3.000  | 3.399   | 3.000   | 4.500   | 1.052    |
| CORRUPTION                               | 313,629 | 3.500  | 4.024   | 4.000   | 5.000   | 1.019    |
| BUREAUCRACY                              | 313,629 | 4.000  | 3.708   | 4.000   | 4.000   | 0.625    |
| Panel C: Propensity score matched sample |         |        |         |         |         |          |
| PATENTS                                  | 55,272  | 0.000  | 20.701  | 0.500   | 4.000   | 129.189  |
| CITATIONS                                | 55,272  | 0.000  | 128.997 | 0.000   | 17.000  | 1092.064 |
| UAI                                      | 55,272  | 46.000 | 57.442  | 46.000  | 86.000  | 20.917   |
| IDV                                      | 55,272  | 46.000 | 74.918  | 91.000  | 91.000  | 22.675   |
| MAS                                      | 55,272  | 62.000 | 68.040  | 62.000  | 66.000  | 17.413   |
| PDI                                      | 55,272  | 40.000 | 46.008  | 40.000  | 54.000  | 14.305   |
| TOR                                      | 55,272  | 12.091 | 55.312  | 74.307  | 74.307  | 27.304   |
| IDG                                      | 55,272  | 41.741 | 60.003  | 68.080  | 68.080  | 12.160   |
| Anglo                                    | 55,272  | 0.000  | 0.643   | 1.000   | 1.000   | 0.479    |
| FarEast                                  | 55,272  | 0.000  | 0.043   | 0.000   | 0.000   | 0.202    |
| Gemanic                                  | 55,272  | 0.000  | 0.023   | 0.000   | 0.000   | 0.151    |
| LatinAmerica                             | 55,272  | 0.000  | 0.002   | 0.000   | 0.000   | 0.040    |
| LatinEurope                              | 55,272  | 0.000  | 0.009   | 0.000   | 0.000   | 0.094    |
| Nordic                                   | 55,272  | 0.000  | 0.025   | 0.000   | 0.000   | 0.156    |
| Confucian                                | 55,272  | 0.000  | 0.254   | 0.000   | 1.000   | 0.436    |
| NearEast                                 | 55,272  | 0.000  | 0.001   | 0.000   | 0.000   | 0.032    |
| Arabic                                   | 55,272  | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| EastEurope                               | 55,272  | 0.000  | 0.000   | 0.000   | 0.000   | 0.000    |
| SIZE                                     | 55,272  | 4.736  | 7.327   | 6.900   | 9.976   | 3.291    |
| R&D                                      | 55,272  | 0.000  | 0.057   | 0.010   | 0.058   | 0.117    |
| ROA                                      | 55,272  | 0.038  | 0.041   | 0.094   | 0.148   | 0.259    |
| PPE                                      | 55,272  | 0.098  | 0.267   | 0.223   | 0.382   | 0.210    |
| LEV                                      | 55,272  | 0.034  | 0.223   | 0.177   | 0.336   | 0.223    |
| CAPIT                                    | 55,272  | 0.017  | 0.050   | 0.036   | 0.065   | 0.050    |
| CASH                                     | 55,272  | 0.042  | 0.198   | 0.118   | 0.269   | 0.217    |
| DIV                                      | 55,272  | 0.000  | 0.010   | 0.003   | 0.012   | 0.018    |
| GDPG                                     | 55,272  | 1.495  | 2.350   | 2.564   | 3.773   | 2.037    |
| STKTURNVER                               | 55,272  | 82.135 | 125.788 | 113.823 | 152.065 | 66.958   |
| CREDIT                                   | 55,272  | 48.968 | 70.902  | 53.349  | 100.028 | 30.248   |
| EDUCATION                                | 55,272  | 11.634 | 15.544  | 19.241  | 19.241  | 5.210    |
| PTY_RIGHTS                               | 55,272  | 4.670  | 4.729   | 4.880   | 4.880   | 0.311    |
| EXP_RISK                                 | 55,272  | 9.740  | 9.808   | 10.000  | 10.000  | 0.455    |
| NANTI_DIR                                | 55,272  | 3.000  | 3.529   | 3.000   | 4.500   | 0.752    |
| CORRUPTION                               | 55,272  | 3.917  | 4.129   | 4.000   | 4.500   | 0.729    |
| BUREAUCRACY                              | 55,272  | 4.000  | 3.931   | 4.000   | 4.000   | 0.275    |

Note: This table presents the summary statistics for all the variables used in the main regression analysis of the impact of cultural values on corporate innovation. The sample period is 1992–2016. Appendix outlines the definitions and data sources for all the variables.

### 3.2. Descriptive statistics

Our innovative firms sample contains all firms with patents filed and eventually granted over the 1992–2016 period. It consists of 27,636 firm-year observations from 27 countries.<sup>10</sup> Table 1 presents the frequency distribution of the total number of patents and citations by year in our sample. It shows a steady increase in the number of innovative firms, as well as patents and citations, over our sample period, with the exception of the 2008–2011 period, during which these numbers decrease. The number of innovative firms in our sample has doubled over the last 25 years. The evolution of the size of our sample over time is representative of the global trend in total patent grants in the world, as reported by the World Intellectual Property Organization (WIPO). In its 2010 annual report, the WIPO reveals a significant yearly increase in the number of patents for the 1995–2008 period and a decrease in the years 2009 and 2010.<sup>11</sup> Its 2020 report shows that, since 2011 and until 2019, the yearly number of patents in the world started increasing again.<sup>12</sup> The country-level distribution of our innovative firms sample reported in Panel A of Table 2 also indicates that this is representative. In fact, country-level data on innovation provided by the 2020 WIPO report show that the top 10 innovative countries are virtually identical to those in our sample. In particular, the top three countries in terms of number of patents and citations are the US, Japan, and Germany. Using a different database, Griffin et al. (2019) also report the same top three innovative countries over the 1976–2006 period. Similar to previous international studies on innovation, our sample shows that US firms significantly dominate. Thus, to

<sup>10</sup> We dropped countries with less than five patents from our sample.

<sup>11</sup> Report available at: [http://www.wipo.int/edocs/pubdocs/en/patents/901/wipo\\_pub\\_901\\_2010.pdf](http://www.wipo.int/edocs/pubdocs/en/patents/901/wipo_pub_901_2010.pdf)

<sup>12</sup> Report available at: [https://www.wipo.int/edocs/pubdocs/en/wipo\\_pub\\_901\\_2020.pdf](https://www.wipo.int/edocs/pubdocs/en/wipo_pub_901_2020.pdf)

address the concern that the US mainly drives our results, we re-estimate our models excluding US firms.

Panel A of [Table 2](#) also shows that US innovative firms are not among the biggest (in terms of their book value), nor are they the most profitable. However, they seem to spend more in R&D, less in property, plant, and equipment, carry more cash, and distribute less dividends than other countries' innovative firms. The distribution of the leverage and capital expenditure, as well as all the firm-level variables, among the other countries seems comparable.

In Panel B of [Table 2](#), for each country in Panel A, we generate a sample containing all firms that are covered by the two Compustat databases with no granted patents during the whole sample period. We report the same statistics for these firms as in Panel A. In Panel C of the same table, we use propensity score matching to match innovative and non-innovative firms, as discussed in Section 3.1. These three samples of innovative, non-innovative, and matched firms are used later in our univariate and multivariate analyses.

The summary statistics for all the variables used in the main regression analysis are shown in [Table 3](#). Panel A reports the innovative firms' means, medians, 25th percentiles, 75th percentiles, and standard deviations. There are about 41 patents and 258 citations per firm and per year on average. The average firm size in the innovation sample, as measured by the logarithm of total assets is about 7.3, with R&D expenditure accounting for about 9% of total assets, profitability for 3% of total assets, property, plant, and equipment for 24% of total assets, leverage for 19% of total assets, capital expenditure for 5% of total assets, cash flow for 23% of total assets, and dividend accounts for 1% of total assets.

Panel B reports the same statistics for the sample of non-innovative firms. In comparison with those shown in Panel A, these statistics suggest that non-innovative firms are, on average, smaller and less profitable than innovative ones. They also spend less in R&D, carry less cash, and pay more dividends. These preliminary results are further investigated in our univariate and multivariate analyses.

Panel C reports the matched sample statistics, which are, as expected, comparable to those of the innovative firms, except for the average number of patents and citations per firm and per year, which are half of those reported in Panel A.

The descriptive statistics of country-level variables are comparable across all panels of [Table 3](#). In particular, the six cultural dimensions (i.e., uncertainty avoidance, individualism, masculinity, power distance, time orientation, and indulgence) have average scores of about 52, 70, 64, 49, 53, and 57, respectively.

[Table 4](#) presents the pair-wise correlations for all the variables used in the main regression analyses. It shows that the three innovation measures are negatively correlated with uncertainty avoidance, masculinity, and power distance, and are positively correlated with individualism, long-term orientation, and indulgence at the 1% significance level. This provides preliminary evidence on the nature of the relationship between corporate innovation and cultural dimensions, which we further investigate in the following sections. [Table 4](#) also shows a significant correlation between the six cultural dimensions, suggesting a potential risk of collinearity; we circumvent this risk in the multivariate analysis by testing the effect of each dimension.

## 4. Results

### 4.1. Univariate analysis

In [Table 5](#), we report and compare the means and the medians of the main firm-level characteristics of innovative firms and non-innovative firms. In Panel A, we compare innovative firms with a matching sample of non-innovative firms. The matching sample is built using propensity matching scores, and innovation is measured using citations per patent. Because firm size is our first firm-level matching variable, it is not surprising that the difference between the two samples is not statistically significant. However, as expected, innovative firms invest more in R&D and less in property, plant, and equipment than non-innovative firms. Moreover, innovative firms tend to rely less on debt, carry more cash, and distribute less dividends than non-innovative firms. This is in line with several empirical studies, which find a strong negative relationship between innovation and debt financing (see, for example, the survey conducted by [Hall and Lerner 2010](#)). [Brown et al. \(2013\)](#) argue that, because of their risky and intangible nature, innovative investments are not easy to finance using debt. These investments often offer no collateral and have a high probability of failure, making them less attractive to creditors. Thus, firms have to rely on stock market financing or internal financing for this type of investment, which is consistent with our results showing that innovative firms tend to hold more cash holding and distribute less dividends. In Panel B, we compare the same variables between innovative firms and the full sample of non-innovative firms. The results are comparable to those in Panel A, except innovative firms are found to be, on average, significantly larger and more profitable than non-innovative firms.

### 4.2. Multivariate analysis

We now extend the univariate analysis by implementing regression tests that control for the impact of other firm- and country-level variables that may also influence innovation. In Panel A of [Table 6](#), we report the marginal effects of a logit regression using the sample that includes innovative and (matched) non-innovative firms, where the dependent variable is the innovation dummy. The results, reported in Column (1) to Column (6), indicate that the six dimensions of culture are associated with the probability of a firm having

**Table 4**  
Correlation coefficients.

| VARIABLES      | LNCITEPAT | LNCITE   | LNCITEPAT_NADJ | UAI      | IDV      | MAS      | PDI      |
|----------------|-----------|----------|----------------|----------|----------|----------|----------|
| LNCITE         | 0.4273*   |          |                |          |          |          |          |
| LNCITEPAT_NADJ | 0.4741*   | 0.5198*  |                |          |          |          |          |
| UAI            | -0.5414*  | -0.2476* | -0.2696*       |          |          |          |          |
| IDV            | 0.7153*   | 0.3549*  | 0.3532*        | -0.6883* |          |          |          |
| MAS            | -0.3468*  | -0.1456* | -0.1741*       | 0.8455*  | -0.4834* |          |          |
| PDI            | -0.4383*  | -0.2236* | -0.2138*       | 0.2629*  | -0.7648* | 0.2050*  |          |
| TOR            | 0.6884*   | 0.3312*  | 0.3429*        | -0.9080* | 0.8477*  | -0.7271* | -0.3931* |
| IDG            | 0.6565*   | 0.3140*  | 0.3259*        | -0.8902* | 0.8808*  | -0.6986* | -0.4942* |
| GDPG           | 0.3083*   | -0.0258* | 0.0268*        | -0.4396* | 0.1511*  | -0.3848* | 0.1189*  |
| STKTURNVER     | 0.1182*   | 0.2107*  | 0.2390*        | -0.0638* | 0.2984*  | 0.004    | -0.2840* |
| CREDIT         | -0.7043*  | -0.3397* | -0.3284*       | 0.5915*  | -0.7897* | 0.4221*  | 0.5019*  |
| EDUCATION      | 0.7209*   | 0.3764*  | 0.3575*        | -0.3746* | 0.8327*  | -0.1029* | -0.6166* |
| PTY_RIGHTS     | 0.5007*   | 0.2712*  | 0.2465*        | -0.0651* | 0.7157*  | 0.0913*  | -0.8297* |
| EXP_RISK       | 0.4358*   | 0.2348*  | 0.2124*        | -0.0469* | 0.7021*  | 0.0992*  | -0.8677* |
| NANTI_DIR      | -0.7164*  | -0.3566* | -0.3539*       | 0.6461*  | -0.8784* | 0.5047*  | 0.6527*  |
| CORRUPTION     | 0.3005*   | 0.0941*  | 0.1069*        | -0.2569* | 0.5054*  | -0.2649* | -0.6423* |
| BUREAUCRACY    | 0.2592*   | 0.1479*  | 0.1243*        | 0.0852*  | 0.4954*  | 0.2705*  | -0.7490* |

Note: This table presents the pair-wise coefficients for all the variables used in the main regression analysis. \* in this table represents significance level of 1%. Appendix outlines the definitions and data sources for all the variables.

patents and citations. In particular, our findings suggest that, the higher the degree of uncertainty avoidance in a particular country (and thus, the higher the ambiguity and risk aversion), the lower the probability of local firms becoming innovative. This result is not surprising, given the fact that innovation is by nature a risky activity with a high probability of failure. On the other hand, the more a national culture encourages individualism, rather than collective achievements, the higher the probability of local firms innovating. This latter result is consistent with [Taylor and Wilson \(2012\)](#), who find that individualism generally helps to increase the rates of technology patenting, while collectivism generally harms innovation rates. Contrary to [Shane \(1993\)](#), who finds no significant relationship between masculinity and country-level innovation rates, the results reported in Column (3) suggest that the probability of local firms innovating is higher in feminine societies that emphasize a low level of conflict, as well as trust and emotional support. The results from Column (4) show that power distance loads negative and is statistically significant. This finding suggests that, in countries where power is spread unequally between individuals, firms are less likely to be innovative, which is in line with [Shane \(1992\)](#). Providing support for Hypothesis 5, the results shown in Column (5) indicate that firms operating in a society with a long-term orientation are more likely to be innovative. This is also in line with [Waarts and van Everdingen \(2005\)](#), who report that European firms with a highly long-term orientation tend to adopt innovative products. The findings reported in Column (6) suggest that corporate innovation is more prominent in countries with an indulgence orientation, which provides support for our prediction outlined in Hypothesis 6. This is also consistent with the study conducted by [Murswieck et al. \(2020\)](#), which shows that indulgence is a key driver of national innovation performance in Europe. Column (7) reports the results of a horse-race regression, in which we include all of Hofstede's cultural dimensions. Some of these dimensions lose statistical significance, however this is not surprising, given the high correlation between these cultural dimensions. In fact, this has been detected in other research related to culture (e.g., Hofstede, 1983; [Zheng et al. 2012](#)). To circumvent misinterpretation caused by multicollinearity, we follow, among others, [Zheng et al. \(2012\)](#) and focus on the effect of each cultural dimension individually. In addition, the effects of Hofstede's cultural dimensions on corporate innovation hold, despite the way in which we control for the 11 cultural zones defined in [Ronen and Shenkar \(2013\)](#). It should be noted that only a few of these cultural zones remained in our regression estimation, due to multicollinearity caused by the very high correlations between them.

The results of the firm-level control variables show that size, profitability, R&D expenses, capital expenditures, and cash holdings are positively and significantly associated with the probability of a firm being innovative. However, dividend distribution, leverage, and property, plant, and equipment expenditures have a statistically significant negative effect on corporation innovation. All these results are stable across the six specifications of our model and are consistent with our univariate analysis and the extant literature.

Among the country-level control variables, we find that stock turnover, level of education, risk of expropriation, and level of corruption are statistically significant across Columns (1) to (6). All of these results are consistent with the extant literature. For instance, with respect to stock turnover, we find that this measure of stock market liquidity increases the probability of a firm being innovative. This is consistent with the results of [Hsu et al. \(2014\)](#), who find that industries that are dependent on external financing sources exhibit a disproportionately higher innovation level in countries with better developed equity markets.

The results in Panel A are generated using the innovation dummy as our dependent variable, which differentiates between firms with patents (innovative firms) and firms without patents (non-innovative firms). To examine the effect of national culture on the

| TOR      | IDG      | GDPG     | STKTURNVER | CREDIT   | EDUCATION | PTY_RIGHTS | EXP_RISK | NANTI_DIR | CORRUPTION |
|----------|----------|----------|------------|----------|-----------|------------|----------|-----------|------------|
| 0.9622*  |          |          |            |          |           |            |          |           |            |
| 0.3643*  | 0.3156*  |          |            |          |           |            |          |           |            |
| 0.1765*  | 0.1562*  | −0.3303* |            |          |           |            |          |           |            |
| −0.7732* | −0.7314* | −0.2155* | −0.2463*   |          |           |            |          |           |            |
| 0.6519*  | 0.6418*  | 0.0662*  | 0.3935*    | −0.7607* |           |            |          |           |            |
| 0.3146*  | 0.3839*  | −0.1707* | 0.3772*    | −0.5380* | 0.8075*   |            |          |           |            |
| 0.2631*  | 0.3573*  | −0.2235* | 0.3284*    | −0.4568* | 0.7299*   | 0.9454*    |          |           |            |
| −0.7927* | −0.7586* | −0.2016* | −0.3353*   | 0.8149*  | −0.7786*  | −0.6216*   | −0.5495* |           |            |
| 0.2890*  | 0.3909*  | 0.0855*  | −0.1052*   | −0.3228* | 0.2954*   | 0.4830*    | 0.5509*  | −0.3693*  |            |
| 0.0818*  | 0.2217*  | −0.2815* | 0.2224*    | −0.2730* | 0.5546*   | 0.8011*    | 0.8881*  | −0.2831*  | 0.5104*    |

quality of innovation, we follow Cumming et al. (2020a), and Griffin et al. (2019), and re-estimate our logit model, where the dependent variable is a dummy variable that equals one for firms with patents and at least one citation during the three subsequent years, and zero otherwise. The results reported in Panel B show that all of the six dimensions of culture influence the quality of corporate innovation in the same way as the likelihood of a firm being innovative (Panel A). The findings shown in Panel C suggest that our results remain qualitatively unchanged if we use a five-year rolling period instead of a three-year rolling period.

As shown in Table 7, our results are not only statistically significant but are also economically sizable. All the six cultural dimensions exhibit a high level of economic significance, with that of power distance being the highest, followed by individualism. For instance, an increase in the level of *PDI* from the minimum to the maximum (which is approximately from the level of power distance in Austria to that in Malaysia) would lead to a 24.8% decrease in the probability that a firm is innovative. A one-standard deviation increase in the level of *IDV* (which is approximately from the level of individualism in Germany to that in India) is associated with a 7.3% increase in the probability that a firm is innovative.

Insert Tables 6 and 7 about here.

We now estimate different specifications of our model using the sample of innovative firms only, using as a dependent variable a continuous measure of innovation based on the adjusted number of citations per patent: *LNCITEPAT*. Our results, reported in Table 8, are consistent with our univariate results. When cultural dimensions are considered individually, the coefficient on each of the six dimensions is statistically significant at the 1% level, even though we control for the 11 cultural zones. However, while uncertainty avoidance, masculinity, and power distance have an adverse effect on the quantity as well as the quality of innovation among innovative firms, individualism, long-term orientation, and indulgence tend to have a positive effect.

The firm-level control variables are statistically significant and load with expected signs. In particular, we find that more innovative firms tend to be larger, spend more on R&D and on capital, spend less on property, plant, and equipment, use less debt, hold less cash, and distribute more dividends. Among the country-level control variables, the coefficients on each country's economic development, stock turnover, education attainment, and patent rights protection are positive and significant at the 1% level in all model specifications. Overall, the anti-director rights, corruption, and bureaucracy variables have negative effects on innovation. The negative effect of the variable *CREDIT* on innovation confirms that innovative firms typically tap equity, rather than debt markets, to finance their innovation. This result is also consistent with the findings of Hsu et al. (2014) regarding the role of developed equity markets in fostering innovation.

#### 4.3. Additional tests

##### 4.3.1. Alternative measures of culture

First, we consider alternative sources of data on cultural values. We use the revised five cultural measures proposed by Tang and Koveos (2008):<sup>13</sup> uncertainty avoidance (*UAI\_TK*), individualism (*IDV\_TK*), masculinity (*MAS\_TK*), power distance (*PDI\_TK*), and long-term orientation (*TOR\_TK*). The results, reported in Table 9, suggest that our results from Table 8 remain qualitatively unchanged when

<sup>13</sup> Tang and Koveos (2008) argue that some of Hofstede's cultural value dimensions may change with economic conditions. They find that national wealth has a curvilinear relationship with individualism and power distance, whereas uncertainty avoidance and masculinity are less likely to change over time.

**Table 5**  
Univariate tests.

| Panel A: Matched Sample. |                  |       |        |                          |       |        |             |           |        |           |
|--------------------------|------------------|-------|--------|--------------------------|-------|--------|-------------|-----------|--------|-----------|
| VARIABLES                | Innovative Firms |       |        | Non-innovative PS Sample |       |        | Differences |           |        |           |
|                          | N                | Mean  | Median | N                        | Mean  | Median | Mean        | t-stat    | Median | z-stat    |
| LNCITEPAT                | 27,636           | 1.313 | 0.749  | 27,636                   | 0.000 | 0.000  | 1.313       | 147.02*** | 0.749  | 176.99*** |
| SIZE                     | 27,636           | 7.315 | 6.841  | 27,636                   | 7.328 | 6.967  | −0.023      | −0.825    | −0.126 | −0.85     |
| R&D                      | 27,636           | 0.085 | 0.033  | 27,636                   | 0.029 | 0.000  | 0.056       | 57.48***  | 0.033  | 107.72*** |
| ROA                      | 27,636           | 0.030 | 0.096  | 27,636                   | 0.053 | 0.092  | −0.024      | −10.74*** | 0.004  | 0.724     |
| PPE                      | 27,636           | 0.242 | 0.210  | 27,636                   | 0.292 | 0.240  | −0.050      | −28.04*** | −0.030 | −15.06*** |
| LEV                      | 27,636           | 0.194 | 0.151  | 27,636                   | 0.252 | 0.209  | −0.058      | −30.82*** | −0.058 | −30.91*** |
| CAPIT                    | 27,636           | 0.057 | 0.038  | 27,636                   | 0.057 | 0.034  | 0.000       | 0.73      | 0.004  | 15.43***  |
| CASH                     | 27,636           | 0.242 | 0.155  | 27,636                   | 0.153 | 0.089  | 0.089       | 49.14***  | 0.066  | 48.75***  |
| DIV                      | 27,636           | 0.009 | 0.002  | 27,636                   | 0.010 | 0.003  | −0.001      | −6.99***  | −0.001 | −7.78***  |

  

| Panel B: Full Sample. |                  |       |        |                            |        |        |             |           |        |           |
|-----------------------|------------------|-------|--------|----------------------------|--------|--------|-------------|-----------|--------|-----------|
| VARIABLES             | Innovative Firms |       |        | Non-innovative Full Sample |        |        | Differences |           |        |           |
|                       | N                | Mean  | Median | N                          | Mean   | Median | Mean        | t-stat    | Median | z-stat    |
| LNCITEPAT             | 27,636           | 2.000 | 1.729  | 313,629                    | 0.000  | 0.000  | 2.000       | 617.65*** | 1.729  | 502.97*** |
| SIZE                  | 27,636           | 7.303 | 6.841  | 313,629                    | 6.234  | 6.201  | 1.069       | 52.12***  | 0.640  | 45.41***  |
| R&D                   | 27,636           | 0.086 | 0.033  | 313,629                    | 0.031  | 0.000  | 0.055       | 85.35***  | 0.033  | 171.97*** |
| ROA                   | 27,636           | 0.019 | 0.096  | 313,629                    | −0.002 | 0.089  | 0.021       | 7.15***   | 0.006  | 6.33***   |
| PPE                   | 27,636           | 0.243 | 0.210  | 313,629                    | 0.297  | 0.243  | −0.054      | −36.11*** | −0.033 | −21.83*** |
| LEV                   | 27,636           | 0.199 | 0.151  | 313,629                    | 0.263  | 0.198  | −0.058      | −33.98*** | −0.047 | −34.59*** |
| CAPIT                 | 27,636           | 0.057 | 0.038  | 313,629                    | 0.050  | 0.035  | 0.007       | 16.65***  | 0.003  | 12.63***  |
| CASH                  | 27,636           | 0.242 | 0.155  | 313,629                    | 0.169  | 0.098  | 0.073       | 58.17***  | 0.057  | 55.66***  |
| DIV                   | 27,636           | 0.010 | 0.002  | 313,629                    | 0.012  | 0.002  | −0.003      | −17.81*** | 0.000  | −8.65***  |

Note: This table reports the mean and median differences of firm characteristics between innovative firms and matched non-innovative firms (Panel A), and between innovative firms and the full sample of non-innovative firms (Panel B). Appendix outlines the definitions and data sources for all the variables. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation.

\*\*\* Significant at the 1% level.

we use the updated version of Hofstede's five cultural dimensions. We do not report results for an updated version of indulgence (*IDG*) because Tang and Koveos (2008) did not develop an update for this dimension.

#### 4.3.2. Alternative measures of innovation

In Table 10 and Table 11, we run our analysis using two alternative measures of innovation. First, we use the number of forward citations received, scaled by the number of citations in the industry. Then, we use the number of forward citations received, scaled by the number of patents filed (and eventually granted), as provided by the latest version of the National Bureau of Economic Research (NBER) patent database (e.g., Brown et al. 2013; Fang et al. 2014; Cornaggia et al. 2015). This database provides information on patents granted by the United States Patent and Trademark Office (USPTO) over the period from 1976 to 2006. Although the database covers only patents granted in the US, Chen et al. (2017) argue that USPTO is a “pseudo international patenting venue”, as companies from outside the US may choose to protect their ideas in the largest market in the world. With these two alternative measures, we obtain the same results regarding the effects of the six cultural dimensions on innovation.

Insert Tables 10 and 11 about here.

#### 4.3.3. Endogeneity

Even though we do not suspect that our results might suffer from any endogeneity bias, given that it is unlikely that patents' citations would affect countries' culture, we repeat our analysis applying a two-stage instrumental variables estimation method and using the historical prevalence of infectious diseases as an instrumental variable, to address any potential reverse causality between innovation and culture.<sup>14</sup> Recent studies in psychology research show that the historical prevalence of infectious disease predicts differences in cultural dimensions, such as individualism and collectivism. As infectious diseases are a source of morbidity, and thus of natural selection (Wolfe et al. 2007; Volk and Atkinson 2013), societies with a high prevalence of such diseases are more likely to develop particular behaviors to avoid them, such as xenophobia, withdrawn actions, and ethnocentrism. As a result, several studies on the relationship between cultural values and economic growth have used the historical prevalence of infectious diseases index as an

<sup>14</sup> We also used the number of foreign-born populations in each country provided by the Organization for Economic Co-operation and Development (OECD) as an alternative instrumental variable for culture, since immigration largely reshapes culture. We obtained similar results that we are not reporting but are available upon request.

Table 6

The impact of cultural value on innovation: Logit model with matched sample.

| Panel A: Innovative firms dummy variable |                        |                        |                        |                        |                        |                        |                        |
|------------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| VARIABLES                                | (1)<br><i>IN_DUMMY</i> | (2)<br><i>IN_DUMMY</i> | (3)<br><i>IN_DUMMY</i> | (4)<br><i>IN_DUMMY</i> | (5)<br><i>IN_DUMMY</i> | (5)<br><i>IN_DUMMY</i> | (7)<br><i>IN_DUMMY</i> |
| <i>UAI</i>                               | −0.002**<br>(−7.92)    |                        |                        |                        |                        |                        | −0.001**<br>(−2.22)    |
| <i>IDV</i>                               |                        | 0.003***<br>(7.82)     |                        |                        |                        |                        | −0.001<br>(0.73)       |
| <i>MAS</i>                               |                        |                        | −0.001**<br>(−2.28)    |                        |                        |                        | −0.001***<br>(−3.08)   |
| <i>PDI</i>                               |                        |                        |                        | −0.003***<br>(−3.72)   |                        |                        | 0.002<br>(1.42)        |
| <i>TOR</i>                               |                        |                        |                        |                        | 0.003***<br>(8.23)     |                        | 0.003<br>(3.15)        |
| <i>IDG</i>                               |                        |                        |                        |                        |                        | 0.004***<br>(6.89)     | 0.000<br>(0.0)         |
| <i>FarEast</i>                           | 0.133***<br>(3.44)     | 0.143***<br>(3.78)     | 0.221***<br>(6.63)     | 0.287***<br>(8.36)     | 0.060<br>(1.38)        | 0.110***<br>(2.6)      | −0.065<br>(−0.89)      |
| <i>Gemanic</i>                           | 0.086***<br>(3.34)     | 0.062**<br>(2.49)      | 0.047<br>(1.55)        | 0.027<br>(1.11)        | 0.152***<br>(5.4)      | 0.108***<br>(3.93)     | 0.104***<br>(2.79)     |
| <i>LatinAmerica</i>                      | 0.163**<br>(2.4)       | 0.040<br>(0.54)        | 0.171***<br>(2.64)     | 0.146**<br>(2.2)       | −0.014<br>(−0.18)      | 0.029<br>(0.38)        | −0.028<br>(−0.32)      |
| <i>LatinEurope</i>                       | 0.163***<br>(3.38)     | 0.055<br>(1.08)        | 0.101*<br>(1.89)       | 0.156***<br>(2.92)     | 0.010*<br>(1.94)       | 0.092*<br>(1.83)       | 0.000<br>(0.00)        |
| <i>SIZE</i>                              | 0.131***<br>(29.55)    | 0.131***<br>(29.56)    | 0.121***<br>(28.16)    | 0.122***<br>(28.27)    | 0.132***<br>(29.85)    | 0.129***<br>(29.29)    | 0.134***<br>(29.89)    |
| <i>R&amp;D</i>                           | 0.144***<br>(25.35)    | 0.143***<br>(25.31)    | 0.144***<br>(25.44)    | 0.143***<br>(25.4)     | 0.144***<br>(25.33)    | 0.143***<br>(25.33)    | 0.143***<br>(25.32)    |
| <i>ROA</i>                               | 0.060***<br>(12.76)    | 0.060***<br>(12.8)     | 0.062***<br>(13.26)    | 0.061***<br>(13.2)     | 0.060***<br>(12.73)    | 0.060***<br>(12.87)    | 0.060***<br>(12.67)    |
| <i>PPE</i>                               | −0.047***<br>(−13.05)  | −0.047***<br>(−13.04)  | −0.047***<br>(−13.02)  | −0.047***<br>(−13.1)   | −0.048***<br>(−13.13)  | −0.048***<br>(−13.16)  | −0.048***<br>(−13.1)   |
| <i>LEV</i>                               | −0.025***<br>(−8.6)    | −0.024***<br>(−8.5)    | −0.023***<br>(−8.2)    | −0.024***<br>(−8.29)   | −0.025***<br>(−8.58)   | −0.024***<br>(−8.42)   | −0.024***<br>(−8.59)   |
| <i>CAPIT</i>                             | 0.027***<br>(8.57)     | 0.026***<br>(8.69)     | 0.026***<br>(8.64)     | 0.026***<br>(8.66)     | 0.026***<br>(8.65)     | 0.026***<br>(8.8)      | 0.026***<br>(8.79)     |
| <i>CASH</i>                              | 0.038***<br>(11.19)    | 0.039***<br>(11.32)    | 0.038***<br>(11.03)    | 0.038***<br>(11.04)    | 0.039***<br>(11.27)    | 0.039***<br>(11.23)    | 0.039***<br>(11.24)    |
| <i>DIV</i>                               | −0.001***<br>(−3.45)   | −0.010***<br>(−3.71)   | −0.007***<br>(−2.67)   | −0.008***<br>(−2.83)   | −0.010***<br>(−3.46)   | −0.010***<br>(−3.34)   | −0.01***<br>(−3.64)    |
| <i>GDPG</i>                              | −0.009<br>(−1.31)      | −0.004<br>(−0.59)      | 0.002<br>(0.22)        | 0.003<br>(0.51)        | −0.005<br>(−0.79)      | −0.002<br>(−0.33)      | −0.009<br>(−1.26)      |
| <i>STKTURNOVER</i>                       | −0.057***<br>(−8.83)   | −0.055***<br>(−8.59)   | −0.054***<br>(−8.33)   | −0.053***<br>(−8.24)   | −0.055***<br>(−8.5)    | −0.053***<br>(−8.15)   | −0.055***<br>(−8.52)   |
| <i>CREDIT</i>                            | −0.0061*<br>(−1.1)     | −0.056<br>(−0.99)      | −0.004<br>(−0.63)      | −0.036<br>(−0.61)      | −0.004<br>(−0.75)      | −0.008<br>(−1.38)      | −0.078<br>(−1.33)      |
| <i>EDUCATION</i>                         | 0.066***<br>(7.3)      | 0.041***<br>(4.66)     | 0.061***<br>(5.85)     | 0.065***<br>(6.67)     | 0.041***<br>(4.7)      | 0.049***<br>(5.59)     | 0.016<br>(1.12)        |
| <i>PTY_RIGHTS</i>                        | 0.025*<br>(1.70)       | 0.021<br>(1.46)        | 0.016<br>(1.03)        | 0.015<br>(1.04)        | 0.044***<br>(2.87)     | 0.025*<br>(1.72)       | 0.019<br>(1.15)        |
| <i>EXP_RISK</i>                          | −0.051**<br>(−2.58)    | −0.072***<br>(−3.52)   | −0.037*<br>(−1.71)     | −0.055**<br>(−2.49)    | −0.070***<br>(−3.39)   | −0.051**<br>(−2.58)    | −0.028<br>(−1.15)      |
| <i>NANTI_DIR</i>                         | 0.006<br>(0.72)        | 0.003<br>(0.42)        | −0.017<br>(−1.62)      | −0.015<br>(−1.52)      | 0.024**<br>(2.37)      | 0.002<br>(0.24)        | 0.010<br>(0.88)        |
| <i>CORRUPTION</i>                        | −0.028***<br>(−5.79)   | −0.028***<br>(−5.81)   | −0.033***<br>(−6.59)   | −0.032***<br>(−6.5)    | −0.024***<br>(−5.04)   | −0.026***<br>(−5.27)   | −0.030***<br>(−6.03)   |
| <i>BUREAUCRACY</i>                       | 0.001<br>(0.13)        | −0.006<br>(−0.53)      | 0.005<br>(0.35)        | 0.001<br>(0.05)        | −0.001<br>(−0.07)      | −0.011<br>(−0.97)      | −0.030<br>(−1.99)      |
| <i>Year effect</i>                       | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| <i>Industry effect</i>                   | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| <i>N</i>                                 | 55,272                 | 55,272                 | 55,272                 | 55,272                 | 55,272                 | 55,272                 | 55,272                 |
| <i>Pseudo R<sup>2</sup></i>              | 0.155                  | 0.155                  | 0.154                  | 0.154                  | 0.155                  | 0.155                  | 0.156                  |

Panel B: Three-years rolling period innovative firms dummy variable

| VARIABLES | (1)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (2)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (3)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (4)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (5)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (5)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> | (7)<br><i>ROLL<sub>t</sub></i><br><i>DUMMY</i> |
|-----------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|
|-----------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|

(continued on next page)

Table 6 (continued)

| Panel B: Three-years rolling period innovative firms dummy variable |                       |                       |                       |                       |                       |                       |                       |
|---------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| VARIABLES                                                           | (1)<br>ROLL_<br>DUMMY | (2)<br>ROLL_<br>DUMMY | (3)<br>ROLL_<br>DUMMY | (4)<br>ROLL_<br>DUMMY | (5)<br>ROLL_<br>DUMMY | (5)<br>ROLL_<br>DUMMY | (7)<br>ROLL_<br>DUMMY |
| UAI                                                                 | −0.002***<br>(−4.39)  |                       |                       |                       |                       |                       | 0.000<br>(−0.11)      |
| IDV                                                                 |                       | 0.003***<br>(4.44)    |                       |                       |                       |                       | 0.000<br>(−0.05)      |
| MAS                                                                 |                       |                       | −0.001**<br>(−2.54)   |                       |                       |                       | 0.001<br>(0.84)       |
| PDI                                                                 |                       |                       |                       | −0.004***<br>(−3.47)  |                       |                       | 0.001<br>(0.51)       |
| TOR                                                                 |                       |                       |                       |                       | 0.003***<br>(5.50)    |                       | 0.004***<br>(3.14)    |
| IDG                                                                 |                       |                       |                       |                       |                       | 0.003***<br>(4.08)    | −0.001<br>(−0.64)     |
| FarEast                                                             | 0.154***<br>(2.76)    | 0.154***<br>(2.80)    | 0.234***<br>(4.94)    | 0.314***<br>(6.99)    | 0.060<br>(0.93)       | 0.119<br>(1.95*)      | −0.020<br>(−0.16)     |
| Gemanic                                                             | 0.063*<br>(1.69)      | 0.044<br>(1.21)       | 0.071<br>(1.58)       | 0.035<br>(0.97)       | 0.140***<br>(3.42)    | 0.094**<br>(2.30)     | 0.129**<br>(2.30)     |
| LatinAmerica                                                        | 0.114<br>(1.05)       | −0.005<br>(−0.05)     | 0.114<br>(1.07)       | 0.050<br>(0.46)       | −0.072<br>(−0.65)     | −0.016<br>(−0.14)     | −0.106<br>(−0.85)     |
| LatinEurope                                                         | 0.053<br>(0.52)       | −0.041<br>(−0.42)     | 0.079<br>(0.70)       | 0.152<br>(1.41)       | −0.002<br>(−0.02)     | −0.006<br>(−0.06)     | −0.113<br>(−0.78)     |
| SIZE                                                                | 0.154***<br>(26.55)   | 0.153***<br>(26.58)   | 0.149***<br>(26.22)   | 0.150***<br>(26.18)   | 0.156***<br>(26.99)   | 0.153***<br>(26.53)   | 0.156***<br>(26.82)   |
| R&D                                                                 | 0.167***<br>(22.00)   | 0.167***<br>(21.98)   | 0.167***<br>(22.03)   | 0.167***<br>(22.02)   | 0.167***<br>(21.99)   | 0.167***<br>(22.00)   | 0.167***<br>(21.99)   |
| ROA                                                                 | 0.054***<br>(10.34)   | 0.054***<br>(10.36)   | 0.054***<br>(10.49)   | 0.054***<br>(10.46)   | 0.053***<br>(10.28)   | 0.054***<br>(10.40)   | 0.053***<br>(10.28)   |
| PPE                                                                 | −0.058***<br>(−12.58) | −0.058***<br>(−12.62) | −0.058***<br>(−12.56) | −0.059***<br>(−12.66) | −0.059***<br>(−12.68) | −0.058***<br>(−12.62) | −0.059***<br>(−12.70) |
| LEV                                                                 | −0.021***<br>(−5.77)  | −0.020***<br>(−5.70)  | −0.020***<br>(−5.57)  | −0.020***<br>(−5.63)  | −0.021***<br>(−5.82)  | −0.020***<br>(−5.67)  | −0.021***<br>(−5.80)  |
| CAPIT                                                               | 0.029***<br>(7.45)    | 0.030***<br>(7.52)    | 0.029***<br>(7.42)    | 0.029***<br>(7.43)    | 0.030***<br>(7.52)    | 0.030***<br>(7.54)    | 0.030***<br>(7.57)    |
| CASH                                                                | 0.040***<br>(8.93)    | 0.040***<br>(8.98)    | 0.040***<br>(8.90)    | 0.040***<br>(8.87)    | 0.041***<br>(9.00)    | 0.040***<br>(8.97)    | 0.041***<br>(8.98)    |
| DIV                                                                 | −0.007**<br>(−2.00)   | −0.007***<br>(−2.15)  | −0.006*<br>(−1.69)    | −0.006*<br>(−1.81)    | −0.007**<br>(−2.13)   | −0.007**<br>(−2.00)   | −0.007**<br>(−2.08)   |
| GDPG                                                                | 0.012<br>(1.30)       | 0.015*<br>(1.68)      | 0.018*<br>(1.99)      | 0.022***<br>(2.40)    | 0.013<br>(1.45)       | 0.016*<br>(1.74)      | 0.013<br>(1.34)       |
| STKTURNOVER                                                         | 0.088***<br>(10.71)   | 0.087***<br>(10.61)   | 0.086***<br>(10.56)   | 0.086***<br>(10.51)   | 0.085***<br>(10.49)   | 0.085***<br>(10.40)   | 0.086***<br>(10.38)   |
| CREDIT                                                              | −0.014*<br>(−1.91)    | −0.014*<br>(−1.89)    | −0.012<br>(−1.62)     | −0.013*<br>(−1.70)    | −0.013*<br>(−1.74)    | −0.014*<br>(−1.85)    | −0.014*<br>(−1.81)    |
| EDUCATION                                                           | 0.038***<br>(2.82)    | 0.019<br>(1.47)       | 0.047***<br>(2.87)    | 0.050***<br>(3.29)    | 0.018<br>(1.42)       | 0.028**<br>(2.12)     | −0.005<br>(−0.20)     |
| PTY_RIGHTS                                                          | 0.030<br>(1.36)       | 0.027<br>(1.24)       | 0.037<br>(1.54)       | 0.028<br>(1.30)       | 0.049**<br>(2.17)     | 0.027<br>(1.26)       | 0.044*<br>(1.72)      |
| EXP_RISK                                                            | −0.014<br>(−0.43)     | −0.034<br>(−1.01)     | −0.033<br>(−0.87)     | −0.051<br>(−1.36)     | −0.035<br>(−1.07)     | −0.017<br>(−0.54)     | −0.003<br>(−0.05)     |
| NANTI_DIR                                                           | −0.010<br>(−0.75)     | −0.010<br>(−0.76)     | −0.020<br>(−1.28)     | −0.022*<br>(−1.67)    | 0.016<br>(1.03)       | −0.014<br>(−1.06)     | 0.016<br>(0.88)       |
| CORRUPTION                                                          | 0.007<br>(1.14)       | 0.007<br>(1.15)       | 0.007<br>(1.17)       | 0.007<br>(1.11)       | 0.003<br>(0.44)       | 0.005<br>(0.87)       | 0.005<br>(0.84)       |
| BUREAUCRACY                                                         | −0.016<br>(−0.77)     | −0.022<br>(−1.05)     | 0.006<br>(0.24)       | −0.003<br>(−0.12)     | −0.019<br>(−0.89)     | −0.026<br>(−1.22)     | −0.039<br>(−1.27)     |
| Year effect                                                         | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Industry effect                                                     | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| N                                                                   | 35,064                | 35,064                | 35,064                | 35,064                | 35,064                | 35,064                | 35,064                |
| Pseudo R <sup>2</sup>                                               | 0.1764                | 0.1764                | 0.1761                | 0.1762                | 0.1767                | 0.1763                | 0.1767                |

Panel C: Five-years rolling period innovative firms dummy variable

| VARIABLES | (1)<br>ROLL_<br>DUMMY | (2)<br>ROLL_<br>DUMMY | (3)<br>ROLL_<br>DUMMY | (4)<br>ROLL_<br>DUMMY | (5)<br>ROLL_<br>DUMMY | (5)<br>ROLL_<br>DUMMY | (7)<br>ROLL_<br>DUMMY |
|-----------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|-----------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|

(continued on next page)

Table 6 (continued)

| Panel C: Five-years rolling period innovative firms dummy variable |                               |                               |                               |                               |                               |                               |                               |
|--------------------------------------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| VARIABLES                                                          | (1)<br><i>ROLL_<br/>DUMMY</i> | (2)<br><i>ROLL_<br/>DUMMY</i> | (3)<br><i>ROLL_<br/>DUMMY</i> | (4)<br><i>ROLL_<br/>DUMMY</i> | (5)<br><i>ROLL_<br/>DUMMY</i> | (5)<br><i>ROLL_<br/>DUMMY</i> | (7)<br><i>ROLL_<br/>DUMMY</i> |
| <i>UAI</i>                                                         | −0.002***<br>(−3.05)          |                               |                               |                               |                               |                               | 0.001<br>(0.64)               |
| <i>IDV</i>                                                         |                               | 0.002***<br>(3.16)            |                               |                               |                               |                               | 0.006<br>(1.17)               |
| <i>MAS</i>                                                         |                               |                               | −0.001**<br>(−2.05)           |                               |                               |                               | −0.003<br>(−1.02)             |
| <i>PDI</i>                                                         |                               |                               |                               | −0.003**<br>(−2.18)           |                               |                               | 0.007<br>(1.55)               |
| <i>TOR</i>                                                         |                               |                               |                               |                               | 0.002***<br>(3.50)            |                               | 0.004***<br>(2.06)            |
| <i>IDG</i>                                                         |                               |                               |                               |                               |                               | 0.003***<br>(2.99)            | −0.007<br>(−1.07)             |
| <i>FarEast</i>                                                     | 0.119<br>(1.55)               | 0.118<br>(1.55)               | 0.199***<br>(2.90)            | 0.252***<br>(3.65)            | 0.036<br>(0.41)               | 0.071<br>(0.84)               | −0.064<br>(−0.37)             |
| <i>Gemanic</i>                                                     | 0.074<br>(1.51)               | 0.060<br>(1.26)               | 0.098*<br>(1.65)              | 0.053<br>(1.10)               | 0.135**<br>(2.44)             | 0.113**<br>(2.06)             | 0.132*<br>(1.66)              |
| <i>LatinEurope</i>                                                 | −0.274***<br>(−2.97)          | −0.324***<br>(−4.48)          | −0.232**<br>(−1.96)           | −0.171<br>(−1.12)             | −0.296***<br>(−3.56)          | −0.301***<br>(−3.71)          | −0.425***<br>(−6.77)          |
| <i>SIZE</i>                                                        | 0.150***<br>(21.57)           | 0.150***<br>(21.65)           | 0.147***<br>(21.47)           | 0.147***<br>(21.32)           | 0.151***<br>(21.77)           | 0.150***<br>(21.61)           | 0.153***<br>(21.79)           |
| <i>R&amp;D</i>                                                     | 0.181***<br>(19.50)           | 0.180***<br>(19.46)           | 0.181***<br>(19.51)           | 0.181***<br>(19.49)           | 0.180***<br>(19.48)           | 0.180***<br>(19.47)           | 0.181***<br>(19.49)           |
| <i>ROA</i>                                                         | 0.057***<br>(9.67)            | 0.057***<br>(9.67)            | 0.057***<br>(9.74)            | 0.057***<br>(9.72)            | 0.057***<br>(9.65)            | 0.057***<br>(9.67)            | 0.057***<br>(9.65)            |
| <i>PPE</i>                                                         | −0.062***<br>(−11.51)         | −0.062***<br>(−11.50)         | −0.062***<br>(−11.49)         | −0.062***<br>(−11.49)         | −0.062***<br>(−11.53)         | −0.062***<br>(−11.49)         | −0.062***<br>(−11.55)         |
| <i>LEV</i>                                                         | −0.013***<br>(−3.15)          | −0.013***<br>(−3.14)          | −0.013***<br>(−3.05)          | −0.013***<br>(−3.07)          | −0.014***<br>(−3.18)          | −0.013***<br>(−3.14)          | −0.014***<br>(−3.16)          |
| <i>CAPIT</i>                                                       | 0.026***<br>(5.66)            | 0.027***<br>(5.70)            | 0.026***<br>(5.64)            | 0.026***<br>(5.64)            | 0.026***<br>(5.69)            | 0.027***<br>(5.70)            | 0.027***<br>(5.74)            |
| <i>CASH</i>                                                        | 0.032***<br>(6.05)            | 0.032***<br>(6.06)            | 0.032***<br>(6.05)            | 0.032***<br>(6.02)            | 0.032***<br>(6.07)            | 0.032***<br>(6.08)            | 0.033***<br>(6.12)            |
| <i>DIV</i>                                                         | 0.003<br>(0.67)               | 0.002<br>(0.56)               | 0.003<br>(0.84)               | 0.003<br>(0.76)               | 0.002<br>(0.61)               | 0.002<br>(0.60)               | 0.002<br>(0.59)               |
| <i>GDPG</i>                                                        | 0.020*<br>(1.70)              | 0.023*<br>(1.93)              | 0.024*<br>(2.04)              | 0.025*<br>(2.13)              | 0.022*<br>(1.89)              | 0.023*<br>(1.98)              | 0.020*<br>(1.66)              |
| <i>STKTURNOVER</i>                                                 | 0.125***<br>(11.80)           | 0.125***<br>(11.79)           | 0.124***<br>(11.71)           | 0.125***<br>(11.77)           | 0.123***<br>(11.66)           | 0.124***<br>(11.74)           | 0.122***<br>(11.47)           |
| <i>CREDIT</i>                                                      | −0.042***<br>(−4.72)          | −0.043***<br>(−4.74)          | −0.041***<br>(−4.52)          | −0.041***<br>(−4.58)          | −0.041***<br>(−4.62)          | −0.042***<br>(−4.62)          | −0.042***<br>(−4.65)          |
| <i>EDUCATION</i>                                                   | 0.001<br>(0.05)               | −0.015<br>(−0.95)             | 0.014<br>(0.66)               | 0.017<br>(0.79)               | −0.012<br>(−0.77)             | −0.005<br>(−0.31)             | −0.058<br>(−1.60)             |
| <i>PTY_RIGHTS</i>                                                  | 0.053<br>(1.53)               | 0.053<br>(1.51)               | 0.067*<br>(1.75)              | 0.040<br>(1.17)               | 0.062*<br>(1.74)              | 0.047<br>(1.38)               | 0.125**<br>(1.99)             |
| <i>EXP_RISK</i>                                                    | 0.054<br>(1.24)               | 0.034<br>(0.74)               | 0.025<br>(0.49)               | 0.015<br>(0.28)               | 0.043<br>(0.95)               | 0.050<br>(1.12)               | 0.035<br>(0.44)               |
| <i>NANTI_DIR</i>                                                   | −0.001<br>(−0.04)             | 0.001<br>(0.07)               | −0.002<br>(−0.09)             | −0.013<br>(−0.77)             | 0.022<br>(1.01)               | −0.001<br>(−0.08)             | 0.061<br>(1.60)               |
| <i>CORRUPTION</i>                                                  | −0.021***<br>(−2.95)          | −0.021***<br>(−2.94)          | −0.021***<br>(−2.89)          | −0.021***<br>(−2.82)          | −0.024***<br>(−3.27)          | −0.023***<br>(−3.08)          | −0.024***<br>(−3.15)          |
| <i>BUREAUCRACY</i>                                                 | −0.078***<br>(−2.76)          | −0.082***<br>(−2.93)          | −0.050<br>(−1.46)             | −0.058*<br>(−1.83)            | −0.080***<br>(−2.87)          | −0.084***<br>(−3.02)          | −0.071<br>(−1.37)             |
| <i>Year effect</i>                                                 | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           |
| <i>Industry effect</i>                                             | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           |
| <i>N</i>                                                           | 25,230                        | 25,230                        | 25,230                        | 25,230                        | 25,230                        | 25,230                        | 25,230                        |
| <i>Pseudo R<sup>2</sup></i>                                        | 0.1862                        | 0.1862                        | 0.1861                        | 0.1861                        | 0.1863                        | 0.1862                        | 0.1864                        |

Note: This table presents the marginal effects obtained by regressing corporate innovation dummy variable *IN\_DUMMY* (Panel A), three-years (Panel B), and five-years (Panel C) rolling period dummy variable *ROLL\_DUMMY*, on cultural values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*), and on controls of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *z* scores are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 7**  
Economic effects on corporate innovation.

|                                 | (1)        | (2)        | (3)        | (4)        | (5)        | (6)        |
|---------------------------------|------------|------------|------------|------------|------------|------------|
| VARIABLES                       | <i>UAI</i> | <i>IND</i> | <i>MAS</i> | <i>PDI</i> | <i>LTO</i> | <i>IDG</i> |
| Change from min to max          | −0.197     | 0.237      | −0.076     | −0.248     | 0.222      | 0.208      |
| One standard deviation increase | −0.049     | 0.073      | −0.015     | −0.039     | 0.078      | 0.048      |

Notes: This table reports economic effects of each of Hofstede's six cultural dimensions on the predicted probability that a firm will undertake innovative activities. The magnitude of the economic effects reported in models (1) to (6) are based on models (1) to (6) in Table 6, respectively. The value represents the change in the predicted probability from being non-innovative to innovative firm resulting from a change in the independent variable from its minimum to its maximum, as well as from a one-half standard deviations below its mean to a one-half standard deviations above its, respectively.

instrument for culture. Thornhill and Fincher (2014) provide a comprehensive review of these studies.<sup>15</sup>

In the first stage, we regress each of the six cultural values (*UAI*, *IDV*, *MAS*, *PDI*, *LOT*, and *IDG*) on the index of the historical prevalence of infectious diseases. Various tests imply that our selected instrumental variable is valid. Specifically, the Wu-Hausman test rejects the null hypothesis that the cultural measures of *UAI*, *IDV*, *MAS*, and *PDI* are exogenous, whereas the test is unable to reject the null hypothesis that the cultural values of *LOT* and *IDG* are exogenous. The high F-statistic and partial  $R^2$  of the instrumental variable indicate that our estimation results do not suffer from weak instruments. Thus, our selected instrumental variable satisfies the relevant conditions necessary to be considered a valid instrument.

In the second stage, we regress our main innovation measure (*LNCITEPAT*) on the estimated cultural values, and the firm-level and country-level control variables. The results of this test, presented in Table 12, show that the five estimated measures of *UAI*, *MAS*, and *PDI* obtained from Stage 1 remain negatively and significantly (at the 1% level) associated with corporate innovation, while those of *IDV*, *LTO*, and *IDG* continue to have positive effects on innovation. The results of the firm-level and country-level control variables are also qualitatively similar to those presented in Table 8. To further alleviate endogeneity concerns, in addition to a two-stage instrumental variable approach, we also implement an instrumental generalized method of moments (GMM) approach, where, as in Arellano and Bover (1995) and Blundell and Bond (1998), the instrumental variables are the lagged first differences of the dependent variable. The results reported in Table 13 show that our main results continue to hold.

Insert Tables 12 and 13 about here.

#### 4.3.4. Country-level regressions and excluding the US

First, estimate the model on a subsample of our innovation data that excludes the US, which accounts for 54.7% of the total firms in our sample of innovative firms. The results, presented in Table 14, show that our main results are not driven by the US.<sup>16</sup> Second, we regress our model at the country level to further mitigate concerns that our results are driven by particular countries. The results, reported in Table 15, suggest that our findings continue to hold at the country level, which should also lessen concerns regarding omitted firm-level variables.

#### 4.3.5. Subperiod analysis

To test the robustness of our results over time, we regress our model over different subperiods. First, we test the effect of the financial crisis on our results. The crisis period is defined as the years 1997 to 1998 (which correspond to the Asian financial crisis), the year 2000 (which corresponds to the collapse of the dotcom bubble), and the years 2007 to 2008 (which correspond to the global financial crisis). Table 16 reports the results for the crisis period (Panel A) and those for the non-crisis period (Panel B). Second, Table 17 reports the results for the period consisting of the year 2000 and before (Panel A), and the period after the year 2000 (Panel B). Recent economic studies show that, after the year 2000, innovators became less concerned with keeping their innovations secret and started publicly disclosing information about their inventions before they were granted a patent (Graham and Hegde 2015). In both cases, the results remain qualitatively similar to those obtained in Table 8.

#### 4.3.6. Uncertainty avoidance and risk taking

Griffin et al. (2019) examine the effect of board gender diversity on corporate innovation across 45 countries. They find that board gender diversity, as measured by the ratio of female to male directors, fosters both the level and the quality of firms' innovation. They

<sup>15</sup> We are thankful to an anonymous reviewer for suggesting this additional analysis.

<sup>16</sup> It is widely believed that the prevalence of infectious diseases influences the formation of personality traits and cultural values among regions (Murray and Schaller 2010; Nikolaev and Salahodjaev 2017). It is, however, unlikely that patents' citations are influenced by the prevalence of these diseases.

**Table 8**  
The impact of cultural values on corporate innovation: Innovation Sample.

| VARIABLES    | (1)                   | (2)                  | (3)                   | (4)                  | (5)                  | (6)                  | (7)                  |
|--------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|
|              | LNCITEPAT_nadj        | LNCITEPAT_nadj       | LNCITEPAT_nadj        | LNCITEPAT_nadj       | LNCITEPAT_nadj       | LNCITEPAT_nadj       | LNCITEPAT_nadj       |
| UAI          | -0.283***<br>(-22.40) |                      |                       |                      |                      |                      | -0.160***<br>(-8.38) |
| IDV          |                       | 0.328***<br>(17.79)  |                       |                      |                      |                      | -0.028<br>(-1.03)    |
| MAS          |                       |                      | -0.185***<br>(-13.13) |                      |                      |                      | 0.039***<br>(2.29)   |
| PDI          |                       |                      |                       | -0.214***<br>(-7.05) |                      |                      | 0.402***<br>(9.47)   |
| TOR          |                       |                      |                       |                      | 0.323***<br>(20.60)  |                      | 0.330***<br>(10.72)  |
| IDG          |                       |                      |                       |                      |                      | 0.452***<br>(17.88)  | 0.029<br>(0.82)      |
| FarEast      | 25.708***<br>(17.99)  | 28.607***<br>(19.11) | 38.663***<br>(23.57)  | 43.710***<br>(20.83) | 17.730***<br>(11.33) | 23.392***<br>(15.11) | -3.006<br>(-1.05)    |
| Germanic     | 13.735***<br>(16.60)  | 9.592***<br>(12.51)  | 12.717***<br>(12.91)  | 5.444***<br>(7.93)   | 20.910***<br>(18.75) | 15.677***<br>(15.86) | 21.750***<br>(15.59) |
| LatinAmerica | 24.289***<br>(13.19)  | 10.996***<br>(5.74)  | 26.243***<br>(12.97)  | 22.673***<br>(10.11) | 3.171<br>(1.59)      | 8.051***<br>(4.16)   | 5.248***<br>(2.09)   |
| LatinEurope  | 10.800***<br>(7.64)   | -3.402***<br>(-2.24) | 7.118***<br>(4.45)    | 4.689***<br>(2.84)   | 2.001<br>(1.15)      | 0.982<br>(0.67)      | -8.088***<br>(-2.63) |
| SIZE         | 15.366***<br>(35.12)  | 15.065***<br>(34.66) | 14.388***<br>(34.63)  | 14.107***<br>(33.82) | 15.368***<br>(35.13) | 15.031***<br>(34.89) | 15.610***<br>(35.26) |
| R&D          | 1.512***<br>(8.38)    | 1.483***<br>(8.23)   | 1.514***<br>(8.36)    | 1.513***<br>(8.35)   | 1.498***<br>(8.29)   | 1.497***<br>(8.29)   | 1.526***<br>(8.44)   |
| ROA          | -0.258<br>(-1.44)     | -0.189<br>(-1.06)    | -0.048<br>(-0.27)     | 0.011<br>(0.06)      | -0.261<br>(-1.46)    | -0.183<br>(-1.02)    | -0.289<br>(-1.61)    |
| PPE          | -1.056***<br>(-5.17)  | -1.047***<br>(-5.12) | -1.040***<br>(-5.09)  | -1.078***<br>(-5.26) | -1.069***<br>(-5.24) | -1.100***<br>(-5.38) | -1.057***<br>(-5.18) |
| LEV          | -0.723***<br>(-4.82)  | -0.652***<br>(-4.35) | -0.579***<br>(-3.88)  | -0.552***<br>(-3.68) | -0.703***<br>(-4.70) | -0.649***<br>(-4.34) | -0.700***<br>(-4.68) |
| CAPIT        | 1.179***<br>(7.26)    | 1.230***<br>(7.55)   | 1.182***<br>(7.23)    | 1.223***<br>(7.46)   | 1.214***<br>(7.49)   | 1.270***<br>(7.81)   | 1.215***<br>(7.50)   |
| CASH         | -0.564***<br>(-2.88)  | -0.525***<br>(-2.69) | -0.543***<br>(-2.77)  | -0.573***<br>(-2.92) | -0.523***<br>(-2.68) | -0.532***<br>(-2.72) | -0.528***<br>(-2.71) |
| DIV          | 2.049***<br>(10.53)   | 2.020***<br>(10.24)  | 2.317***<br>(11.82)   | 2.345***<br>(11.88)  | 2.080***<br>(10.68)  | 2.113***<br>(10.81)  | 2.054***<br>(10.44)  |
| GDPG         | 1.604***<br>(6.47)    | 2.332***<br>(9.42)   | 2.674***<br>(10.70)   | 3.192***<br>(11.92)  | 2.061***<br>(9.02)   | 2.381***<br>(10.32)  | 1.244***<br>(5.75)   |
| STKTURNVER   | 1.621***<br>(7.31)    | 1.419***<br>(6.32)   | 1.337***<br>(5.93)    | 1.222***<br>(5.43)   | 1.293***<br>(5.92)   | 1.128***<br>(5.10)   | 1.507***<br>(6.85)   |
| CREDIT       | -1.132***<br>(-5.52)  | -1.047***<br>(-5.08) | -0.871***<br>(-3.63)  | -0.773***<br>(-4.06) | -0.952***<br>(-4.71) | -1.301***<br>(-6.27) | -1.255***<br>(-6.19) |
| EDUCATION    | 8.301***<br>(24.63)   | 5.401***<br>(17.13)  | 8.903***<br>(21.81)   | 7.441***<br>(21.51)  | 5.372***<br>(17.24)  | 6.221***<br>(19.50)  | 3.512***<br>(6.63)   |
| PTY_RIGHTS   | 4.337***<br>(10.11)   | 3.531***<br>(7.67)   | 4.700***<br>(9.91)    | 2.556***<br>(5.57)   | 6.209***<br>(13.34)  | 4.289***<br>(8.98)   | 5.139***<br>(10.35)  |

(continued on next page)

Table 8 (continued)

|                 | (1)                  | (2)                   | (3)                  | (4)                   | (5)                  | (6)                  | (7)                   |
|-----------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|-----------------------|
| VARIABLES       | LNCITEPAT_nadj       | LNCITEPAT_nadj        | LNCITEPAT_nadj       | LNCITEPAT_nadj        | LNCITEPAT_nadj       | LNCITEPAT_nadj       | LNCITEPAT_nadj        |
| EXP_RISK        | -0.529<br>(-0.95)    | -1.881***<br>(-3.07)  | -0.788<br>(-1.20)    | 1.031<br>(1.60)       | -2.329***<br>(-3.40) | -0.315<br>(-0.52)    | 2.293**<br>(2.44)     |
| NANTI_DIR       | -1.255***<br>(-4.21) | -2.348***<br>(-7.90)  | -2.691***<br>(-7.20) | -4.637***<br>(-15.65) | 0.446<br>(1.24)      | -2.127***<br>(-6.66) | 0.020<br>(0.06)       |
| CORRUPTION      | -1.682***<br>(-9.29) | -1.902***<br>(-10.38) | -1.844***<br>(-9.59) | -2.380***<br>(-12.30) | -1.414***<br>(-7.56) | -1.550***<br>(-8.23) | -1.782***<br>(-9.01)  |
| BUREAUCRACY     | -1.764***<br>(-5.58) | -2.925***<br>(-7.94)  | 0.020<br>(0.05)      | -2.503***<br>(-7.25)  | -2.146***<br>(-5.32) | -3.452***<br>(-9.73) | -3.743***<br>(-5.87)  |
| Constant        | 32.705***<br>(12.69) | -8.121***<br>(-2.98)  | 27.835***<br>(10.36) | 24.739***<br>(8.59)   | -0.712<br>(-0.28)    | -9.754***<br>(-3.55) | -14.489***<br>(-2.80) |
| Year effect     | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  | Yes                   |
| Industry effect | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                  | Yes                   |
| Observations    | 27,636               | 27,636                | 27,636               | 27,636                | 27,636               | 27,636               | 27,636                |
| R-squared       | 0.221                | 0.219                 | 0.216                | 0.214                 | 0.221                | 0.219                | 0.223                 |

Note: This table presents the estimation results obtained by regressing corporate innovation (*LNCITEPAT*) on cultural values (*UAI*, *CLT*, *TOR*, and *IDG*) and on controls of firm characteristics and country variables, respectively. The dependent variable is *LNCITEPAT*. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. The industry controls are based on the 12 Fama-French industry classification codes. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

Table 9

Alternative measures of culture values.

|                        | (1)                   | (2)                   | (3)                   | (4)                   | (5)                  | (6)                   |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|
|                        | LNCITEPAT_nadj        | LNCITEPAT_nadj        | LNCITEPAT_nadj        | LNCITEPAT_nadj        | LNCITEPAT_nadj       | LNCITEPAT_nadj        |
| VARIABLES              |                       |                       |                       |                       |                      |                       |
| <i>UAI_TK</i>          | −0.395***<br>(−21.52) |                       |                       |                       |                      | −0.349***<br>(−6.82)  |
| <i>IND_TK</i>          |                       | 0.563***<br>(11.51)   |                       |                       |                      | −0.534***<br>(−7.74)  |
| <i>MAS_TK</i>          |                       |                       | −0.087***<br>(−3.21)  |                       |                      | 0.046<br>(1.18)       |
| <i>PDI_TK</i>          |                       |                       |                       | −0.414***<br>(−9.82)  |                      | −0.443***<br>(−10.76) |
| <i>LTO_TK</i>          |                       |                       |                       |                       | 0.269***<br>(19.41)  | 0.140***<br>(3.98)    |
| <i>FarEast</i>         | 28.792***<br>(16.18)  | 35.860***<br>(20.76)  | 42.021***<br>(19.92)  | 47.274***<br>(23.78)  | 16.191***<br>(9.82)  | 32.384***<br>(15.60)  |
| <i>Gemanic</i>         | 9.278***<br>(12.46)   | 13.041***<br>(12.14)  | 6.817***<br>(5.55)    | 2.317***<br>(3.07)    | 6.642***<br>(9.41)   | −1.464<br>(−0.67)     |
| <i>LatinAmerica</i>    | 27.882***<br>(13.16)  | 19.634***<br>(8.85)   | 29.882***<br>(11.56)  | 29.265***<br>(11.10)  | 13.538***<br>(7.20)  | 32.956***<br>(12.97)  |
| <i>LatinEurope</i>     | 12.557***<br>(8.19)   | 0.049<br>(0.03)       | −2.050<br>(−1.19)     | 1.704<br>(0.92)       | 0.573<br>(0.42)      | 13.033***<br>(6.46)   |
| <i>SIZE</i>            | 15.547***<br>(35.21)  | 14.357***<br>(34.59)  | 13.838***<br>(34.44)  | 13.888***<br>(34.63)  | 15.248***<br>(34.84) | 15.655***<br>(35.22)  |
| <i>RD</i>              | 1.497***<br>(8.30)    | 1.519***<br>(8.40)    | 1.541***<br>(8.48)    | 1.537***<br>(8.47)    | 1.477***<br>(8.19)   | 1.492***<br>(8.27)    |
| <i>ROA</i>             | −0.293<br>(−1.63)     | −0.017<br>(−0.09)     | 0.087<br>(0.49)       | 0.081<br>(0.45)       | −0.238<br>(−1.33)    | −0.325*<br>(−1.80)    |
| <i>PPE</i>             | −1.006***<br>(−4.92)  | −1.067***<br>(−5.20)  | −1.051***<br>(−5.11)  | −1.016***<br>(−4.94)  | −1.035***<br>(−5.07) | −0.952***<br>(−4.64)  |
| <i>LEV</i>             | −0.712***<br>(−4.75)  | −0.527***<br>(−3.54)  | −0.482***<br>(−3.24)  | −0.476***<br>(−3.20)  | −0.690***<br>(−4.60) | −0.740***<br>(−4.93)  |
| <i>CAPIT</i>           | 1.182***<br>(7.28)    | 1.258***<br>(7.68)    | 1.221***<br>(7.42)    | 1.248***<br>(7.61)    | 1.223***<br>(7.53)   | 1.179***<br>(7.25)    |
| <i>CASH</i>            | −0.514***<br>(−2.63)  | −0.558***<br>(−2.85)  | −0.573***<br>(−2.91)  | −0.560***<br>(−2.85)  | −0.506***<br>(−2.59) | −0.496***<br>(−2.54)  |
| <i>DIV</i>             | 1.955***<br>(9.95)    | 2.257***<br>(11.37)   | 2.458***<br>(12.41)   | 2.415***<br>(12.22)   | 2.017***<br>(10.29)  | 1.944***<br>(9.88)    |
| <i>GDPG</i>            | 1.380***<br>(5.71)    | 2.709***<br>(10.68)   | 2.909***<br>(11.33)   | 2.953***<br>(11.56)   | 2.420***<br>(10.20)  | 1.291***<br>(5.41)    |
| <i>STKTURNOVER</i>     | 1.897***<br>(8.37)    | 1.368***<br>(5.89)    | 1.545***<br>(6.65)    | 1.715***<br>(7.39)    | 1.386***<br>(6.23)   | 2.276***<br>(9.71)    |
| <i>CREDIT</i>          | −1.198***<br>(−5.84)  | −1.268***<br>(−5.97)  | −1.205***<br>(−5.58)  | −0.914***<br>(−4.12)  | −0.704***<br>(−3.43) | −0.805***<br>(−3.87)  |
| <i>EDUCATION</i>       | 4.154***<br>(13.45)   | 4.238***<br>(11.43)   | 7.366***<br>(15.89)   | 5.509***<br>(16.55)   | 6.456***<br>(20.68)  | 5.401***<br>(7.98)    |
| <i>PTY_RIGHTS</i>      | 5.217***<br>(11.78)   | 1.586***<br>(3.36)    | 1.462***<br>(2.98)    | 1.327***<br>(2.78)    | 5.510***<br>(12.25)  | 6.314***<br>(13.52)   |
| <i>EXP_RISK</i>        | −0.916<br>(−1.33)     | −2.689***<br>(−2.71)  | 5.273***<br>(7.38)    | 1.727***<br>(2.07)    | −1.573***<br>(−2.66) | 1.763<br>(1.59)       |
| <i>NANTI_DIR</i>       | −1.715***<br>(−5.14)  | −3.427***<br>(−10.23) | −4.670***<br>(−11.32) | −4.437***<br>(−13.02) | −1.938***<br>(−6.22) | −0.749<br>(−1.29)     |
| <i>CORRUPTION</i>      | −1.863***<br>(−9.89)  | −2.800***<br>(−14.59) | −3.084***<br>(15.92)  | −3.496***<br>(−17.37) | −1.483***<br>(−7.90) | −2.112***<br>(−10.70) |
| <i>BUREAUCRACY</i>     | −0.463<br>(−1.26)     | −3.864***<br>(−8.54)  | −3.327***<br>(−7.90)  | −3.946***<br>(−8.91)  | −0.872**<br>(−2.52)  | −0.310<br>(−0.76)     |
| <i>Intercept</i>       | 34.793***<br>(13.21)  | −37.837***<br>(−7.33) | 18.578***<br>(6.54)   | 21.521***<br>(8.27)   | 2.571<br>(1.03)      | 80.660***<br>(9.17)   |
| <i>Year effect</i>     | Yes                   | Yes                   | Yes                   | Yes                   | Yes                  | Yes                   |
| <i>Industry effect</i> | Yes                   | Yes                   | Yes                   | Yes                   | Yes                  | Yes                   |
| <i>N</i>               | 27,573                | 27,573                | 27,573                | 27,573                | 27,636               | 27,573                |
| <i>R<sup>2</sup></i>   | 0.222                 | 0.216                 | 0.214                 | 0.215                 | 0.220                | 0.223                 |

Note: This table presents the estimation results obtained by regressing corporate innovation (*LNCITEPAT*) on alternative measures of cultural values (*UAI\_TK*, *CLT\_TK*, *MAS\_TK*, *PDI\_TK*, and *TOR\_TK*) and on controls of firm characteristics and country variables, respectively. The dependent variable is *LNCITEPAT*. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 10**  
Alternative measures of innovation.

| VARIABLES           | (1)                   | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  | (9)                  | (10)                 | (11)                 | (12)                 | (13)                 | (14)                 |
|---------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                     | LNците                | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               | LNците               |
| <i>UAI</i>          | -0.010***<br>(-12.81) |                      |                      |                      |                      |                      | -0.006***<br>(-7.23) | -0.002***<br>(-9.19) |                      |                      |                      |                      |                      | -0.001***<br>(-5.73) |
| <i>IDV</i>          |                       | 0.011***<br>(10.13)  |                      |                      |                      |                      |                      |                      | 0.002***<br>(7.48)   |                      |                      |                      |                      | -0.000<br>(-1.04)    |
| <i>MAS</i>          |                       |                      | -0.008***<br>(-9.38) |                      |                      |                      |                      |                      |                      | -0.001***<br>(-8.12) |                      |                      |                      | -0.000<br>(-1.59)    |
| <i>PDI</i>          |                       |                      |                      | -0.011***<br>(-6.85) |                      |                      | 0.006***<br>(3.37)   |                      |                      |                      | -0.002***<br>(-5.46) |                      |                      | 0.001***<br>(3.19)   |
| <i>TOR</i>          |                       |                      |                      |                      | 0.011***<br>(11.78)  |                      | 0.007***<br>(5.41)   |                      |                      |                      |                      | 0.002***<br>(8.53)   |                      | 0.001***<br>(4.46)   |
| <i>IDG</i>          |                       |                      |                      |                      |                      | 0.016***<br>(11.23)  |                      |                      |                      |                      |                      |                      | 0.003***<br>(7.92)   | 0.000<br>(0.88)      |
| <i>ForEast</i>      | 0.697***<br>(7.59)    | 0.796***<br>(7.77)   | 1.157***<br>(11.37)  | 1.449***<br>(11.09)  | 0.439***<br>(4.38)   | 0.601***<br>(6.13)   | 0.213***<br>(2.27)   | 0.104***<br>(6.35)   | 0.121***<br>(6.46)   | 0.183***<br>(8.50)   | 0.230***<br>(8.05)   | 0.062***<br>(3.56)   | 0.088***<br>(5.21)   | 0.027<br>(1.45)      |
| <i>Gemanic</i>      | 0.282***<br>(5.77)    | 0.141***<br>(3.00)   | 0.337***<br>(5.19)   | 0.039***<br>(0.84)   | 0.514***<br>(8.23)   | 0.365***<br>(6.66)   | 0.532***<br>(7.74)   | 0.051***<br>(4.51)   | 0.026***<br>(2.49)   | 0.063***<br>(4.97)   | 0.009***<br>(0.89)   | 0.088***<br>(6.18)   | 0.064***<br>(4.80)   | 0.095***<br>(6.41)   |
| <i>LatinAmerica</i> | 0.732***<br>(7.11)    | 0.283***<br>(2.70)   | 0.818***<br>(8.41)   | 0.648***<br>(4.92)   | 0.031<br>(0.28)      | 0.160<br>(1.56)      | 0.305***<br>(3.20)   | 0.136***<br>(7.12)   | 0.062***<br>(3.43)   | 0.152***<br>(7.67)   | 0.122***<br>(5.36)   | 0.019<br>(1.05)      | 0.040***<br>(2.36)   | 0.071***<br>(3.73)   |
| <i>LatinEurope</i>  | 0.072<br>(0.72)       | -0.409***<br>(-3.66) | 0.056<br>(0.62)      | 0.037<br>(0.38)      | -0.230***<br>(-2.00) | -0.254***<br>(-2.51) | -0.189***<br>(-1.96) | 0.026<br>(1.63)      | -0.055***<br>(-3.16) | 0.028<br>(1.66)      | 0.019<br>(1.02)      | -0.025<br>(-1.40)    | -0.029*<br>(-1.78)   | -0.013<br>(-0.70)    |
| <i>SIZE</i>         | 0.383***<br>(15.15)   | 0.372***<br>(14.74)  | 0.355***<br>(14.78)  | 0.345***<br>(14.51)  | 0.381***<br>(15.02)  | 0.373***<br>(14.89)  | 0.389***<br>(15.12)  | 0.060***<br>(10.01)  | 0.058***<br>(9.81)   | 0.056***<br>(9.81)   | 0.054***<br>(9.63)   | 0.060***<br>(9.92)   | 0.058***<br>(9.84)   | 0.061***<br>(10.00)  |
| <i>R&amp;D</i>      | 0.047***<br>(6.05)    | 0.046***<br>(5.93)   | 0.046***<br>(6.00)   | 0.047***<br>(5.98)   | 0.047***<br>(5.98)   | 0.046***<br>(5.97)   | 0.047***<br>(5.97)   | 0.005***<br>(2.99)   | 0.005***<br>(2.90)   | 0.005***<br>(2.96)   | 0.005***<br>(2.95)   | 0.005***<br>(2.94)   | 0.005***<br>(2.93)   | 0.005***<br>(2.99)   |
| <i>ROA</i>          | 0.010<br>(1.29)       | 0.013<br>(1.59)      | 0.016***<br>(2.04)   | 0.018***<br>(2.27)   | 0.010<br>(1.31)      | 0.013<br>(1.57)      | 0.009<br>(1.15)      | 0.003<br>(1.37)      | 0.003<br>(1.60)      | 0.004*<br>(1.87)     | 0.004*<br>(2.07)     | 0.004*<br>(1.39)     | 0.003<br>(1.58)      | 0.002<br>(1.28)      |
| <i>PPE</i>          | -0.047***<br>(-4.04)  | -0.047***<br>(-4.00) | -0.046***<br>(-3.95) | -0.048***<br>(-4.09) | -0.048***<br>(-4.07) | -0.049***<br>(-4.16) | -0.047***<br>(-4.04) | -0.008***<br>(-3.16) | -0.008***<br>(-3.13) | -0.008***<br>(-3.09) | -0.008***<br>(-3.20) | -0.008***<br>(-3.18) | -0.008***<br>(-3.25) | -0.008***<br>(-3.15) |
| <i>LEV</i>          | -0.004<br>(-0.50)     | -0.001<br>(-0.17)    | 0.000<br>(0.05)      | 0.001<br>(0.13)      | -0.003<br>(-0.39)    | -0.001<br>(-0.19)    | -0.004<br>(-0.49)    | -0.000<br>(-0.22)    | 0.000<br>(0.01)      | 0.000<br>(0.13)      | 0.000<br>(0.21)      | -0.000<br>(-0.14)    | -0.000<br>(-0.01)    | -0.000<br>(-0.21)    |
| <i>CAPIT</i>        | 0.042***<br>(6.43)    | 0.044***<br>(6.63)   | 0.042***<br>(6.30)   | 0.043***<br>(6.50)   | 0.043***<br>(6.60)   | 0.045***<br>(6.84)   | 0.043***<br>(6.51)   | 0.007***<br>(4.81)   | 0.007***<br>(4.96)   | 0.007***<br>(4.73)   | 0.007***<br>(4.89)   | 0.007***<br>(4.94)   | 0.008***<br>(5.11)   | 0.007***<br>(4.85)   |
| <i>CASH</i>         | -0.009<br>(-1.01)     | -0.008<br>(-0.86)    | -0.008<br>(-0.89)    | -0.009<br>(-1.03)    | -0.008<br>(-0.86)    | -0.008<br>(-0.88)    | -0.008<br>(-0.90)    | -0.003<br>(-1.21)    | -0.002<br>(-1.11)    | -0.002<br>(-1.12)    | -0.003<br>(-1.22)    | -0.002<br>(-1.11)    | -0.002<br>(-1.12)    | -0.002<br>(-1.13)    |
| <i>DIV</i>          | 0.049***<br>(4.75)    | 0.048***<br>(4.55)   | 0.057***<br>(5.39)   | 0.057***<br>(5.38)   | 0.050***<br>(4.84)   | 0.050***<br>(4.87)   | 0.049***<br>(4.71)   | 0.011***<br>(3.79)   | 0.011***<br>(3.70)   | 0.013***<br>(4.16)   | 0.013***<br>(4.16)   | 0.012***<br>(3.87)   | 0.012***<br>(3.87)   | 0.011***<br>(3.77)   |
| <i>GDPG</i>         | 0.009<br>(1.30)       | 0.033***<br>(4.23)   | 0.040***<br>(5.17)   | 0.063***<br>(6.82)   | 0.025***<br>(3.79)   | 0.034***<br>(4.87)   | 0.004<br>(0.77)      | 0.001<br>(1.01)      | 0.006***<br>(3.57)   | 0.006***<br>(4.38)   | 0.010***<br>(5.77)   | 0.004***<br>(3.11)   | 0.004***<br>(4.02)   | 0.000<br>(0.41)      |
| <i>STKTURNVER</i>   | 0.008<br>(1.09)       | 0.015***<br>(1.96)   | 0.016***<br>(2.24)   | 0.022***<br>(2.98)   | 0.019***<br>(2.73)   | 0.025***<br>(3.39)   | 0.013***<br>(1.83)   | 0.003***<br>(1.71)   | 0.004***<br>(2.40)   | 0.004***<br>(2.60)   | 0.005***<br>(3.21)   | 0.005***<br>(3.00)   | 0.005***<br>(3.52)   | 0.003***<br>(2.23)   |
| <i>CREDIT</i>       | -0.003<br>(-0.44)     | 0.000<br>(0.03)      | 0.011*<br>(1.67)     | 0.008<br>(1.13)      | 0.003<br>(0.55)      | -0.009<br>(-1.30)    | -0.003<br>(-0.50)    | 0.002<br>(1.21)      | 0.002<br>(1.54)      | 0.002<br>(2.88)      | 0.003***<br>(2.35)   | 0.003***<br>(1.97)   | 0.001<br>(0.42)      | 0.002<br>(1.30)      |
| <i>EDUCATION</i>    | 0.236***<br>(14.63)   | 0.138***<br>(8.98)   | 0.284***<br>(12.73)  | 0.234***<br>(12.18)  | 0.137***<br>(9.25)   | 0.166***<br>(10.95)  | 0.165***<br>(10.73)  | 0.040***<br>(10.73)  | 0.023***<br>(7.19)   | 0.049***<br>(10.46)  | 0.039***<br>(9.54)   | 0.023***<br>(7.27)   | 0.028***<br>(8.53)   | 0.030***<br>(6.93)   |
| <i>PTY_RIGHTS</i>   | 0.142***<br>(6.94)    | 0.115***<br>(4.66)   | 0.186***<br>(6.34)   | 0.104***<br>(3.79)   | 0.202***<br>(8.48)   | 0.145***<br>(6.01)   | 0.190***<br>(7.64)   | 0.022***<br>(3.32)   | 0.017***<br>(2.45)   | 0.030***<br>(4.11)   | 0.015***<br>(2.07)   | 0.032***<br>(4.41)   | 0.022***<br>(3.05)   | 0.031***<br>(4.48)   |
| <i>EXP_RISK</i>     | 0.010                 | -0.035               | -0.048               | -0.001               | -0.046               | 0.011                | 0.016                | -0.001               | -0.008               | -0.013               | -0.002               | -0.010               | -0.000               | -0.001               |

(continued on next page)

Table 10 (continued)

| VARIABLES       | (1)                            | (2)                             | (3)                             | (4)                             | (5)                        | (6)                            | (7)                       | (8)                             | (9)                             | (10)                            | (11)                            | (12)                         | (13)                            | (14)                         |
|-----------------|--------------------------------|---------------------------------|---------------------------------|---------------------------------|----------------------------|--------------------------------|---------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|
|                 | LNCFITE                        | LNCFITE                         | LNCFITE                         | LNCFITE                         | LNCFITE                    | LNCFITE                        | LNCFITE                   | LNCFITE                         | LNCFITE                         | LNCFITE                         | LNCFITE                         | LNCFITE                      | LNCFITE                         | LNCFITE                      |
| NANTI_DIR       | (0.40)<br>-0.046***<br>(-3.33) | (-1.07)<br>-0.083***<br>(-5.86) | (-1.35)<br>-0.063***<br>(-3.04) | (-0.03)<br>-0.135***<br>(-7.75) | (-1.33)<br>0.007<br>(0.47) | (0.38)<br>-0.070***<br>(-4.81) | (0.41)<br>0.004<br>(0.24) | (-0.10)<br>-0.010***<br>(-3.43) | (-0.97)<br>-0.017***<br>(-5.32) | (-1.54)<br>-0.012***<br>(-3.26) | (-0.25)<br>-0.026***<br>(-7.16) | (-1.20)<br>-0.002<br>(-0.53) | (-0.03)<br>-0.015***<br>(-4.52) | (-0.13)<br>-0.001<br>(-0.40) |
| CORRUPTION      | -0.022***<br>(-3.45)           | -0.029***<br>(-4.48)            | -0.018***<br>(-2.75)            | -0.037***<br>(-5.53)            | -0.013***<br>(-2.07)       | -0.015***<br>(-2.42)           | -0.014***<br>(-2.17)      | -0.003***<br>(-2.59)            | -0.005***<br>(-3.53)            | -0.002***<br>(-1.68)            | -0.006***<br>(-4.23)            | -0.002<br>(-1.58)            | -0.002*<br>(-1.79)              | -0.002<br>(-1.32)            |
| BUREAUCRACY     | -0.063***<br>(-2.34)           | -0.103***<br>(-3.34)            | 0.031<br>(1.28)                 | -0.070***<br>(-2.68)            | -0.077***<br>(-2.46)       | -0.120***<br>(-4.24)           | -0.075***<br>(-3.07)      | -0.009***<br>(-2.48)            | -0.016***<br>(-3.59)            | 0.008*<br>(1.90)                | -0.010***<br>(-2.63)            | -0.011***<br>(-2.56)         | -0.019***<br>(-4.61)            | -0.009***<br>(-2.16)         |
| Intercept       | 1.134***<br>(12.44)            | -0.246***<br>(-2.22)            | 1.096***<br>(10.25)             | 1.058***<br>(9.34)              | 0.013<br>(0.14)            | -0.338***<br>(-3.17)           | 0.292<br>(1.34)           | 0.313***<br>(6.21)              | 0.082<br>(1.51)                 | 0.312***<br>(5.97)              | 0.299***<br>(5.77)              | 0.125***<br>(2.39)           | 0.066<br>(1.22)                 | 0.179***<br>(2.85)           |
| Year effect     | Yes                            | Yes                             | Yes                             | Yes                             | Yes                        | Yes                            | Yes                       | Yes                             | Yes                             | Yes                             | Yes                             | Yes                          | Yes                             | Yes                          |
| Industry effect | Yes                            | Yes                             | Yes                             | Yes                             | Yes                        | Yes                            | Yes                       | Yes                             | Yes                             | Yes                             | Yes                             | Yes                          | Yes                             | Yes                          |
| N               | 27,636                         | 27,636                          | 27,636                          | 27,636                          | 27,636                     | 27,636                         | 27,636                    | 27,636                          | 27,636                          | 27,636                          | 27,636                          | 27,636                       | 27,636                          | 27,636                       |
| R <sup>2</sup>  | 0.317                          | 0.311                           | 0.307                           | 0.303                           | 0.316                      | 0.312                          | 0.319                     | 0.262                           | 0.258                           | 0.257                           | 0.254                           | 0.261                        | 0.259                           | 0.263                        |

Note: This table presents the estimation results obtained by regressing corporate innovation (LNCFITE and LNCFITEPAT\_NADD) on cultural values (UAL, CLT, MAS, PDI, TOR, and IDG) and on controls of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 11**  
NBER Patents and Citations database.

| VARIABLES          | (1)<br><i>LNCITEPAT</i> | (2)<br><i>LNCITEPAT</i> | (3)<br><i>LNCITEPAT</i> | (4)<br><i>LNCITEPAT</i> | (5)<br><i>LNCITEPAT</i> | (6)<br><i>LNCITEPAT</i> | (7)<br><i>LNCITEPAT</i> |
|--------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>UAI</i>         | −0.014***<br>(−4.65)    |                         |                         |                         |                         |                         | −0.015**<br>(−2.36)     |
| <i>IDV</i>         |                         | 0.009**<br>(2.28)       |                         |                         |                         |                         | 0.002<br>(0.34)         |
| <i>MAS</i>         |                         |                         | −0.010**<br>(−2.41)     |                         |                         |                         | −0.008<br>(−1.60)       |
| <i>PDI</i>         |                         |                         |                         | −0.012**<br>(−2.39)     |                         |                         | −0.014<br>(−1.19)       |
| <i>TOR</i>         |                         |                         |                         |                         | 0.014***<br>(4.38)      |                         | 0.002<br>(0.27)         |
| <i>IDG</i>         |                         |                         |                         |                         |                         | 0.009*<br>(1.66)        | −0.024***<br>(−3.23)    |
| <i>FarEast</i>     | −0.433<br>(−1.47)       | −0.294<br>(−0.92)       | −0.702**<br>(−2.10)     | −0.160<br>(−0.48)       | −0.439<br>(−1.38)       | −0.196<br>(−0.56)       | −0.407<br>(−0.64)       |
| <i>Gemanic</i>     | 0.071<br>(0.40)         | 0.029<br>(0.15)         | 0.411<br>(1.60)         | −0.106<br>(−0.60)       | −0.028<br>(−0.16)       | 0.005<br>(0.02)         | −0.017<br>(−0.06)       |
| <i>LatinEurope</i> | 0.577**<br>(2.48)       | 0.123<br>(0.55)         | 0.243<br>(1.08)         | 0.216<br>(0.97)         | 0.260<br>(1.11)         | 0.260<br>(0.89)         | 0.962**<br>(2.11)       |
| <i>SIZE</i>        | 0.657***<br>(53.36)     | 0.656***<br>(53.19)     | 0.655***<br>(53.25)     | 0.655***<br>(53.22)     | 0.657***<br>(53.33)     | 0.656***<br>(53.22)     | 0.658***<br>(53.30)     |
| <i>R&amp;D</i>     | 0.114***<br>(12.54)     | 0.114***<br>(12.45)     | 0.114***<br>(12.45)     | 0.113***<br>(12.39)     | 0.114***<br>(12.51)     | 0.113***<br>(12.39)     | 0.114***<br>(12.52)     |
| <i>ROA</i>         | −0.058***<br>(−5.89)    | −0.058***<br>(−5.84)    | −0.058***<br>(−5.87)    | −0.058***<br>(−5.90)    | −0.058***<br>(−5.89)    | −0.058***<br>(−5.82)    | −0.058***<br>(−5.88)    |
| <i>PPE</i>         | −0.144***<br>(−14.34)   | −0.143***<br>(−14.23)   | −0.144***<br>(−14.28)   | −0.143***<br>(−14.25)   | −0.144***<br>(−14.30)   | −0.143***<br>(−14.21)   | −0.145***<br>(−14.38)   |
| <i>LEV</i>         | −0.038***<br>(−5.63)    | −0.038***<br>(−5.62)    | −0.038***<br>(−5.61)    | −0.038***<br>(−5.65)    | −0.038***<br>(−5.64)    | −0.038***<br>(−5.64)    | −0.037***<br>(−5.56)    |
| <i>CAPIT</i>       | 0.145***<br>(18.31)     | 0.145***<br>(18.30)     | 0.144***<br>(18.25)     | 0.144***<br>(18.29)     | 0.145***<br>(18.32)     | 0.145***<br>(18.33)     | 0.145***<br>(18.33)     |
| <i>CASH</i>        | −0.015*<br>(−1.74)      | −0.014*<br>(−1.71)      | −0.014*<br>(−1.72)      | −0.014*<br>(−1.73)      | −0.014*<br>(−1.71)      | −0.014*<br>(−1.68)      | −0.014*<br>(−1.69)      |
| <i>DIV</i>         | 0.026***<br>(3.36)      | 0.026***<br>(3.37)      | 0.027***<br>(3.54)      | 0.027***<br>(3.47)      | 0.026***<br>(3.33)      | 0.027***<br>(3.45)      | 0.026***<br>(3.36)      |
| <i>GDPG</i>        | 0.007<br>(0.20)         | 0.045<br>(1.26)         | 0.022<br>(0.60)         | 0.039<br>(1.12)         | −0.002<br>(−0.07)       | 0.030<br>(0.86)         | −0.036<br>(−0.94)       |
| <i>STKTURNOVER</i> | 0.092**<br>(2.48)       | 0.121***<br>(3.19)      | 0.116***<br>(2.90)      | 0.130***<br>(3.53)      | 0.095**<br>(2.53)       | 0.121***<br>(3.18)      | 0.030<br>(0.72)         |
| <i>CREDIT</i>      | 0.037<br>(1.15)         | −0.028<br>(−0.99)       | −0.006<br>(−0.19)       | −0.018<br>(−0.59)       | 0.051<br>(1.36)         | −0.040<br>(−1.35)       | 0.073*<br>(1.86)        |
| <i>EDUCATION</i>   | 0.327***<br>(10.24)     | 0.328***<br>(9.46)      | 0.400***<br>(9.51)      | 0.315***<br>(9.46)      | 0.218***<br>(4.81)      | 0.329***<br>(8.17)      | 0.453***<br>(4.69)      |
| <i>PTY_RIGHTS</i>  | −0.196***<br>(−4.92)    | −0.201***<br>(−4.89)    | −0.169***<br>(−4.39)    | −0.163***<br>(−4.05)    | −0.213***<br>(−5.17)    | −0.178***<br>(−3.73)    | −0.290***<br>(−4.78)    |
| <i>EXP_RISK</i>    | −0.083***<br>(−2.62)    | −0.077**<br>(−2.04)     | −0.027<br>(−1.02)       | −0.033<br>(−1.25)       | −0.022<br>(−0.83)       | −0.047<br>(−1.44)       | −0.040<br>(−0.83)       |
| <i>NANTI_DIR</i>   | −0.014<br>(−0.44)       | −0.011<br>(−0.33)       | 0.099**<br>(2.23)       | 0.016<br>(0.56)         | 0.011<br>(0.36)         | 0.009<br>(0.30)         | 0.033<br>(0.62)         |
| <i>CORRUPTION</i>  | 0.044<br>(1.50)         | 0.040<br>(1.24)         | 0.051*<br>(1.71)        | 0.072**<br>(2.52)       | 0.055*<br>(1.90)        | 0.045<br>(1.42)         | 0.054<br>(1.59)         |
| <i>BUREAUCRACY</i> | −0.541***<br>(−2.59)    | −0.476**<br>(−2.34)     | −0.769***<br>(−2.75)    | −0.731***<br>(−3.13)    | −0.495**<br>(−2.04)     | −0.516**<br>(−2.19)     | −0.440**<br>(−2.06)     |
| Constant           | 3.284***<br>(3.99)      | 1.582<br>(1.62)         | 4.177***<br>(3.26)      | 3.870***<br>(3.72)      | 1.435<br>(1.46)         | 1.902*<br>(1.96)        | 5.305***<br>(3.45)      |
| Year effect        | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Industry effect    | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Observations       | 15,807                  | 15,807                  | 15,807                  | 15,807                  | 15,807                  | 15,786                  | 15,786                  |
| R-squared          | 0.405                   | 0.404                   | 0.404                   | 0.404                   | 0.404                   | 0.404                   | 0.405                   |

Note: This table uses NBER patents and citations database and presents the results by regressing corporate innovation (*LNCITEPAT*) on culture values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) and on controls of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 12**  
Addressing Endogeneity: Instrumental Variable Approach.

| Panel A: First-stage results   |                         |                          |                         |                         |                         |                          |
|--------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|-------------------------|--------------------------|
| VARIABLES                      | (1)<br>UAI              | (2)<br>IDV               | (3)<br>MAS              | (4)<br>PDI              | (5)<br>LTO              | (6)<br>IDG               |
| <i>Diseases index</i>          | 21.978***<br>(53.98)    | −15.696***<br>(−44.15)   | 16.297***<br>(32.34)    | 7.626***<br>(47.45)     | −17.620***<br>(−34.68)  | −11.258***<br>(−47.61)   |
| <i>FarEast</i>                 | −36.725***<br>(−8.65)   | 23.036***<br>(7.79)      | 12.902***<br>(3.64)     | 34.384***<br>(19.31)    | 56.859***<br>(18.24)    | 28.272***<br>(15.31)     |
| <i>Gemanic</i>                 | 29.036***<br>(17.53)    | −13.459***<br>(−11.71)   | 44.374***<br>(32.93)    | 6.443***<br>(6.87)      | −48.071***<br>(−23.49)  | −23.284***<br>(−14.21)   |
| <i>LatinAmerica</i>            | 12.403<br>(1.43)        | 31.639***<br>(8.38)      | 20.172***<br>(2.80)     | −2.734<br>(−1.09)       | 55.306***<br>(13.88)    | 29.550***<br>(10.39)     |
| <i>LatinEurope</i>             | 9.561<br>(1.49)         | 28.678***<br>(5.33)      | 28.608***<br>(7.10)     | 26.084***<br>(12.35)    | 15.722*<br>(1.76)       | 10.841***<br>(3.03)      |
| <i>SIZE</i>                    | 1.110***<br>(7.98)      | −0.680***<br>(−7.51)     | −0.212*<br>(−1.79)      | −0.231***<br>(−3.01)    | −1.286***<br>(−8.86)    | −0.443***<br>(−5.60)     |
| <i>RD</i>                      | 0.113***<br>(3.04)      | 0.019<br>(0.62)          | 0.040<br>(0.98)         | −0.023<br>(−1.15)       | −0.040<br>(−1.07)       | −0.016<br>(−0.85)        |
| <i>ROA</i>                     | −0.192***<br>(−4.66)    | 0.107***<br>(2.96)       | 0.044<br>(0.95)         | 0.012<br>(0.47)         | 0.250***<br>(5.28)      | 0.069***<br>(2.67)       |
| <i>PPE</i>                     | 0.081<br>(0.84)         | −0.092<br>(−1.42)        | 0.189**<br>(2.06)       | −0.024<br>(−0.48)       | −0.027<br>(−0.31)       | 0.050<br>(0.97)          |
| <i>LEV</i>                     | −0.180***<br>(−3.58)    | 0.035<br>(1.17)          | −0.004<br>(−0.09)       | −0.067***<br>(−2.89)    | 0.144***<br>(3.54)      | 0.022<br>(0.94)          |
| <i>CAPIT</i>                   | −0.081<br>(−1.08)       | −0.068<br>(−1.58)        | −0.195***<br>(−3.30)    | −0.008<br>(−0.26)       | −0.030<br>(−0.57)       | −0.139***<br>(−4.09)     |
| <i>CASH</i>                    | −0.072<br>(−1.36)       | −0.073**<br>(−2.03)      | 0.089*<br>(1.86)        | −0.028<br>(−0.94)       | −0.072<br>(−1.43)       | −0.039<br>(−1.46)        |
| <i>DIV</i>                     | −0.133<br>(−1.58)       | 0.384***<br>(5.11)       | 0.293***<br>(3.42)      | 0.029<br>(0.59)         | 0.110<br>(0.99)         | 0.080<br>(1.64)          |
| <i>GDPG</i>                    | −2.307***<br>(−6.74)    | 0.251<br>(1.37)          | −0.293<br>(−1.14)       | 1.214***<br>(7.62)      | 0.837***<br>(3.50)      | 0.093<br>(0.62)          |
| <i>STKTURNOVER</i>             | 1.226***<br>(6.42)      | 0.458***<br>(2.99)       | 0.423*<br>(1.68)        | 0.141<br>(0.90)         | 0.065<br>(0.33)         | 0.311**<br>(2.45)        |
| <i>CREDIT</i>                  | −0.722***<br>(−2.63)    | 0.353**<br>(2.35)        | 0.898***<br>(3.80)      | 0.341***<br>(2.62)      | 0.069<br>(0.25)         | 0.818***<br>(5.86)       |
| <i>EDUCATION</i>               | 5.237***<br>(8.02)      | 3.939***<br>(6.85)       | 13.333***<br>(26.00)    | 5.475***<br>(15.38)     | 4.285***<br>(4.92)      | 1.026***<br>(2.96)       |
| <i>PTY_RIGHTS</i>              | 0.960<br>(0.70)         | 0.244<br>(0.24)          | 10.754***<br>(10.81)    | 2.023***<br>(2.76)      | −7.300***<br>(−5.92)    | −1.557<br>(−1.32)        |
| <i>EXP_RISK</i>                | 7.301***<br>(2.99)      | 1.465<br>(0.79)          | −9.454***<br>(−6.29)    | −6.868***<br>(−8.32)    | 0.954<br>(0.32)         | −2.261<br>(−1.41)        |
| <i>NANTI_DIR</i>               | 3.556***<br>(5.07)      | −1.670***<br>(−3.62)     | 7.900***<br>(9.51)      | 1.566***<br>(4.52)      | −9.311***<br>(−14.19)   | −1.779***<br>(−4.56)     |
| <i>CORRUPTION</i>              | 1.182***<br>(7.52)      | −0.913***<br>(−7.27)     | −1.427***<br>(−8.63)    | −0.267***<br>(−3.52)    | 0.174<br>(1.61)         | 0.148**<br>(2.21)        |
| <i>BUREAUCRACY</i>             | −6.422***<br>(−4.30)    | 7.239***<br>(5.84)       | 9.591***<br>(10.61)     | 0.153<br>(0.32)         | 5.897***<br>(3.04)      | 6.346***<br>(7.90)       |
| Constant                       | 57.420***<br>(42.91)    | 74.179***<br>(70.06)     | 66.054***<br>(52.02)    | 44.403***<br>(64.37)    | 52.685***<br>(47.73)    | 57.406***<br>(42.17)     |
| <i>Year effect</i>             | Yes                     | Yes                      | Yes                     | Yes                     | Yes                     | Yes                      |
| <i>Industry effect</i>         | Yes                     | Yes                      | Yes                     | Yes                     | Yes                     | Yes                      |
| <i>N</i>                       | 27,636                  | 27,636                   | 27,636                  | 27,636                  | 27,636                  | 27,636                   |
| <i>R<sup>2</sup></i>           | 0.954                   | 0.981                    | 0.943                   | 0.972                   | 0.974                   | 0.963                    |
| <i>Wu-Hausman F-statistics</i> | 14.2462<br>(p = 0.0002) | 0.273365<br>(p = 0.6011) | 60.9951<br>(p = 0.0000) | 126.942<br>(p = 0.0000) | 2.26945<br>(p = 0.1320) | 0.551136<br>(p = 0.4579) |
| <i>F-statistics</i>            | 62,479.9                | 66,457.4                 | 40,480                  | 26,748.8                | 41,976.4                | 61,350.1                 |
| <i>Partial R<sup>2</sup></i>   | 0.6938                  | 0.7067                   | 0.5948                  | 0.4924                  | 0.6035                  | 0.6899                   |
| Panel B: Second-stage results  |                         |                          |                         |                         |                         |                          |
| VARIABLES                      | (1)<br>LNCITEPAT        | (2)<br>LNCITEPAT         | (3)<br>LNCITEPAT        | (4)<br>LNCITEPAT        | (5)<br>LNCITEPAT        | (6)<br>LNCITEPAT         |
| <i>Predicted UAI</i>           | −0.240***<br>(−11.66)   |                          |                         |                         |                         |                          |
| <i>Predicted IDV</i>           |                         | 0.336***<br>(11.64)      |                         |                         |                         |                          |
| <i>Predicted MAS</i>           |                         |                          | −0.323***               |                         |                         |                          |

(continued on next page)

Table 12 (continued)

| Panel B: Second-stage results |                      |                        |                      |                       |                        |                        |
|-------------------------------|----------------------|------------------------|----------------------|-----------------------|------------------------|------------------------|
| VARIABLES                     | (1)                  | (2)                    | (3)                  | (4)                   | (5)                    | (6)                    |
|                               | LNCITEPAT            | LNCITEPAT              | LNCITEPAT            | LNCITEPAT             | LNCITEPAT              | LNCITEPAT              |
| Predicted PDI                 |                      |                        | (−11.61)             | −0.691***<br>(−11.58) |                        |                        |
| Predicted LTO                 |                      |                        |                      |                       | 0.299***<br>(11.66)    |                        |
| Predicted IDG                 |                      |                        |                      |                       |                        | 0.468***<br>(11.64)    |
| FarEast                       | 27.340***<br>(11.38) | 28.411***<br>(11.94)   | 40.319***<br>(17.48) | 59.910***<br>(19.60)  | 19.139***<br>(7.05)    | 22.911***<br>(8.96)    |
| Gemanic                       | 12.186***<br>(7.42)  | 9.743***<br>(6.24)     | 19.574***<br>(9.70)  | 9.677***<br>(6.17)    | 19.602***<br>(9.75)    | 16.124***<br>(8.85)    |
| LatinAmerica                  | 24.266***<br>(5.88)  | 10.669**<br>(2.49)     | 27.815***<br>(6.70)  | 19.402***<br>(4.65)   | 4.749<br>(1.07)        | 7.458*<br>(1.71)       |
| LatinEurope                   | 8.537***<br>(2.70)   | −3.385<br>(−1.14)      | 15.496***<br>(4.52)  | 24.272***<br>(6.28)   | 1.542<br>(0.51)        | 1.169<br>(0.39)        |
| SIZE                          | 15.134***<br>(56.00) | 15.096***<br>(56.05)   | 14.799***<br>(56.73) | 14.708***<br>(56.76)  | 15.252***<br>(55.54)   | 15.075***<br>(56.12)   |
| RD                            | 1.515***<br>(6.61)   | 1.482***<br>(6.45)     | 1.501***<br>(6.52)   | 1.472***<br>(6.38)    | 1.500***<br>(6.54)     | 1.496***<br>(6.51)     |
| ROA                           | −0.206<br>(−0.93)    | −0.196<br>(−0.88)      | −0.146<br>(−0.65)    | −0.152<br>(−0.68)     | −0.235<br>(−1.05)      | −0.193<br>(−0.86)      |
| PPE                           | −1.058***<br>(−4.86) | −1.046***<br>(−4.80)   | −1.016***<br>(−4.64) | −1.094***<br>(−4.99)  | −1.069***<br>(−4.91)   | −1.101***<br>(−5.04)   |
| LEV                           | −0.687***<br>(−4.27) | −0.656***<br>(−4.07)   | −0.645***<br>(−4.00) | −0.690***<br>(−4.26)  | −0.687***<br>(−4.26)   | −0.655***<br>(−4.06)   |
| CAPIT                         | 1.187***<br>(6.72)   | 1.229***<br>(6.95)     | 1.143***<br>(6.44)   | 1.201***<br>(6.76)    | 1.215***<br>(6.88)     | 1.271***<br>(7.19)     |
| CASH                          | −0.566***<br>(−2.76) | −0.524**<br>(−2.55)    | −0.520**<br>(−2.52)  | −0.568***<br>(−2.75)  | −0.527**<br>(−2.57)    | −0.530***<br>(−2.58)   |
| DIV                           | 2.107***<br>(13.30)  | 2.010***<br>(12.54)    | 2.234***<br>(14.18)  | 2.159***<br>(13.60)   | 2.106***<br>(13.30)    | 2.101***<br>(13.24)    |
| GDPG                          | 1.843***<br>(4.65)   | 2.311***<br>(5.97)     | 2.301***<br>(5.92)   | 3.234***<br>(8.47)    | 2.145***<br>(5.51)     | 2.352***<br>(6.08)     |
| STKTURNOVER                   | 1.564***<br>(4.21)   | 1.423***<br>(3.83)     | 1.407***<br>(3.78)   | 1.172***<br>(3.14)    | 1.289***<br>(3.48)     | 1.124***<br>(3.03)     |
| CREDIT                        | −1.104***<br>(−3.34) | −1.050***<br>(−3.17)   | −0.641*<br>(−1.92)   | −0.696**<br>(−2.09)   | −0.952***<br>(−2.88)   | −1.314***<br>(−3.95)   |
| EDUCATION                     | 7.962***<br>(15.12)  | 5.384***<br>(10.65)    | 11.018***<br>(16.72) | 10.490***<br>(16.59)  | 5.425***<br>(10.76)    | 6.226***<br>(12.40)    |
| PTY_RIGHTS                    | 3.896***<br>(4.38)   | 3.584***<br>(4.05)     | 7.143***<br>(7.17)   | 5.064***<br>(5.48)    | 5.849***<br>(6.20)     | 4.395***<br>(4.87)     |
| EXP_RISK                      | 0.207<br>(0.17)      | −2.036<br>(−1.61)      | −4.601***<br>(−3.35) | −6.291***<br>(−4.30)  | −1.829<br>(−1.46)      | −0.486<br>(−0.40)      |
| NANTI_DIR                     | −1.967***<br>(−3.55) | −2.259***<br>(−4.19)   | −0.265<br>(−0.40)    | −1.737***<br>(−3.05)  | −0.034<br>(−0.05)      | −1.986***<br>(−3.59)   |
| CORRUPTION                    | −1.856***<br>(−6.57) | −1.879***<br>(−6.66)   | −1.111***<br>(−3.60) | −1.388***<br>(−4.65)  | −1.520***<br>(−5.20)   | −1.503***<br>(−5.13)   |
| BUREAUCRACY                   | −2.021***<br>(−2.84) | −2.912***<br>(−4.13)   | 2.620***<br>(2.99)   | −0.375<br>(−0.50)     | −2.245***<br>(−3.16)   | −3.452***<br>(−4.91)   |
| Intercept                     | 9.593***<br>(6.55)   | −28.899***<br>(−12.70) | 17.619***<br>(8.50)  | 27.340***<br>(9.55)   | −20.035***<br>(−12.58) | −31.458***<br>(−12.68) |
| Year effect                   | Yes                  | Yes                    | Yes                  | Yes                   | Yes                    | Yes                    |
| Industry effect               | Yes                  | Yes                    | Yes                  | Yes                   | Yes                    | Yes                    |
| N                             | 27,636               | 27,636                 | 27,636               | 27,636                | 27,636                 | 27,636                 |
| R <sup>2</sup>                | 0.221                | 0.219                  | 0.214                | 0.210                 | 0.221                  | 0.219                  |

Note: This table presents the two-stage estimation results. In the first stage, we regress culture value (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) on the index of historical prevalence of infectious diseases (*DISEASES\_IDX*) (Murray and Schaller 2010), respectively. In the second stage, we regress corporate innovation (*LNCITEPAT*) on the predicted culture values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) and controls of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 13**  
Addressing Endogeneity: GMM Approach.

| VARIABLES             | (1)<br><i>LNCITEPAT</i> | (2)<br><i>LNCITEPAT</i> | (3)<br><i>LNCITEPAT</i> | (4)<br><i>LNCITEPAT</i> | (5)<br><i>LNCITEPAT</i> | (6)<br><i>LNCITEPAT</i> | (7)<br><i>LNCITEPAT</i> |
|-----------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>Lag(LNCITEPAT)</i> | 0.399***<br>(35.00)     | 0.400***<br>(35.08)     | 0.402***<br>(35.20)     | 0.401***<br>(35.13)     | 0.404***<br>(35.29)     | 0.400***<br>(35.13)     | 0.428***<br>(33.51)     |
| <i>UAI</i>            | −0.140***<br>(−2.92)    |                         |                         |                         |                         |                         | 0.564**<br>(2.49)       |
| <i>IDV</i>            |                         | 0.166***<br>(2.70)      |                         |                         |                         |                         | −3.044***<br>(−9.84)    |
| <i>MAS</i>            |                         |                         | −0.080***<br>(−2.60)    |                         |                         |                         | 0.300**<br>(2.29)       |
| <i>PDI</i>            |                         |                         |                         | −0.271**<br>(−2.27)     |                         |                         | 5.109***<br>(5.72)      |
| <i>TOR</i>            |                         |                         |                         |                         | 0.263***<br>(5.09)      |                         | 3.299***<br>(10.18)     |
| <i>IDG</i>            |                         |                         |                         |                         |                         | 0.315***<br>(3.22)      | 4.451***<br>(7.33)      |
| <i>FarEast</i>        | −32.674***<br>(−2.85)   | −32.347***<br>(−2.74)   | −24.669***<br>(−3.10)   | −9.977<br>(−1.20)       | −55.526***<br>(−4.53)   | −36.922***<br>(−3.11)   | −436.415***<br>(−7.01)  |
| <i>Gemanic</i>        | 8.333**<br>(2.23)       | 8.225**<br>(2.19)       | 0.110<br>(0.03)         | 3.781<br>(1.04)         | 15.678***<br>(3.79)     | 14.343***<br>(3.14)     | 224.262***<br>(8.54)    |
| <i>LatinAmerica</i>   | 368.175*<br>(1.87)      | 400.095**<br>(1.98)     | 347.577*<br>(1.81)      | 230.299<br>(1.21)       | 274.982<br>(1.44)       | 343.374*<br>(1.77)      | −1874.514***<br>(−5.73) |
| <i>LatinEurope</i>    | 32.725**<br>(2.49)      | 27.631**<br>(2.22)      | 20.832*<br>(1.88)       | 28.003**<br>(2.13)      | 36.038***<br>(3.02)     | 32.185**<br>(2.54)      | −116.254***<br>(−5.19)  |
| <i>SIZE</i>           | 0.413***<br>(6.33)      | 0.414***<br>(6.33)      | 0.457***<br>(7.13)      | 0.411***<br>(6.28)      | 0.422***<br>(6.46)      | 0.408***<br>(6.28)      | 0.091<br>(1.27)         |
| <i>R&amp;D</i>        | 0.025<br>(1.32)         | 0.025<br>(1.30)         | 0.030<br>(1.57)         | 0.026<br>(1.34)         | 0.025<br>(1.32)         | 0.026<br>(1.33)         | 0.012<br>(0.61)         |
| <i>ROA</i>            | −0.023<br>(−1.15)       | −0.022<br>(−1.12)       | −0.020<br>(−1.03)       | −0.021<br>(−1.09)       | −0.023<br>(−1.18)       | −0.022<br>(−1.13)       | −0.016<br>(−0.80)       |
| <i>PPE</i>            | 0.030<br>(1.06)         | 0.031<br>(1.09)         | 0.035<br>(1.26)         | 0.030<br>(1.08)         | 0.030<br>(1.06)         | 0.030<br>(1.07)         | −0.015<br>(−0.52)       |
| <i>LEV</i>            | −0.020<br>(−1.38)       | −0.021<br>(−1.42)       | −0.022<br>(−1.51)       | −0.021<br>(−1.43)       | −0.019<br>(−1.32)       | −0.020<br>(−1.39)       | −0.016<br>(−1.10)       |
| <i>CAPIT</i>          | 0.020*<br>(1.90)        | 0.020*<br>(1.91)        | 0.020*<br>(1.90)        | 0.020*<br>(1.95)        | 0.020*<br>(1.88)        | 0.020*<br>(1.92)        | 0.025**<br>(2.33)       |
| <i>CASH</i>           | 0.079***<br>(4.10)      | 0.079***<br>(4.09)      | 0.079***<br>(4.11)      | 0.078***<br>(4.07)      | 0.080***<br>(4.16)      | 0.079***<br>(4.11)      | 0.075***<br>(3.85)      |
| <i>DIV</i>            | −0.015<br>(−0.97)       | −0.015<br>(−0.95)       | −0.013<br>(−0.80)       | −0.015<br>(−0.97)       | −0.016<br>(−1.00)       | −0.015<br>(−0.98)       | −0.032**<br>(−2.03)     |
| <i>GDPG</i>           | −0.003<br>(−0.20)       | −0.004<br>(−0.22)       | −0.004<br>(−0.25)       | −0.005<br>(−0.30)       | −0.005<br>(−0.26)       | −0.006<br>(−0.34)       | −0.000<br>(−0.02)       |
| <i>STKTURNVER</i>     | 0.145***<br>(9.31)      | 0.146***<br>(9.33)      | 0.144***<br>(9.24)      | 0.145***<br>(9.31)      | 0.142***<br>(9.11)      | 0.144***<br>(9.24)      | 0.135***<br>(8.48)      |
| <i>CREDIT</i>         | −0.104***<br>(−2.59)    | −0.104***<br>(−2.59)    | −0.106***<br>(−2.63)    | −0.103**<br>(−2.57)     | −0.105***<br>(−2.62)    | −0.104**<br>(−2.58)     | −0.100**<br>(−2.46)     |
| <i>EDUCATION</i>      | 0.700<br>(0.69)         | −0.483<br>(−0.45)       | 0.865<br>(0.85)         | 0.230<br>(0.23)         | −1.814<br>(−1.64)       | −0.490<br>(−0.46)       | −22.157***<br>(−6.79)   |
| <i>PTY_RIGHTS</i>     | 5.412***<br>(3.01)      | 6.028***<br>(3.29)      | −0.388<br>(−0.26)       | 5.501***<br>(3.04)      | 6.742***<br>(3.69)      | 8.366***<br>(4.04)      | 46.282***<br>(8.62)     |
| <i>EXP_RISK</i>       | −5.490***<br>(−4.96)    | −8.312***<br>(−6.23)    | −7.236***<br>(−6.17)    | −7.025***<br>(−6.16)    | −8.111***<br>(−7.08)    | −9.203***<br>(−6.44)    | −29.776***<br>(−5.66)   |
| <i>NANTI_DIR</i>      | 2.622*<br>(1.69)        | 1.673<br>(1.24)         | 0.473<br>(0.41)         | 0.562<br>(0.49)         | 5.440***<br>(3.50)      | 2.673*<br>(1.83)        | 29.583***<br>(6.31)     |
| <i>CORRUPTION</i>     | −0.008<br>(−0.54)       | −0.008<br>(−0.53)       | −0.013<br>(−0.83)       | −0.008<br>(−0.51)       | −0.010<br>(−0.64)       | −0.008<br>(−0.48)       | 0.011<br>(0.67)         |
| <i>BUREAUCRACY</i>    | 0.025<br>(0.15)         | 0.019<br>(0.11)         | −0.156<br>(−0.93)       | 0.032<br>(0.19)         | 0.017<br>(0.10)         | 0.031<br>(0.18)         | −0.001<br>(−0.00)       |
| Constant              | 11.198***<br>(3.79)     | −9.274**<br>(−2.06)     | 8.312***<br>(3.98)      | 14.735***<br>(2.72)     | −10.825***<br>(−4.03)   | −15.654***<br>(−2.72)   | −489.075***<br>(−6.05)  |
| Year effect           | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Industry effect       | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Observations          | 18,503                  | 18,503                  | 18,503                  | 18,503                  | 18,503                  | 18,503                  | 18,503                  |
| Number of groups      | 3427                    | 3427                    | 3427                    | 3427                    | 3427                    | 3427                    | 3427                    |

Note: This table presents the GMM estimation results obtained by regressing corporate innovation (*LNCITEPAT*) on lagged corporate innovation (*lag(LNCITEPAT)*), cultural values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) and on controls of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1%

and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 14**

Innovation subsample analysis excluding U.S.

| VARIABLES           | (1)<br><i>LNCITEPAT</i> | (2)<br><i>LNCITEPAT</i> | (3)<br><i>LNCITEPAT</i> | (4)<br><i>LNCITEPAT</i> | (5)<br><i>LNCITEPAT</i> | (6)<br><i>LNCITEPAT</i> | (7)<br><i>LNCITEPAT</i> |
|---------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>UAI</i>          | −0.005**<br>(−3.82)     |                         |                         |                         |                         |                         | 0.002<br>(1.64)         |
| <i>IDV</i>          |                         | 0.004**<br>(2.53)       |                         |                         |                         |                         | 0.008***<br>(2.66)      |
| <i>MAS</i>          |                         |                         | −0.011***<br>(−6.50)    |                         |                         |                         | −0.012***<br>(−6.21)    |
| <i>PDI</i>          |                         |                         |                         | −0.007***<br>(−3.62)    |                         |                         | −0.003<br>(−1.21)       |
| <i>TOR</i>          |                         |                         |                         |                         | 0.001<br>(0.52)         |                         | −0.008***<br>(−4.46)    |
| <i>IDG</i>          |                         |                         |                         |                         |                         | 0.004**<br>(2.17)       | −0.003<br>(−0.80)       |
| <i>FarEast</i>      | −0.471***<br>(−3.38)    | −0.617***<br>(−4.34)    | −0.264**<br>(−2.24)     | −0.311***<br>(−2.94)    | −0.642***<br>(−4.52)    | −0.654***<br>(−4.57)    | −0.071<br>(−0.71)       |
| <i>Gemanic</i>      | 1.032***<br>(6.76)      | 1.014***<br>(6.72)      | 1.381***<br>(7.58)      | 1.012***<br>(6.79)      | 1.026***<br>(6.11)      | 1.060***<br>(7.49)      | 1.169***<br>(6.85)      |
| <i>LatinAmerica</i> | −0.282*<br>(−1.71)      | −0.593***<br>(−3.64)    | −0.243*<br>(−1.68)      | −0.503***<br>(−3.22)    | −0.539***<br>(−3.29)    | −0.594***<br>(−3.52)    | −0.206<br>(−1.43)       |
| <i>LatinEurope</i>  | −0.555***<br>(−3.13)    | −0.677***<br>(−3.94)    | −0.184<br>(−1.33)       | −0.386**<br>(−2.50)     | −0.627***<br>(−3.59)    | −0.629***<br>(−3.54)    | −0.094<br>(−1.12)       |
| <i>SIZE</i>         | 0.210***<br>(6.50)      | 0.206***<br>(6.15)      | 0.219***<br>(7.06)      | 0.198***<br>(6.62)      | 0.194***<br>(5.97)      | 0.201***<br>(6.80)      | 0.212***<br>(6.23)      |
| <i>R&amp;D</i>      | 0.150***<br>(3.52)      | 0.149***<br>(3.53)      | 0.148***<br>(3.47)      | 0.153***<br>(3.50)      | 0.159***<br>(3.72)      | 0.155***<br>(3.54)      | 0.140***<br>(3.37)      |
| <i>ROA</i>          | 0.023<br>(0.87)         | 0.023<br>(0.87)         | 0.024<br>(0.91)         | 0.022<br>(0.84)         | 0.021<br>(0.81)         | 0.021<br>(0.80)         | 0.022<br>(0.85)         |
| <i>PPE</i>          | −0.045***<br>(−3.77)    | −0.044***<br>(−3.68)    | −0.040***<br>(−3.38)    | −0.044***<br>(−3.74)    | −0.044***<br>(−3.68)    | −0.045***<br>(−3.77)    | −0.037***<br>(−3.25)    |
| <i>LEV</i>          | −0.023**<br>(−2.13)     | −0.022**<br>(−2.07)     | −0.023**<br>(−2.17)     | −0.024**<br>(−2.27)     | −0.021**<br>(−2.00)     | −0.021**<br>(−2.04)     | −0.024**<br>(−2.27)     |
| <i>CAPIT</i>        | 0.036***<br>(3.54)      | 0.039***<br>(3.74)      | 0.030***<br>(3.03)      | 0.037***<br>(3.64)      | 0.038***<br>(3.72)      | 0.040***<br>(3.81)      | 0.030***<br>(3.02)      |
| <i>CASH</i>         | −0.074***<br>(−4.35)    | −0.071***<br>(−4.19)    | −0.069***<br>(−4.09)    | −0.075***<br>(−4.37)    | −0.076***<br>(−4.49)    | −0.073***<br>(−4.23)    | −0.066***<br>(−3.95)    |
| <i>DIV</i>          | 0.006<br>(0.51)         | 0.005<br>(0.40)         | 0.009<br>(0.79)         | 0.008<br>(0.69)         | 0.011<br>(0.97)         | 0.009<br>(0.73)         | 0.004<br>(0.32)         |
| <i>GDPG</i>         | −0.058*<br>(−1.92)      | −0.047<br>(−1.61)       | −0.054*<br>(−1.86)      | −0.035<br>(−1.22)       | −0.042<br>(−1.46)       | −0.045<br>(−1.57)       | −0.048<br>(−1.59)       |
| <i>STKTURNOVER</i>  | 0.091***<br>(2.69)      | 0.074**<br>(2.29)       | 0.067**<br>(2.15)       | 0.074**<br>(2.36)       | 0.067*<br>(1.96)        | 0.082**<br>(2.41)       | 0.022<br>(0.58)         |
| <i>CREDIT</i>       | 0.042***<br>(3.06)      | 0.049***<br>(3.53)      | 0.057***<br>(4.13)      | 0.055***<br>(4.01)      | 0.051***<br>(3.69)      | 0.046***<br>(3.36)      | 0.066***<br>(5.04)      |
| <i>EDUCATION</i>    | 0.233***<br>(7.45)      | 0.152***<br>(6.53)      | 0.315***<br>(8.48)      | 0.197***<br>(6.39)      | 0.151***<br>(6.70)      | 0.157***<br>(6.35)      | 0.295***<br>(6.58)      |
| <i>PTY_RIGHTS</i>   | 0.009<br>(0.27)         | −0.026<br>(−0.81)       | 0.129***<br>(2.76)      | −0.012<br>(−0.39)       | −0.031<br>(−0.88)       | −0.024<br>(−0.78)       | 0.053<br>(1.20)         |
| <i>EXP_RISK</i>     | −0.004<br>(−0.09)       | −0.047<br>(−0.90)       | −0.188***<br>(−3.81)    | −0.089*<br>(−1.96)      | −0.019<br>(−0.41)       | −0.022<br>(−0.49)       | −0.280***<br>(−4.35)    |
| <i>NANTI_DIR</i>    | 0.031<br>(1.35)         | 0.056**<br>(2.32)       | 0.139***<br>(4.33)      | 0.064**<br>(2.56)       | 0.055*<br>(2.17)        | 0.055*<br>(2.38)        | 0.145***<br>(3.88)      |
| <i>CORRUPTION</i>   | −0.042**<br>(−2.29)     | −0.062***<br>(−3.65)    | −0.037*<br>(−1.66)      | −0.053***<br>(−2.76)    | −0.072***<br>(−3.48)    | −0.058***<br>(−2.61)    | −0.005<br>(−0.18)       |
| <i>BUREAUCRACY</i>  | −0.236***<br>(−4.92)    | −0.235***<br>(−4.95)    | −0.054<br>(−1.54)       | −0.206***<br>(−4.68)    | −0.230***<br>(−4.82)    | −0.238***<br>(−4.86)    | −0.018<br>(−0.50)       |
| <i>Intercept</i>    | 1.504***<br>(9.23)      | 0.795***<br>(4.82)      | 1.949***<br>(10.77)     | 1.394***<br>(8.82)      | 1.008***<br>(8.19)      | 0.838***<br>(6.41)      | 1.826***<br>(6.44)      |
| <i>Year effect</i>  | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |

(continued on next page)

Table 14 (continued)

|                 | (1)       | (2)       | (3)       | (4)       | (5)       | (6)       | (7)       |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| VARIABLES       | LNCITEPAT | LNCITEPAT | LNCITEPAT | LNCITEPAT | LNCITEPAT | LNCITEPAT | LNCITEPAT |
| Industry effect | Yes       | Yes       | Yes       | Yes       | Yes       | Yes       | Yes       |
| N               | 10,579    | 10,579    | 10,579    | 10,579    | 10,579    | 10,579    | 10,579    |
| R <sup>2</sup>  | 0.093     | 0.092     | 0.100     | 0.093     | 0.092     | 0.092     | 0.101     |

Note: This table presents the results by regressing corporate innovation (*LNCITEPAT*) on cultural values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) and on controls of firm characteristics and country variables, respectively. The results are based on innovation subsample excluding U.S. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. The industry controls are based on the 12 Fama-French industry classification codes. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors, and clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

also find that this positive effect of board gender diversity on innovation is stronger in countries with more masculine cultures.

In line with Griffin et al. (2019), we investigate in this section the effect of corporate risk taking on firms' innovation and test whether this effect depends on the uncertainty avoidance level of the country to which they belong.<sup>17</sup> As discussed before, innovation is a complex process that involves risk-taking behavior, which might be easier to undertake in societies where risk tolerance is high. Li et al. (2013) find that uncertainty avoidance has a negative impact on risk taking. Following Boubakri et al. (2013), we use three different proxies of corporate risk-taking behavior to examine its interaction effect with uncertainty avoidance on corporate innovation. Column (1) of Table 18 reports the results, where risk-taking is measured using the volatility of a firm's earnings over four years (*RISK\_TAKING1*). In Column (2), *RISK\_TAKING2* is the difference between the maximum and minimum *ROA* reported over the five-year interval. In Column (3) of the same table, *RISK\_TAKING3* is the country-level average of all the firm-level observations of *RISK\_TAKING1*. In line with our prediction, the results reported in the three columns of Table 18 show that the positive effect of corporate risk-taking on innovation is less pronounced in countries with higher levels of uncertainty avoidance.

## 5. Conclusion

We examine the impact of national culture (an informal institution) on firm-level innovation rates in a cross-country sample. Our study is motivated by the importance of innovation to productivity and economic growth, and by the sparse literature on the cultural determinants of innovation at the country level.

Using a newly available comprehensive database on innovation around the world, over a period spanning from 1992 to 2016, we gather a sample of innovative, non-innovative, and propensity-matched non-innovative firms. Drawing on the cultural dimensions from Hofstede (1980), Hofstede and Bond (1988), and Hofstede et al. (2010), we find that, after controlling for the impact of formal institutions, and firm-level and country-level variables, culture has relevance for innovation. Specifically, we find that the probability of a firm innovating is higher in individualistic societies, as well as in indulgent and long-term oriented cultures with less power distance, less uncertainty avoidance, and less masculinity. When we focus on the innovative firms subsample, we continue to find a significant impact of culture on the innovation performance and quality of innovation firms. Additional tests suggest that firm-level risk taking is a channel for the negative effect of uncertainty avoidance on innovation. These results suggest that culture indeed affects innovation because it shapes the attitudes and beliefs with which individuals deal with novelty, risk, and individual initiatives.

Our results remain robust to additional checks, different model specifications, alternative measures of culture and innovation, different subsample analyses, and to addressing endogeneity concerns. Our evidence adds to the recent trend of the literature on culture and finance by showing that culture affects corporate strategic choices and outcomes such as innovation.

All in all, we show that culture has a first-order effect on innovation, implying that, to enhance innovation and entrepreneurship in a country, policy makers and managers alike need to do more than increase R&D investments. Changing attitudes to foster innovation at the country and corporate levels is the best strategy, as it constitutes a channel to economic growth.

<sup>17</sup> We also performed a battery of additional subsampling analyses. For example, we reran our model on a subsample of purely domestic firms (excluding firms with a nonzero foreign sales) to check if our results are driven by large multinational firms that are weakly influenced by their headquarter country's culture. We also divided our sample by each country's median of (i) gross domestic expenditure on research and development growth, and (ii) GDP growth, to test if the culture's effect is conditional on these fast-changing economic variables. In both cases, we obtained statistically similar results to those shown in Table 8. The tables are not reported for the sake of brevity but are available upon request.

Table 15

The impact of culture value on corporate innovation: Country-level estimation.

| VARIABLES          | (1)<br><i>LNCITEPAT</i> | (2)<br><i>LNCITEPAT</i> | (3)<br><i>LNCITEPAT</i> | (4)<br><i>LNCITEPAT</i> | (5)<br><i>LNCITEPAT</i> | (6)<br><i>LNCITEPAT</i> | (7)<br><i>LNCITEPAT</i> |
|--------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>UAI</i>         | −0.010***<br>(−7.34)    |                         |                         |                         |                         |                         | −0.008***<br>(−7.44)    |
| <i>IDV</i>         |                         | −0.001<br>(−0.84)       |                         |                         |                         |                         | −0.011***<br>(−5.49)    |
| <i>MAS</i>         |                         |                         | −0.004***<br>(−2.65)    |                         |                         |                         | −0.002*<br>(−1.73)      |
| <i>PDI</i>         |                         |                         |                         | −0.005**<br>(−2.56)     |                         |                         | −0.004**<br>(−2.20)     |
| <i>TOR</i>         |                         |                         |                         |                         | 0.007***<br>(4.80)      |                         | 0.009***<br>(5.29)      |
| <i>IDG</i>         |                         |                         |                         |                         |                         | 0.005**<br>(2.43)       | 0.000<br>(0.05)         |
| <i>FarEast</i>     | 0.871***<br>(7.22)      | 1.110***<br>(8.15)      | 1.092***<br>(8.33)      | 1.275***<br>(7.74)      | 0.949***<br>(7.45)      | 1.046***<br>(7.98)      | 1.056***<br>(7.52)      |
| <i>Gemanic</i>     | 0.402***<br>(5.17)      | 0.189**<br>(2.52)       | 0.351***<br>(3.73)      | 0.175**<br>(2.40)       | 0.385***<br>(4.93)      | 0.235***<br>(3.19)      | 0.541***<br>(5.14)      |
| <i>LatinEurope</i> | 0.609***<br>(5.13)      | 0.776***<br>(5.96)      | 0.722***<br>(5.74)      | 0.720***<br>(5.82)      | 0.497***<br>(4.02)      | 0.604***<br>(4.73)      | 0.668***<br>(5.66)      |
| <i>SIZE</i>        | −0.015<br>(−0.64)       | −0.064***<br>(−2.50)    | −0.051**<br>(−2.05)     | −0.062**<br>(−2.45)     | −0.011<br>(−0.41)       | −0.040<br>(−1.47)       | 0.020<br>(0.76)         |
| <i>R&amp;D</i>     | 0.001<br>(0.06)         | 0.000<br>(0.01)         | 0.002<br>(0.08)         | 0.001<br>(0.02)         | 0.015<br>(0.52)         | 0.006<br>(0.23)         | 0.004<br>(0.18)         |
| <i>ROA</i>         | 0.025<br>(0.87)         | −0.026<br>(−0.91)       | −0.037<br>(−1.23)       | −0.034<br>(−1.15)       | −0.029<br>(−1.02)       | −0.022<br>(−0.76)       | 0.001<br>(0.03)         |
| <i>PPE</i>         | −0.033<br>(−1.03)       | −0.011<br>(−0.32)       | 0.004<br>(0.13)         | −0.005<br>(−0.16)       | −0.016<br>(−0.48)       | −0.012<br>(−0.35)       | −0.048<br>(−1.46)       |
| <i>LEV</i>         | 0.031<br>(1.39)         | 0.027<br>(1.20)         | 0.029<br>(1.28)         | 0.025<br>(1.08)         | 0.024<br>(1.04)         | 0.024<br>(1.08)         | 0.023<br>(1.13)         |
| <i>CAPIT</i>       | −0.039<br>(−1.33)       | −0.027<br>(−0.92)       | −0.028<br>(−0.93)       | −0.016<br>(−0.51)       | −0.005<br>(−0.19)       | −0.008<br>(−0.28)       | −0.033<br>(−1.22)       |
| <i>CASH</i>        | 0.051**<br>(2.20)       | 0.101***<br>(4.29)      | 0.108***<br>(4.63)      | 0.112***<br>(4.79)      | 0.112***<br>(4.71)      | 0.101***<br>(4.32)      | 0.047**<br>(2.07)       |
| <i>DIV</i>         | −0.053**<br>(−2.37)     | 0.002<br>(0.09)         | 0.005<br>(0.22)         | −0.004<br>(−0.15)       | −0.036<br>(−1.56)       | −0.021<br>(−0.88)       | −0.012<br>(−0.52)       |
| <i>GDPG</i>        | −0.028<br>(−1.10)       | 0.037<br>(1.45)         | 0.045*<br>(1.75)        | 0.049*<br>(1.85)        | 0.050**<br>(2.07)       | 0.045*<br>(1.81)        | −0.010<br>(−0.40)       |
| <i>STKTURNOVER</i> | 0.117***<br>(3.09)      | 0.157***<br>(3.70)      | 0.128***<br>(2.83)      | 0.152***<br>(3.67)      | 0.147***<br>(3.80)      | 0.162***<br>(3.87)      | 0.075*<br>(1.89)        |
| <i>CREDIT</i>      | −0.136***<br>(−5.71)    | −0.122***<br>(−4.69)    | −0.110***<br>(−4.36)    | −0.107***<br>(−4.19)    | −0.087***<br>(−3.77)    | −0.118***<br>(−4.79)    | −0.108***<br>(−4.95)    |
| <i>EDUCATION</i>   | 0.481***<br>(11.55)     | 0.418***<br>(9.90)      | 0.452***<br>(10.29)     | 0.421***<br>(10.41)     | 0.389***<br>(10.09)     | 0.402***<br>(10.07)     | 0.524***<br>(11.45)     |
| <i>PTY_RIGHTS</i>  | 0.208***<br>(4.65)      | 0.163***<br>(3.75)      | 0.199***<br>(4.08)      | 0.200***<br>(4.34)      | 0.211***<br>(4.54)      | 0.170***<br>(3.92)      | 0.343***<br>(6.34)      |
| <i>EXP_RISK</i>    | 0.219***<br>(3.82)      | 0.322***<br>(4.86)      | 0.281***<br>(4.63)      | 0.278***<br>(4.50)      | 0.249***<br>(4.14)      | 0.283***<br>(4.66)      | 0.340***<br>(5.25)      |
| <i>NANTI_DIR</i>   | −0.138***<br>(−5.59)    | −0.187***<br>(−6.56)    | −0.168***<br>(−5.93)    | −0.158***<br>(−5.40)    | −0.154***<br>(−6.08)    | −0.167***<br>(−6.06)    | −0.161***<br>(−6.13)    |
| <i>CORRUPTION</i>  | −0.154***<br>(−3.11)    | −0.109**<br>(−2.09)     | −0.170***<br>(−2.90)    | −0.140***<br>(−2.73)    | −0.164***<br>(−3.18)    | −0.150***<br>(−2.83)    | −0.214***<br>(−3.89)    |
| <i>BUREAUCRACY</i> | −0.229***<br>(−4.75)    | −0.144***<br>(−3.25)    | −0.142***<br>(−3.26)    | −0.185***<br>(−3.81)    | −0.141***<br>(−3.21)    | −0.148***<br>(−3.37)    | −0.270***<br>(−5.36)    |
| Constant           | 0.594***<br>(4.99)      | 0.120<br>(0.99)         | 0.201**<br>(2.05)       | 0.296**<br>(2.29)       | −0.258*<br>(−1.94)      | −0.211<br>(−1.36)       | 0.968***<br>(4.07)      |
| Year effect        | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     | Yes                     |
| Observations       | 416                     | 416                     | 416                     | 416                     | 416                     | 416                     | 416                     |
| R-squared          | 0.684                   | 0.639                   | 0.644                   | 0.644                   | 0.661                   | 0.642                   | 0.719                   |

Note: This table presents the country-level estimation results by regressing country average corporate innovation (*LNCITEPAT*) on culture values (*UAI*, *CLT*, *MAS*, *PDI*, *TOR*, and *IDG*) and on controls of country average of firm characteristics and country variables, respectively. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, year control is included, but the results are not tabulated. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. *t*-statistics based on robust standard errors are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 16**  
Subsample analysis: Crisis vs. Non-crisis period.

| Panel A: Crisis period |                       |                       |                      |                       | Panel B: Non-crisis period |                       |                       |                       |                      |                       |                      |                      |                      |                      |
|------------------------|-----------------------|-----------------------|----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|
| VARIABLES              | (1)                   | (2)                   | (3)                  | (4)                   | (5)                        | (6)                   | (7)                   | (8)                   | (9)                  | (10)                  | (11)                 | (12)                 | (13)                 | (14)                 |
|                        | LNCITEPAT             | LNCITEPAT             | LNCITEPAT            | LNCITEPAT             | LNCITEPAT                  | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT            | LNCITEPAT             | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            |
| UAI                    | -0.579***<br>(-13.24) |                       |                      |                       |                            |                       | -0.249***<br>(-5.58)  | -0.253***<br>(-19.17) |                      |                       |                      |                      |                      | -0.176***<br>(-7.32) |
| IDV                    |                       | 0.626***<br>(9.64)    |                      |                       |                            |                       | 0.029<br>(0.39)       |                       | 0.285***<br>(14.73)  |                       |                      |                      |                      | -0.030<br>(-1.07)    |
| MAS                    |                       |                       | -0.247***<br>(-4.47) |                       |                            |                       | -0.054<br>(-0.76)     |                       |                      | -0.164***<br>(-11.19) |                      |                      |                      | 0.034*<br>(1.94)     |
| PDI                    |                       |                       |                      | -0.077<br>(-0.87)     |                            |                       | 0.555***<br>(6.66)    |                       |                      |                       | -0.250***<br>(-7.87) |                      |                      | 0.276***<br>(4.95)   |
| TOR                    |                       |                       |                      |                       | 0.754***<br>(12.31)        |                       | 0.578***<br>(6.96)    |                       |                      |                       |                      | 0.276***<br>(17.32)  |                      | 0.260***<br>(7.71)   |
| IDG                    |                       |                       |                      |                       |                            | 0.969**<br>(11.25)    | 0.241***<br>(2.98)    |                       |                      |                       |                      |                      | 0.380***<br>(14.50)  | -0.031<br>(-0.75)    |
| FarEast                | 10.952***<br>(2.69)   | 22.407***<br>(6.17)   | 42.134***<br>(9.20)  | 41.748***<br>(8.04)   | -5.389<br>(-1.12)          | 13.808***<br>(3.54)   | -30.006***<br>(-4.72) | 26.587***<br>(16.79)  | 29.127***<br>(17.48) | 37.991***<br>(21.17)  | 44.499***<br>(19.29) | 20.233***<br>(12.03) | 24.601***<br>(14.52) | 6.299*<br>(1.74)     |
| Germanic               | 21.441***<br>(8.56)   | 12.523***<br>(5.60)   | 15.203***<br>(4.46)  | 3.149<br>(1.61)       | 40.248***<br>(10.51)       | 22.972***<br>(7.67)   | 43.180***<br>(9.34)   | 12.089***<br>(14.13)  | 8.227***<br>(10.35)  | 10.883***<br>(6.83)   | 4.943***<br>(6.00)   | 17.933***<br>(16.00) | 13.523***<br>(13.22) | 17.935***<br>(10.71) |
| LatinAmerica           | 24.975***<br>(5.18)   | 2.325<br>(0.55)       | 32.791***<br>(6.52)  | 29.531***<br>(6.67)   | -21.676***<br>(-3.93)      | -4.261<br>(-0.90)     | -15.791***<br>(-2.43) | 23.822***<br>(11.03)  | 12.504***<br>(5.52)  | 26.152***<br>(11.06)  | 22.312***<br>(8.31)  | 6.515***<br>(2.82)   | 10.119***<br>(4.47)  | 11.296***<br>(3.78)  |
| LatinEurope            | 6.022<br>(1.14)       | -25.929***<br>(-3.84) | -0.901<br>(-0.17)    | -14.749***<br>(-2.18) | -17.211***<br>(-2.04)      | -17.937***<br>(-2.93) | -27.833***<br>(-2.12) | 9.842***<br>(6.79)    | -2.476<br>(-1.60)    | 6.368***<br>(3.76)    | 6.788***<br>(4.03)   | 2.477<br>(1.55)      | 1.369<br>(0.91)      | -2.209<br>(-0.77)    |
| SIZE                   | 20.317***<br>(18.81)  | 20.083***<br>(18.52)  | 18.402***<br>(18.28) | 18.341***<br>(18.17)  | 20.714***<br>(18.85)       | 20.031***<br>(18.69)  | 21.290***<br>(17.94)  | 14.439***<br>(30.08)  | 14.123***<br>(29.67) | 13.637***<br>(29.69)  | 13.470***<br>(29.12) | 14.365***<br>(30.03) | 14.053***<br>(29.81) | 14.534***<br>(30.09) |
| R&D                    | 1.770***<br>(4.05)    | 1.706***<br>(3.91)    | 1.777***<br>(4.06)   | 1.762***<br>(4.03)    | 1.771***<br>(4.04)         | 1.772***<br>(4.05)    | 1.794***<br>(4.08)    | 1.440***<br>(7.42)    | 1.414***<br>(7.29)   | 1.435***<br>(7.36)    | 1.433***<br>(7.35)   | 1.420***<br>(7.31)   | 1.421***<br>(7.32)   | 1.448***<br>(7.45)   |
| ROA                    | -0.096<br>(-0.23)     | -0.037<br>(-0.09)     | 0.366<br>(0.88)      | 0.382<br>(0.92)       | -0.176<br>(-0.42)          | -0.007<br>(-0.02)     | -0.298<br>(-0.70)     | -0.365*<br>(-1.85)    | -0.294<br>(-1.50)    | -0.198<br>(-1.01)     | -0.164<br>(-0.83)    | -0.357*<br>(-1.81)   | -0.285<br>(-1.45)    | -0.375*<br>(-1.90)   |
| PPE                    | -2.121***<br>(-3.96)  | -2.129***<br>(-3.95)  | -2.158***<br>(-4.00) | -2.263***<br>(-4.19)  | -2.059***<br>(-3.85)       | -2.210***<br>(-4.12)  | -1.943***<br>(-3.63)  | -0.894***<br>(-4.09)  | -0.890***<br>(-4.07) | -0.887***<br>(-4.05)  | -0.917***<br>(-4.19) | -0.910***<br>(-4.16) | -0.926***<br>(-4.23) | -0.900***<br>(-4.12) |
| LEV                    | -0.762*<br>(-1.89)    | -0.648<br>(-1.61)     | -0.453<br>(-1.13)    | -0.441<br>(-1.10)     | -0.706*<br>(-1.76)         | -0.576<br>(-1.44)     | -0.738*<br>(-1.84)    | -0.694***<br>(-4.40)  | -0.627***<br>(-3.98) | -0.579***<br>(-3.68)  | -0.562***<br>(-3.57) | -0.678***<br>(-4.30) | -0.630***<br>(-4.00) | -0.685***<br>(-4.34) |
| CAPIT                  | 1.966***<br>(5.22)    | 2.032***<br>(5.38)    | 1.933***<br>(5.09)   | 1.985***<br>(5.23)    | 1.959***<br>(5.22)         | 2.110***<br>(5.59)    | 1.988***<br>(5.30)    | 1.022***<br>(5.73)    | 1.071***<br>(5.99)   | 1.032***<br>(5.75)    | 1.071***<br>(5.95)   | 1.059***<br>(5.95)   | 1.102***<br>(6.16)   | 1.043***<br>(5.85)   |
| CASH                   | -0.109<br>(-0.23)     | -0.027<br>(-0.06)     | -0.122<br>(-0.26)    | -0.136<br>(-0.29)     | 0.000<br>(0.00)            | -0.034<br>(-0.07)     | 0.056<br>(0.12)       | -0.622***<br>(-2.96)  | -0.589***<br>(-2.80) | -0.594***<br>(-2.82)  | -0.615***<br>(-2.92) | -0.590***<br>(-2.81) | -0.599***<br>(-2.85) | -0.615***<br>(-2.93) |
| DIV                    | 2.770***<br>(5.92)    | 2.607***<br>(5.47)    | 3.399***<br>(7.08)   | 3.369***<br>(6.98)    | 2.712***<br>(5.80)         | 2.802***<br>(5.97)    | 2.615***<br>(5.55)    | 1.896***<br>(8.90)    | 1.892***<br>(8.77)   | 2.103***<br>(9.81)    | 2.123***<br>(9.86)   | 1.939***<br>(9.09)   | 1.974***<br>(9.22)   | 1.911***<br>(8.88)   |
| GDPG                   | 3.037***<br>(5.10)    | 4.262***<br>(6.58)    | 4.680***<br>(6.90)   | 5.551***<br>(8.31)    | 3.222***<br>(5.33)         | 4.051***<br>(6.55)    | 1.762***<br>(3.17)    | 0.661***<br>(2.38)    | 1.410***<br>(4.94)   | 1.806***<br>(6.38)    | 2.132***<br>(6.89)   | 1.254***<br>(4.79)   | 1.537***<br>(5.75)   | 0.542***<br>(2.15)   |
| STKTURNOVER            | 3.197***<br>(7.96)    | 2.865***<br>(7.08)    | 2.345***<br>(6.90)   | 2.147***<br>(5.31)    | 2.432***<br>(6.25)         | 2.133***<br>(5.49)    | 2.666***<br>(5.55)    | 2.039***<br>(6.11)    | 1.878***<br>(5.52)   | 1.875***<br>(5.62)    | 1.687***<br>(5.10)   | 1.784***<br>(5.43)   | 1.524***<br>(4.54)   | 2.010***<br>(5.96)   |
| CREDIT                 | -3.028***<br>(-8.51)  | -3.087***<br>(-8.53)  | -2.819***<br>(-7.76) | -2.894***<br>(-7.92)  | -3.274***<br>(-8.92)       | -3.580***<br>(-9.68)  | -3.580***<br>(-9.46)  | -0.045<br>(-0.17)     | 0.137<br>(0.52)      | 0.564***<br>(1.99)    | 0.349<br>(1.23)      | 0.404<br>(1.56)      | -0.137<br>(-0.50)    | 0.072<br>(0.28)      |
| EDUCATION              | 8.973***<br>(11.13)   | 4.249***<br>(4.00)    | 11.458***<br>(10.02) | 9.142***<br>(9.75)    | 2.038***<br>(2.07)         | 5.035***<br>(6.66)    | 0.148<br>(0.09)       | 8.402***<br>(22.55)   | 5.794***<br>(17.11)  | 8.807***<br>(19.98)   | 7.815***<br>(20.67)  | 5.862***<br>(17.57)  | 6.481***<br>(18.66)  | 5.101***<br>(9.06)   |
| PTY_RIGHTS             | 3.851***<br>(4.44)    | 2.899***<br>(3.13)    | 5.269***<br>(5.15)   | 2.893***<br>(3.32)    | 7.117***<br>(7.60)         | 3.603***<br>(3.55)    | 5.782***<br>(4.87)    | 4.456***<br>(9.06)    | 3.690***<br>(6.92)   | 4.706***<br>(8.61)    | 2.908***<br>(5.42)   | 6.203***<br>(11.63)  | 4.309***<br>(7.97)   | 5.271***<br>(8.90)   |

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Table 16 (continued)

| VARIABLES       | Panel A: Crisis period |                       |                     |                      |                       | Panel B: Non-crisis period |                       |                       |                       |                       |                       |                      |                       |                       |
|-----------------|------------------------|-----------------------|---------------------|----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|-----------------------|
|                 | (1)                    | (2)                   | (3)                 | (4)                  | (5)                   | (6)                        | (7)                   | (8)                   | (9)                   | (10)                  | (11)                  | (12)                 | (13)                  | (14)                  |
|                 | LNCITEPAT              | LNCITEPAT             | LNCITEPAT           | LNCITEPAT            | LNCITEPAT             | LNCITEPAT                  | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT            | LNCITEPAT             | LNCITEPAT             |
| EXP_RISK        | 5.787***<br>(3.11)     | 3.705*<br>(1.86)      | 4.276**<br>(2.28)   | 8.060***<br>(3.77)   | 5.061**<br>(2.00)     | 8.850***<br>(4.27)         | 10.369***<br>(2.74)   | -1.607***<br>(-2.78)  | -2.745***<br>(-4.26)  | -1.710**<br>(-2.43)   | -0.907<br>(-1.34)     | -3.323***<br>(-4.88) | -1.431**<br>(-2.27)   | 0.154<br>(0.18)       |
| NANTI_DIR       | 0.903<br>(1.21)        | -0.152<br>(-0.22)     | -0.070<br>(-0.06)   | -2.730***<br>(-3.59) | 6.835***<br>(6.42)    | 0.876<br>(1.13)            | 5.679***<br>(5.83)    | -2.366***<br>(-7.58)  | -3.586***<br>(-11.28) | -4.123***<br>(-10.61) | -5.510***<br>(-16.58) | -1.319***<br>(-3.65) | -3.461***<br>(-10.14) | -1.398***<br>(-3.55)  |
| CORRUPTION      | -2.889***<br>(-5.50)   | -0.925<br>(-1.45)     | -1.456**<br>(-2.04) | -3.807***<br>(-5.60) | -4.232***<br>(-6.36)  | -2.853***<br>(-4.37)       | -5.821***<br>(-7.05)  | -2.415***<br>(-11.36) | -2.519***<br>(-11.82) | -2.260***<br>(-10.03) | -2.591***<br>(-11.78) | -2.122***<br>(-9.84) | -2.293***<br>(-10.57) | -2.506***<br>(-10.85) |
| BUREAUCRACY     | -4.737***<br>(-4.13)   | -7.511***<br>(-5.05)  | -1.865<br>(-1.51)   | -5.963***<br>(-4.73) | -7.343***<br>(-4.08)  | -9.620***<br>(-6.77)       | -8.741***<br>(-3.08)  | -1.790***<br>(-5.41)  | -2.750***<br>(-7.18)  | -0.116<br>(-0.26)     | -2.121***<br>(-5.86)  | -1.934***<br>(-4.97) | -3.125***<br>(-8.46)  | -2.907***<br>(-5.23)  |
| Intercept       | 46.963***<br>(9.35)    | -32.899***<br>(-5.11) | 29.735***<br>(5.17) | 17.632***<br>(3.12)  | -27.833***<br>(-5.23) | -44.434***<br>(-6.78)      | -40.780***<br>(-3.70) | 26.159***<br>(9.83)   | -7.595***<br>(-2.61)  | 24.815***<br>(8.87)   | 26.239***<br>(8.98)   | -2.328<br>(-0.88)    | -8.715***<br>(-2.99)  | -3.115<br>(-0.48)     |
| Year effect     | Yes                    | Yes                   | Yes                 | Yes                  | Yes                   | Yes                        | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                  | Yes                   | Yes                   |
| Industry effect | Yes                    | Yes                   | Yes                 | Yes                  | Yes                   | Yes                        | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                  | Yes                   | Yes                   |
| N               | 5803                   | 5803                  | 5803                | 5803                 | 5803                  | 5803                       | 5803                  | 21,833                | 21,833                | 21,833                | 21,833                | 21,833               | 21,833                | 21,833                |
| R <sup>2</sup>  | 0.253                  | 0.250                 | 0.242               | 0.241                | 0.256                 | 0.251                      | 0.260                 | 0.216                 | 0.214                 | 0.211                 | 0.211                 | 0.216                | 0.213                 | 0.217                 |

Note: This table presents the estimation results obtained by regressing corporate innovation (LNCITEPAT) on cultural values (UAL, IDV, MAS, PDI, LTO, and IDG) and on controls of firm characteristics and country variables, respectively. The crisis period is defined as years 1997–98, 2000, and 2007–08. The regression tests in Panel A are for crisis period whereas those in Panel B cover non-crisis period. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. The industry controls are based on the 12 Fama-French industry classification codes. t-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 17**  
Subsample analysis: Before vs. after year 2000.

| Panel A: Year 2000 and before |                       |                      |                      |                      |                      |                      |                      |                       |                      |                      |                      |                      |                      |                       | Panel B: After year 2000 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|-------------------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|--------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| VARIABLES                     | (1)                   | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                   | (9)                  | (10)                 | (11)                 | (12)                 | (13)                 | (14)                  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                               | LNCITEPAT             | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT             | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT            | LNCITEPAT             |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| UAI                           | -1.048***<br>(-14.50) |                      |                      |                      |                      |                      | -1.010***<br>(-2.96) | -0.203***<br>(-17.58) |                      |                      |                      |                      |                      | -0.192***<br>(-10.04) |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| IDV                           |                       | 0.963***<br>(9.17)   |                      |                      |                      |                      | -0.001<br>(-0.00)    |                       | 0.218***<br>(12.39)  |                      |                      |                      |                      | -0.040***<br>(-1.97)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| MAS                           |                       |                      | -0.840***<br>(-7.32) |                      |                      |                      | 0.024<br>(0.14)      |                       |                      | -0.099***<br>(-8.64) |                      |                      |                      | 0.072***<br>(5.55)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PDI                           |                       |                      |                      | -0.055<br>(-0.34)    |                      |                      | 0.651***<br>(3.26)   |                       |                      |                      | -0.192***<br>(-7.50) |                      |                      | 0.031<br>(0.68)       |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| TOR                           |                       |                      |                      |                      | 0.831***<br>(11.71)  |                      | 0.393***<br>(3.62)   |                       |                      |                      |                      | 0.201***<br>(15.13)  |                      | 0.099***<br>(3.72)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| IDG                           |                       |                      |                      |                      |                      | 1.212***<br>(11.48)  | -0.319<br>(-1.33)    |                       |                      |                      |                      |                      | 0.305***<br>(12.41)  | 0.017<br>(0.54)       |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| FarEast                       | 10.908<br>(1.26)      | 51.816***<br>(5.40)  | 125.307***<br>(8.99) | 101.051***<br>(9.18) | 28.384***<br>(2.60)  | 51.778***<br>(6.23)  | -24.909<br>(-0.94)   | 15.027***<br>(12.10)  | 17.488***<br>(12.27) | 26.142***<br>(18.01) | 30.856***<br>(17.08) | 12.537***<br>(9.25)  | 13.786***<br>(9.45)  | 8.671***<br>(3.18)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Germanic                      | -3.552<br>(-0.81)     | 1.593<br>(0.32)      | 45.302***<br>(7.59)  | 14.137***<br>(3.05)  | 38.020***<br>(8.08)  | -3.092<br>(-0.61)    | 14.909***<br>(2.00)  | 8.211***<br>(13.10)   | 4.974***<br>(8.65)   | 6.137***<br>(8.36)   | 2.644***<br>(4.96)   | 12.168***<br>(14.54) | 9.418***<br>(12.11)  | 9.279***<br>(6.81)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| LatinAmerica                  | 69.839***<br>(8.16)   | 38.074***<br>(3.24)  | 152.297***<br>(8.95) | 114.405***<br>(9.92) | 21.716*<br>(1.72)    | 48.874***<br>(5.27)  | 50.900***<br>(3.28)  | 12.359***<br>(7.26)   | 5.025***<br>(2.71)   | 17.035***<br>(9.40)  | 14.058***<br>(7.04)  | 2.412<br>(1.35)      | 2.360<br>(1.24)      | 7.505***<br>(3.38)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| LatinEurope                   |                       |                      |                      |                      |                      |                      |                      | 6.152***<br>(5.73)    | -2.663***<br>(-2.33) | 4.406***<br>(3.01)   | 5.678***<br>(4.16)   | 1.127<br>(1.01)      | 0.259<br>(0.23)      | 1.996<br>(1.03)       |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SIZE                          | 30.391***<br>(28.68)  | 30.113***<br>(28.50) | 29.565***<br>(28.30) | 29.223***<br>(28.18) | 30.154***<br>(28.57) | 30.063***<br>(28.53) | 30.490***<br>(28.68) | 9.943***<br>(23.17)   | 9.627***<br>(22.76)  | 9.291***<br>(22.67)  | 9.243***<br>(22.51)  | 9.812***<br>(22.97)  | 9.632***<br>(22.81)  | 9.957***<br>(23.01)   |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| R&D                           | 2.842***<br>(7.89)    | 2.833***<br>(7.88)   | 2.848***<br>(7.91)   | 2.838***<br>(7.90)   | 2.891***<br>(8.01)   | 2.827***<br>(7.86)   | 2.868***<br>(7.96)   | 1.042***<br>(5.57)    | 1.025***<br>(5.48)   | 1.071***<br>(5.69)   | 1.062***<br>(5.65)   | 1.034***<br>(5.52)   | 1.036***<br>(5.53)   | 1.046***<br>(5.58)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| ROA                           | -0.949***<br>(-2.70)  | -0.884***<br>(-2.52) | -0.815***<br>(-2.32) | -0.752***<br>(-2.15) | -0.898***<br>(-2.56) | -0.875***<br>(-2.50) | -0.954***<br>(-2.71) | 0.151<br>(0.80)       | 0.231<br>(1.23)      | 0.326*<br>(1.74)     | 0.327*<br>(1.74)     | 0.186<br>(0.99)      | 0.230<br>(1.22)      | 0.150<br>(0.79)       |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PPE                           | -4.103***<br>(-7.81)  | -4.046***<br>(-7.70) | -4.027***<br>(-7.65) | -4.082***<br>(-7.70) | -4.138***<br>(-7.87) | -4.090***<br>(-7.76) | -4.015***<br>(-7.59) | -0.601***<br>(-3.26)  | -0.625***<br>(-3.39) | -0.683***<br>(-3.70) | -0.694***<br>(-3.76) | -0.642***<br>(-3.48) | -0.657***<br>(-3.56) | -0.616***<br>(-3.34)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| LEV                           | -0.712*<br>(-1.87)    | -0.731*<br>(-1.92)   | -0.645*<br>(-1.70)   | -0.726*<br>(-1.91)   | -0.703*<br>(-1.85)   | -0.677*<br>(-1.78)   | -0.675*<br>(-1.77)   | -0.504***<br>(-3.45)  | -0.437***<br>(-3.00) | -0.396***<br>(-2.73) | -0.392***<br>(-2.69) | -0.484***<br>(-3.32) | -0.452***<br>(-3.10) | -0.508***<br>(-3.48)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| CAPIT                         | 2.293***<br>(7.12)    | 2.300***<br>(7.13)   | 2.216***<br>(6.86)   | 2.255***<br>(6.96)   | 2.273***<br>(7.05)   | 2.362***<br>(7.30)   | 2.251***<br>(6.95)   | 0.951***<br>(5.95)    | 1.020***<br>(6.37)   | 1.010***<br>(6.28)   | 1.034***<br>(6.43)   | 1.012***<br>(6.34)   | 1.040***<br>(6.50)   | 0.981***<br>(6.13)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| CASH                          | -0.892***<br>(-2.12)  | -0.895***<br>(-2.13) | -0.915***<br>(-2.17) | -0.988***<br>(-2.34) | -0.908***<br>(-2.16) | -0.892***<br>(-2.12) | -0.842***<br>(-2.00) | -0.351***<br>(-1.75)  | -0.327***<br>(-1.63) | -0.366***<br>(-1.82) | -0.368***<br>(-1.82) | -0.339***<br>(-1.69) | -0.334***<br>(-1.66) | -0.364***<br>(-1.81)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| DIV                           | 1.507***<br>(3.58)    | 1.487***<br>(3.51)   | 1.963***<br>(4.61)   | 1.883***<br>(4.42)   | 1.748***<br>(4.15)   | 1.568***<br>(3.72)   | 1.577***<br>(3.70)   | 1.630***<br>(8.10)    | 1.599***<br>(7.86)   | 1.729***<br>(8.50)   | 1.730***<br>(8.50)   | 1.642***<br>(8.13)   | 1.651***<br>(8.16)   | 1.640***<br>(8.06)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| GDPG                          | 0.970<br>(1.50)       | 1.771***<br>(2.46)   | 1.493***<br>(2.16)   | 2.596***<br>(3.17)   | 1.458***<br>(2.22)   | 1.555***<br>(2.31)   | 0.382<br>(0.61)      | -0.616***<br>(-2.99)  | 0.112<br>(0.54)      | 0.375*<br>(1.83)     | 0.537*<br>(2.55)     | 0.085<br>(0.43)      | 0.181<br>(0.91)      | -0.533***<br>(-2.60)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| STKTURNOVER                   | 2.548***<br>(2.30)    | 2.883***<br>(2.58)   | 3.945***<br>(3.06)   | 4.809***<br>(3.58)   | 3.689***<br>(3.28)   | 2.664***<br>(2.43)   | 1.991***<br>(2.31)   | 1.398***<br>(6.89)    | 1.255***<br>(6.13)   | 1.308***<br>(6.43)   | 1.193***<br>(5.88)   | 1.273***<br>(6.31)   | 1.018***<br>(4.99)   | 1.366***<br>(6.56)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| CREDIT                        | 4.782***<br>(5.04)    | 1.160<br>(1.00)      | -2.181***<br>(-2.24) | -6.419***<br>(-5.76) | -1.038<br>(-1.33)    | -2.132<br>(-1.62)    | 2.442<br>(1.89)      | -1.445***<br>(-5.11)  | -0.891***<br>(-3.09) | 0.041<br>(0.14)      | -0.209<br>(-0.66)    | -0.501*<br>(-1.82)   | -1.274***<br>(-4.19) | -1.321***<br>(-4.04)  |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| EDUCATION                     | 15.401***<br>(17.80)  | 4.176***<br>(3.69)   | 23.221***<br>(11.48) | 11.164***<br>(10.38) | 5.271***<br>(5.75)   | 8.784***<br>(10.39)  | 9.147***<br>(3.38)   | 6.850***<br>(20.70)   | 5.208***<br>(16.26)  | 7.268***<br>(19.46)  | 6.967***<br>(20.08)  | 5.248***<br>(16.92)  | 5.455***<br>(17.07)  | 5.506***<br>(13.10)   |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PTY_RIGHTS                    | 0.976<br>(0.97)       | 4.325***<br>(4.20)   | 9.282***<br>(6.32)   | 3.108***<br>(2.88)   | 7.459***<br>(7.33)   | 2.070*<br>(1.79)     | 1.602<br>(0.64)      | 2.649***<br>(6.66)    | 1.726***<br>(3.97)   | 2.802***<br>(6.34)   | 1.568***<br>(3.59)   | 4.054***<br>(9.49)   | 2.531***<br>(5.98)   | 2.520***<br>(4.73)    |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| (continued on next page)      |                       |                      |                      |                      |                      |                      |                      |                       |                      |                      |                      |                      |                      |                       |                          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

(continued on next page)

Table 17 (continued)

|                 | Panel A: Year 2000 and before |                       |                     |                      |                       |                       | Panel B: After year 2000 |                       |                       |                       |                       |                       |                       |                       |
|-----------------|-------------------------------|-----------------------|---------------------|----------------------|-----------------------|-----------------------|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                 | (1)                           | (2)                   | (3)                 | (4)                  | (5)                   | (6)                   | (7)                      | (8)                   | (9)                   | (10)                  | (11)                  | (12)                  | (13)                  | (14)                  |
| VARIABLES       | LNCITEPAT                     | LNCITEPAT             | LNCITEPAT           | LNCITEPAT            | LNCITEPAT             | LNCITEPAT             | LNCITEPAT                | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             | LNCITEPAT             |
| EXP_RISK        | 9.292***<br>(4.66)            | 10.229***<br>(4.43)   | 17.420***<br>(7.31) | 21.640***<br>(7.97)  | 10.979***<br>(4.84)   | 22.098***<br>(8.64)   | 10.783***<br>(3.63)      | -2.067***<br>(-4.69)  | -2.867***<br>(-5.69)  | -2.307***<br>(-3.93)  | -2.126***<br>(-4.08)  | -3.301***<br>(-6.56)  | -2.111***<br>(-4.44)  | -0.769<br>(-1.29)     |
| NANTI_DIR       | -2.833***<br>(-2.48)          | -1.803<br>(-1.21)     | 16.484***<br>(8.77) | 7.348***<br>(6.32)   | 8.258***<br>(7.57)    | 2.886***<br>(2.53)    | -0.518<br>(-0.19)        | -1.174***<br>(-4.34)  | -2.361***<br>(-7.72)  | -3.926***<br>(-12.15) | -4.387***<br>(-13.55) | -0.958***<br>(-3.19)  | -2.180***<br>(-6.55)  | -1.072***<br>(-3.25)  |
| CORRUPTION      | -0.800<br>(-1.36)             | -2.686***<br>(-4.37)  | -1.006<br>(-1.41)   | -5.512***<br>(-7.64) | -0.090<br>(-0.13)     | -1.023<br>(-1.62)     | -0.167<br>(-0.20)        | -2.476***<br>(-13.06) | -2.510***<br>(-13.12) | -2.340***<br>(-11.70) | -2.481***<br>(-12.71) | -2.235***<br>(-11.62) | -2.357***<br>(-12.13) | -2.565***<br>(-12.53) |
| BUREAUCRACY     | -8.586***<br>(-4.43)          | -13.647***<br>(-4.95) | 2.709<br>(1.11)     | -2.723<br>(-1.19)    | -8.363***<br>(-3.75)  | -15.592***<br>(-5.99) | -6.998*<br>(-2.49)       | -1.499***<br>(-5.95)  | -2.005***<br>(-6.96)  | -0.000<br>(-0.00)     | -1.185***<br>(-4.13)  | -1.416***<br>(-5.10)  | -2.311***<br>(-8.21)  | -2.751***<br>(-7.04)  |
| Intercept       | 79.482***<br>(12.75)          | -51.394***<br>(-5.40) | 73.453***<br>(8.93) | 24.903***<br>(3.09)  | -25.377***<br>(-3.98) | -51.129***<br>(-6.32) | 53.124<br>(1.60)         | 7.080***<br>(4.09)    | -21.191***<br>(-8.95) | 1.335<br>(0.76)       | 3.141*<br>(1.66)      | -16.187***<br>(-7.89) | -23.262***<br>(-9.27) | -3.154<br>(-0.62)     |
| Year effect     | Yes                           | Yes                   | Yes                 | Yes                  | Yes                   | Yes                   | Yes                      | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Industry effect | Yes                           | Yes                   | Yes                 | Yes                  | Yes                   | Yes                   | Yes                      | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| N               | 8221                          | 8221                  | 8221                | 8221                 | 8221                  | 8221                  | 8221                     | 19,415                | 19,415                | 19,415                | 19,415                | 19,415                | 19,415                | 19,415                |
| R <sup>2</sup>  | 0.287                         | 0.285                 | 0.282               | 0.279                | 0.285                 | 0.285                 | 0.288                    | 0.190                 | 0.186                 | 0.184                 | 0.184                 | 0.188                 | 0.186                 | 0.190                 |

Note: This table presents the estimation results obtained by regressing corporate innovation (LNCITEPAT) on cultural values (UAL, IDV, MAS, PDI, LTO, and IDG) and on controls of firm characteristics and country variables, respectively. The regression tests in Panel A include the sample period of year 2000 and before whereas those in Panel B cover the sample period after year 2000. The appendix outlines the definitions and data sources for the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. The industry controls are based on the 12 Fama-French industry classification codes. t-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

**Table 18**

Uncertainty avoidance and risk taking.

| VARIABLES                        | (1)<br><i>LNCITEPAT</i> | (2)<br><i>LNCITEPAT</i> | (3)<br><i>LNCITEPAT</i> |
|----------------------------------|-------------------------|-------------------------|-------------------------|
| <i>UAI</i>                       | −0.001<br>(−0.36)       | −0.000<br>(−0.25)       | −0.002<br>(−0.93)       |
| <i>UAI</i> × <i>RISK_TAKING1</i> | −0.060*<br>(−1.95)      |                         |                         |
| <i>RISK_TAKING1</i>              | 2.768*<br>(1.94)        |                         |                         |
| <i>UAI</i> × <i>RISK_TAKING2</i> |                         | −0.015**<br>(−2.19)     |                         |
| <i>RISK_TAKING2</i>              |                         | 0.908***<br>(2.80)      |                         |
| <i>UAI</i> × <i>RISK_TAKING3</i> |                         |                         | −0.410***<br>(−10.88)   |
| <i>RISK_TAKING3</i>              |                         |                         | 19.360***<br>(7.96)     |
| <i>FarEast</i>                   | 0.295<br>(1.19)         | 0.554**<br>(2.22)       | −0.234*<br>(−1.89)      |
| <i>Gemanic</i>                   | 0.411***<br>(3.66)      | 0.360***<br>(3.52)      | 0.809***<br>(14.58)     |
| <i>LatinAmerica</i>              | 0.107<br>(0.57)         | 0.127<br>(0.72)         | 1.124***<br>(8.40)      |
| <i>LatinEurope</i>               | −0.983***<br>(−3.66)    | −0.824***<br>(−3.36)    | −0.087<br>(−0.57)       |
| <i>SIZE</i>                      | 1.135***<br>(60.52)     | 1.034***<br>(59.96)     | 0.733***<br>(56.57)     |
| <i>R&amp;D</i>                   | 0.111***<br>(9.76)      | 0.103***<br>(10.22)     | 0.092***<br>(9.71)      |
| <i>ROA</i>                       | −0.044***<br>(−3.63)    | −0.006<br>(−0.55)       | 0.028***<br>(3.09)      |
| <i>PPE</i>                       | −0.161***<br>(−11.01)   | −0.143***<br>(−10.90)   | −0.081***<br>(−8.54)    |
| <i>LEV</i>                       | −0.042***<br>(−4.64)    | −0.034***<br>(−4.22)    | −0.026***<br>(−3.84)    |
| <i>CAPIT</i>                     | 0.132***<br>(12.70)     | 0.125***<br>(13.30)     | 0.089***<br>(12.08)     |
| <i>CASH</i>                      | 0.007<br>(0.69)         | 0.009<br>(0.97)         | 0.013<br>(1.46)         |
| <i>DIV</i>                       | 0.075***<br>(7.56)      | 0.075***<br>(8.13)      | 0.087***<br>(11.65)     |
| <i>GDPG</i>                      | 0.089***<br>(3.25)      | 0.100***<br>(3.47)      | 0.025*<br>(1.67)        |
| <i>STKTURNOVER</i>               | 0.110***<br>(3.47)      | 0.122***<br>(3.91)      | 0.035***<br>(2.71)      |
| <i>CREDIT</i>                    | −0.270***<br>(−12.44)   | −0.239***<br>(−11.23)   | −0.085***<br>(−8.64)    |
| <i>EDUCATION</i>                 | 0.795***<br>(20.30)     | 0.720***<br>(19.62)     | 0.542***<br>(11.17)     |
| <i>PTY_RIGHTS</i>                | −0.045<br>(−0.77)       | 0.002<br>(0.04)         | 0.312***<br>(12.31)     |
| <i>EXP_RISK</i>                  | 0.293***<br>(3.27)      | 0.224***<br>(2.79)      | 0.088*<br>(1.95)        |
| <i>NANTI_DIR</i>                 | −0.246***<br>(−6.91)    | −0.259***<br>(−8.28)    | −0.260***<br>(−15.57)   |
| <i>CORRUPTION</i>                | −0.160***<br>(−7.61)    | −0.159***<br>(−8.09)    | −0.143***<br>(−14.45)   |
| <i>BUREAUCRACY</i>               | −0.349***<br>(−5.83)    | −0.284***<br>(−5.10)    | −0.326***<br>(−10.09)   |
| Constant                         | 2.541***<br>(12.75)     | 2.050***<br>(9.65)      | 1.077***<br>(5.58)      |
| Year effect                      | Yes                     | Yes                     | Yes                     |
| Industry effect                  | Yes                     | Yes                     | Yes                     |
| Observations                     | 16,042                  | 18,188                  | 26,498                  |
| R-squared                        | 0.427                   | 0.407                   | 0.503                   |

Note: This table presents the estimation results obtained by regressing corporate innovation (*LNCITEPAT*) on *UAI* cultural value, three measures of risk-taking and their interaction with *UAI*, and on controls of firm characteristics and country variables, respectively. In column (1), risk-taking is measured using the volatility of a firm's earnings over four-years (*RISK\_TAKING1*). In Column (2), *RISK\_TAKING2* is the difference between the maximum and minimum *ROA* reported over the 5-year interval. In Column (3), *RISK\_TAKING3* is the country-level average of all the firm-level observations of *RISK\_TAKING1*. The appendix outlines the definitions and data sources for

the regression variables. For the sake of brevity, industry and year controls are included, but the results are not tabulated. We winsorize all variables at the 1% and 99% levels and standardize all control variables to zero mean and one standard deviation. The industry controls are based on the 12 Fama-French industry classification codes. *t*-statistics based on robust standard errors, clustered at the firm level, are reported in parentheses.

\*\*\* Significant at the 1% level.

\*\* Significant at the 5% level.

\* Significant at the 10% level.

## Appendix A

### A.1. Variable definitions and data sources

| Variables                                                                                                              | Descriptions                                                                                                                                                                                                                                                                                                                                                     | Sources                                                                |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Panel A: Measures of Innovation and Cultural Variables                                                                 |                                                                                                                                                                                                                                                                                                                                                                  |                                                                        |
| $IN\_DUMMY_{t+1}$                                                                                                      | Dummy variable equals one if the firm has patents and citations in year $t + 1$ , and zero otherwise.                                                                                                                                                                                                                                                            | Authors' calculation from the Derwent Innovation database.             |
| $ROLL\_DUMMY$                                                                                                          | Dummy variable equals one if the firm has patents and at least one citation during the three (then five) subsequent years, and zero otherwise.                                                                                                                                                                                                                   | Authors' calculation from the Derwent Innovation database.             |
| $PATENTS_{t+1}$<br>$CITATIONS_{t+1}$                                                                                   | $PATENTS$ is firms' total number of patents filed (and eventually granted) in year $t + 1$ , whereas $CITATIONS$ is firms' total number of citations received in year $t + 1$ .                                                                                                                                                                                  | Authors' calculation from the Derwent Innovation database.             |
| $LNCITEPAT_{t+1}$                                                                                                      | $LNCITEPAT$ is the natural logarithm of one plus a firm's total number of citations received, scaled by the average patent count (filed and eventually granted) of all patents in the same country and industry class in year $t + 1$ .                                                                                                                          | Authors' calculation from the Derwent Innovation database.             |
| $LNCITE_{t+1}$                                                                                                         | $LNCITE$ is the natural logarithm of one plus a firm's total number of citations received, scaled by the average citation count of all patents in the same industry class and year in year $t + 1$ .                                                                                                                                                             | Authors' calculation from the Derwent Innovation database.             |
| $LNCITEPAT\_NADJ_{t+1}$                                                                                                | $LNCITEPAT\_NADJ$ is the natural logarithm of one plus a firm's total number of citations received, scaled by the number of the patents filed (and eventually granted) in year $t + 1$ .                                                                                                                                                                         | Authors' calculation from the Derwent Innovation database.             |
| $LNCITEPAT\_NBER_{t+1}$                                                                                                | $PATENTS$ is firms' total number of patents granted in year $t + 1$ , whereas $CITATIONS$ is firms' total number of citations received in year $t + 1$ . $LNCITEPAT$ is the natural logarithm of one plus a firm's total number of citations received, scaled by the average number of the patents filed in a country (and eventually granted) in year $t + 1$ . | Authors' calculation based on the NBER patents and citations database. |
| $UAI$                                                                                                                  | Hofstede's cultural index of uncertainty avoidance.                                                                                                                                                                                                                                                                                                              | Hofstede (2001).                                                       |
| $IDV$                                                                                                                  | Hofstede's cultural index of individualism.                                                                                                                                                                                                                                                                                                                      | Hofstede (2001).                                                       |
| $MAS$                                                                                                                  | Hofstede's cultural index of masculinity.                                                                                                                                                                                                                                                                                                                        | Hofstede (2001).                                                       |
| $PDI$                                                                                                                  | Hofstede's cultural index of power distance.                                                                                                                                                                                                                                                                                                                     | Hofstede (2001).                                                       |
| $TOR$                                                                                                                  | Hofstede and Bond's cultural index of time orientation.                                                                                                                                                                                                                                                                                                          | Hofstede and Bond (1988).                                              |
| $IDG$                                                                                                                  | Hofstede, Hofstede, and Minkov's cultural index of indulgence.                                                                                                                                                                                                                                                                                                   | Hofstede et al. (2010).                                                |
| $UAI\_TK$                                                                                                              | Revised Hofstede's cultural index of uncertainty avoidance.                                                                                                                                                                                                                                                                                                      | Tang and Koveos (2008).                                                |
| $IDV\_TK$                                                                                                              | Revised Hofstede's cultural index of individualism.                                                                                                                                                                                                                                                                                                              | Tang and Koveos (2008).                                                |
| $MAS\_TK$                                                                                                              | Revised Hofstede's cultural index of masculinity.                                                                                                                                                                                                                                                                                                                | Tang and Koveos (2008).                                                |
| $PDI\_TK$                                                                                                              | Revised Hofstede's cultural index of power distance.                                                                                                                                                                                                                                                                                                             | Tang and Koveos (2008).                                                |
| $TOR\_TK$                                                                                                              | Revised Hofstede and Bond's cultural index of time orientation.                                                                                                                                                                                                                                                                                                  | Tang and Koveos (2008).                                                |
| African, Anglo, Arabic, Nordic, Germanic, Latin America, Near East, Latin Europe, East Europe, Far East, and Confucian | Ecocultural global clusters identified by Ronen and Shenkar (2013).                                                                                                                                                                                                                                                                                              | Ronen and Shenkar (2013).                                              |
| Panel B: Firm-Level Control Variables                                                                                  |                                                                                                                                                                                                                                                                                                                                                                  |                                                                        |
| $SIZE_t$                                                                                                               | Natural logarithm of total book assets ( $AT$ ) in year $t$ .                                                                                                                                                                                                                                                                                                    | Calculated from Compustat NA and Global.                               |
| $R\&D_t$                                                                                                               | Ratio of research and development expenditure to total book assets in year $t$ .                                                                                                                                                                                                                                                                                 | Calculated from Compustat NA and Global.                               |
| $ROA_t$                                                                                                                | Ratio of operating income before depreciation to total book assets in year $t$ .                                                                                                                                                                                                                                                                                 | Calculated from Compustat NA and Global.                               |
| $PPE_t$                                                                                                                | Ratio of property, plant, and equipment expenditure to total book assets in year $t$ .                                                                                                                                                                                                                                                                           | Calculated from Compustat NA and Global.                               |
| $LEV_t$                                                                                                                | Ratio of book value of debt (long-term debt plus debt in current liabilities) to total book assets in year $t$ .                                                                                                                                                                                                                                                 | Calculated from Compustat NA and Global.                               |
| $CAPIT_t$                                                                                                              | Ratio of capital expenditures to total book assets in year $t$ .                                                                                                                                                                                                                                                                                                 | Calculated from Compustat NA and Global.                               |
| $CASH_t$                                                                                                               | Ratio of cash and short-term investments to total book assets in year $t$ .                                                                                                                                                                                                                                                                                      | Calculated from Compustat NA and Global.                               |
| $DIV_t$                                                                                                                | Ratio of common dividends to total book assets in year $t$ .                                                                                                                                                                                                                                                                                                     | Calculated from Compustat NA and Global.                               |
| Panel C: Country-Level Control Variables                                                                               |                                                                                                                                                                                                                                                                                                                                                                  |                                                                        |

(continued on next page)

(continued)

| Variables                                             | Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                         | Sources                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <i>GDPG</i>                                           | A country's annual GDP growth.                                                                                                                                                                                                                                                                                                                                                                                                       | World Development Indicators (WDI).                        |
| <i>STKTURNOVER</i>                                    | Stocks' turnover ratio traded in a country's equity markets.                                                                                                                                                                                                                                                                                                                                                                         | WDI.                                                       |
| <i>CREDIT</i>                                         | Domestic credit through the financial sector scaled by a country's GDP. The financial sector includes monetary authorities, deposit money banks, and other financial corporations, such as finance and leasing companies, moneylenders, insurance corporations, pension funds, and foreign exchange companies.                                                                                                                       | WDI.                                                       |
| <i>EDUCATION</i>                                      | A country's educational attainment, measured by the percentage of its population that has completed tertiary education.                                                                                                                                                                                                                                                                                                              | Barro and Lee (2013).                                      |
| <i>PTY_RIGHT</i>                                      | International patent protection index measures a country's adoption of national patent laws and composition of patent rights. The protection scores range from zero to five, with higher values indicating the adoption of stronger patent laws.                                                                                                                                                                                     | Ginarte and Park (1997), and Park (2008).                  |
| <i>EXP_RISK</i>                                       | Risk of expropriation measures the quality of market institutions, ranging from zero to 10, with higher scores representing better institutions and lower scores indicating the risk of confiscation and forced nationalization.                                                                                                                                                                                                     | McArthur and Sachs (2001), and Tebaldi and Elmslie (2008). |
| <i>NANTI_DIR</i>                                      | Revised anti-director rights index. An aggregate index of shareholder rights, formed by summing: (1) votes by mail; (2) shares not deposited; (3) cumulative voting; (4) oppressed minorities; (5) pre-emptive rights; and (6) capital required to call a meeting.                                                                                                                                                                   | Djankov et al. (2008).                                     |
| <i>CORRUPTION</i>                                     | Political and financial corruption under a country's juridical system, ranging from zero to six, with a higher score representing a lower level of actual or potential corruption.                                                                                                                                                                                                                                                   | International Country Risk Guide (ICRG).                   |
| <i>BUREAUCRACY</i>                                    | A country's quality of bureaucracy and institutions, ranging from zero to six, with higher values reflecting a better quality of institutions.                                                                                                                                                                                                                                                                                       | ICRG.                                                      |
| Panel D: Instrumental Variable<br><i>DISEASES_IDX</i> | An index of the prevalence of nine infectious diseases. These diseases are leishmaniasis, schistosomes, trypanosomes, leprosy, malaria, typhus, filariae, dengue fever, and tuberculosis. The index is built on a four-point coding scheme: zero is completely absent or never reported, one is rarely reported, two is sporadically or moderately reported, and three is present at severe levels or epidemic levels at least once. | Murray and Schaller 2010.                                  |

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